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BIOLOGICAL SCIENCE

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Aim and Scope of the periodical:

Each year, the Honours students at Rhodes University produce wonderfully written and researched assignments, including writing reviews of various topics and media types and conducting full research projects. Within the Zoology & Entomology Honours course, these submissions are handed in at various points during the year, and this periodical (compiled twice a year) aims to compile the Review Article, Podcast Review, Book Review, and final Research Project manuscript submissions of each Honours class. This periodical is not intended to replace the publication of work by students, but rather to provide students with the experience of having their work "published", and to bring their work to a wider audience. We hope to showcase the work of the Honours students, to hopefully make the journey from graduating Honours to obtaining a job or a postgraduate position a little easier!

BIOLOGICAL SCIENCES

Volume 1, Issue 1 (August 2022)

CONTENTS:

Review Articles

W.T. Sexton and C. Owen: Invasive species: Nature's resilience? Or nature's demise? [2-10]

E. Boon and P.W. Froneman: Why is biology unique? [11-15]

S. Ngxekisa and B. Smit: Current theories on evolutionary drivers for high body temperatures in birds and mammals [16-22]

S. Banoo and C. Owen: Are the Shell seismic surveys off South Africa's Wild Coast really as damaging as social media portrays? [23-29]

B. de la Fontaine and S. Edwards: Future Imperfect: The Ethics of Human Gene Editing with CRISPR-Cas9 [30-39]

M. Faul and N. Mggatsa: Rewilding; are we doing it the right way and does it work? [40-44]

G.K. Nicolau, L. Verburgt and N. Mggatsa: Can informally protected areas provide the safe haven required for South African reptiles? An assessment of the total conservation estate in South Africa and its potential for reptile conservation [45-57]

K. Lumphondo and B. Smit: Why do bats carry so many zoonotic viruses? In the height of the current Covid-19 pandemic, this has been a hot topic [58-63]

S. Manqele, S. Edwards, and R. Tandlich: Do Animals Have Rights? [64-68]

S.S. Samy, P.W. Froneman, C.D. McQuaid, and K.C.K. Ma: A review of decapod biological invasions of polar waters [69-74]

Podcast Reviews

K. Lumphondo [75-82]

M. Faul [83-90]

S. Samy [91-97]

S. Ngxekisa [98-107]

G.K. Nicolau [108-114]

E. Boon [115-118]

S. Manqele [119-126]

W. Sexton [127-133]

B. de la Fontaine [134-141]

S. Banoo [147-148]

Invasive species: Nature's resilience? Or nature's demise?

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ABSTRACT

Biological invasions are said to be one of the leading causes of biodiversity loss. However, as the field of invasion biology has progressed, skeptics have found flaws within definitions and ideas with the field of invasion biology. To date, conversations between skeptics and invasion biologists have been lacking and needs to be addressed. In this review we looked at some of the controversy around invasion biology and focused on definitions of “invasive”, “native”, “pristine”. We also spent time focusing on whether invasions cause extinctions as well as the idea of why we as humans spend time trying to consistently “control” nature in our view. We end off with answering whether we should be controlling invasive species.

Keywords: controversy, ecosystems, anthropogenic disturbance.

I. INTRODUCTION

Biological invasions are said to be one of the leading causes of biodiversity loss (figure 1) (Reaser et al. 2007, Mazor et al. 2018, Wald et al. 2019). This had led to worldwide attention and because of this, invasion biology has become a well-studied and respected field in ecology and conservation biology (Simberloff & Stiling 1996). Invasive species, as shown by many studies, can have catastrophic impacts on environments that stem into damaging the local economy and communities, and can act as a vector for new diseases (Nalepa & Schloesser 1992, Wilson et al. 2007, Coetzee & Hill 2012, Coetzee et al. 2014). It is for these reasons, conservationists and invasion biologists have tried to find ways to combat invasions and implement control strategies to better cope with them (Larson et al. 2011, Ormsby & Brenton-Rule 2017).

However, throughout the years of this evolving field of invasion biology, there have been some skeptics (Richardson & Ricciardi 2013). For example, some skeptics beg to question, is it in our right to eradicate invasive species and element control strategies? Are invasive species a symptom of nature repairing itself after degradation through a variety of anthropogenic factors? Part of the issue for skeptics is that the use of adjectives like ‘invasive’, ‘alien’, and ‘exotic’ could be subconsciously associated with other terms that could lead to different interpretation and confusion in ecological theory (Colautti & MacIsaac 2004). There are various philosophical and religious beliefs

around invasive species and ecosystem functioning that require addressing.

Support from the public and local communities is a crucial factor in the maintenance of management programmes to remove or control invasive species (Bremner & Park 2007). At present, there is a lacking number of reviews around this topic and the skepticism surrounding invasion biology is important to address to improve public support. This review aims at assessing some arguments around invasion biology and questioning whether they really do require control by humans or whether they are a natural process that should be left without intervention. It achieves this by discussing different aspects and terminology frequently used in invasion biology, providing an example of where these definitions might not fully hold, and then answering the questions raised by skeptics.

II. THE SPECIES: NATIVE VS INVASIVE

Invasive species are usually described as ‘non-native’. “Native” organisms are those that occur in a given area where they have evolved with their local environment and are “within their natural range” (Webb 1985). Conversely, invasive species are described as those inhabiting new environments (Mooney et al. 2005). They are usually successful in this new environment because they are free of their natural enemies, competitors, parasites and diseases that usually control their populations in their native range (figure 2) (Keane 2002, Lovell et al. 2006).

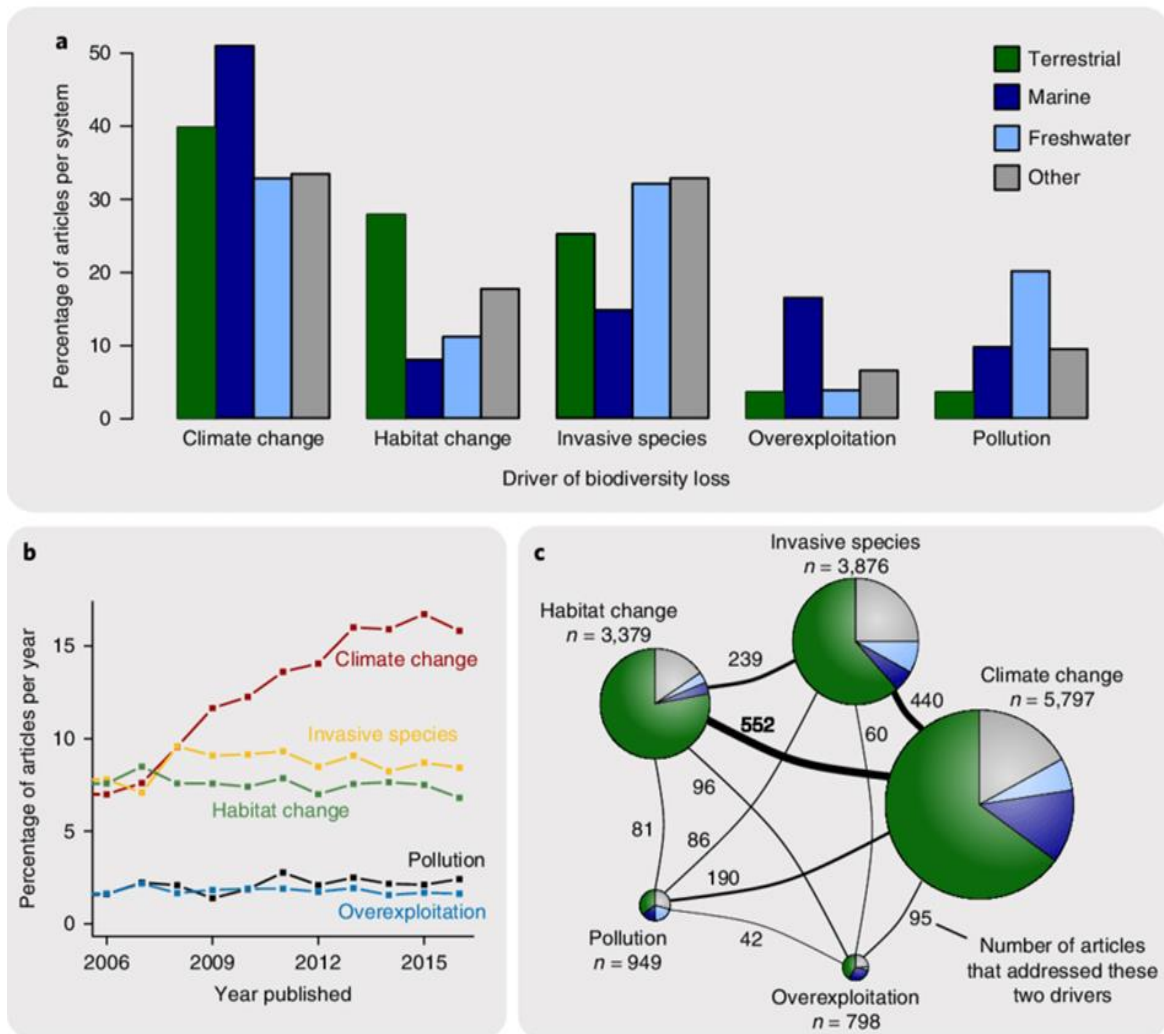


Figure 1: The amount of articles showing the relevant drivers of biodiversity loss. (Mazor et al. 2018).

Invasive species are also often able to exploit resources better than some of the local species they interact with (David et al. 2017). Often, these invasive species are able to dedicate more resources to growth and reproduction and hence can reduce or eliminate populations of native species through predation, competition, the disruption of community assemblages, or causing economic or social damage (Holmes et al. 2009, Coetzee & Hill 2012). However, for some this delineation between invasive species and native species is somewhat skewed in the literature.

European honeybees (EHB) (*Apis mellifera*), which occur today throughout the globe and are used heavily in agriculture for their production of honey (Southwick & Southwick 1992, vanEngelsdorp & Meixner 2010), are a good

example of the contention surrounding the definitions of native versus invasive. Honey is a pricey export for many countries and contributes to many local economies (Southwick & Southwick 1992), highlighting the importance of the species for economies. This species fits the term “native” within European borders but is described as “indigenized” in other environments around the world. If a species is said to be “indigenized”, it describes a species as integrated in the environment because there is either an economical or environmental benefit to it being present in the environment. EHB are generalist pollinators that can pollinate a range of plants (Williams 2002, Donkersley et al. 2017).

This generalist behavior is put to use by packing thousands of domesticated bees in boxes and transporting them all across the U.S.A to pollinate large monocultures such as almond plantations (Ollerton et al. 2012).

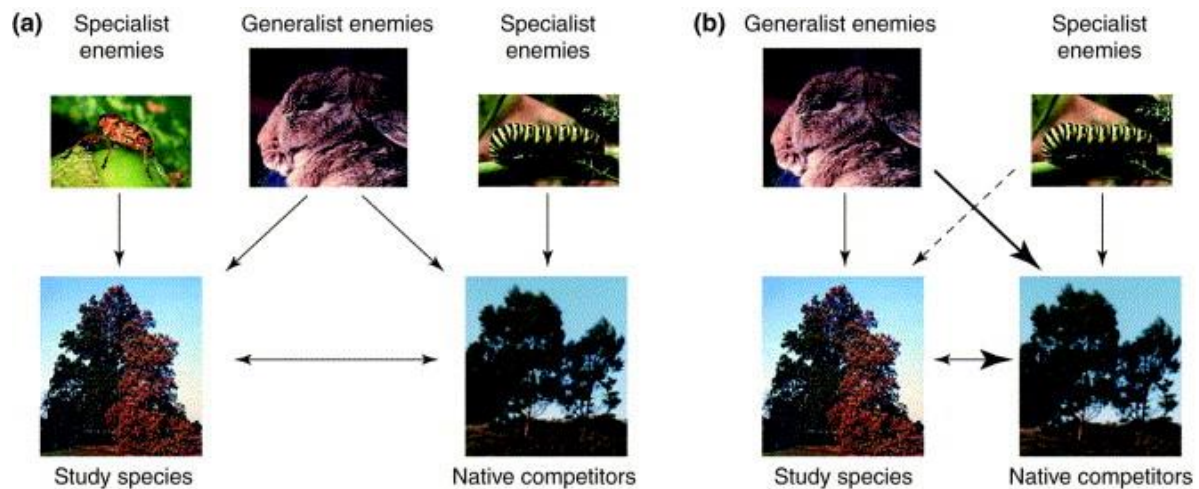


Figure 2: The enemy release hypothesis shows a.) specialist enemies and generalist enemies in a native community whilst b.) the same interactions in an exotic community (Keane 2002).

Many of the bees that are packed up and sent back to their original sites after providing the pollination service have now interacted with other wild bee colonies and may have come into contact with pathogens and parasites, which can spread all around the world as honeybees are traded to farmers around the globe (figure 3) (Ellis & Munn, 2005). This can have drastic effects on colonies and other “native” bees (Ellis & Munn 2005, Mutinelli 2011) and displays the overuse of the EHB, which is causing a rapid decline in other “native” bee species (Ellis & Munn 2005). In Australia, a study showed that the presence of EHB in an environment lowered the fecundity of a local Australian native bee (*Hylaeus alcyoneus*) (Paini & Roberts 2005). A similar trend has been recorded across many other countries where foraging efficiency and thus population sizes of native bees have been affected (Freitas et al. 2009). Factors such as the spread of pathogens and diseases from EHB mentioned above, as well as competition for resources are reasons for native bee declines (Thomson 2004). Apart from the introduction of the EHB, one other “non-native” honeybee is spreading its way through the American landscapes, which contributes to the complexity of the example (figure 4). Sixty years ago in Sao Paulo, Brazil, a colony of African honeybees (*Apis mellifera scutellata*) came into contact with a EHB population, which resulted in a hybrid now known as the African killer bee (AKB) (Calfee et al. 2020). As this new hybrid species moved north from Brazil, they invaded the agricultural colonies of the EHB, killing the queen of the colonies and causing a functional collapse of them (Winston 1992). The main issue with this is that these bees do not produce much honey that could be profitable for farmers

(Howard 2012). The AKB is currently deemed an invasive threat.

This example illustrates the possible contradiction and tension in the definitions of native and invasive. If the EHB has become indigenized and accepted as part of the ecosystem, then why is the AKB regarded as invasive? Both bee species cause ecological damage through their impacts on native bee species. North America alone is said have around 4000 species of native bee species (Woodard et al. 2020). Is it not important to protect these species from harm from both the EHB and the AKB? Based on the definition of an invasive species given above, why is the EHB classed as indigenized, while the AKB is called invasive? Are our definitions of natural and invasive species purely for protecting economies or the natural environment?

III. THE HABITAT: PRISTINE VS INVADED HABITATS

The next area of contention are the definitions of the habitats themselves. All countries have environments and reserves that are deemed “pristine” and thus are worth protecting (Duffey 1982, Parker & Goerck 1997). A pristine environment is one that is described as an undisturbed natural environment that exists with organisms that have evolved alongside it throughout the years (Cronon 1996). One of the main concerns in a pristine environment is the threat of an invasive species that has the potential to colonize and damage the ecosystem functioning (Stachowicz & Tilman 2005).

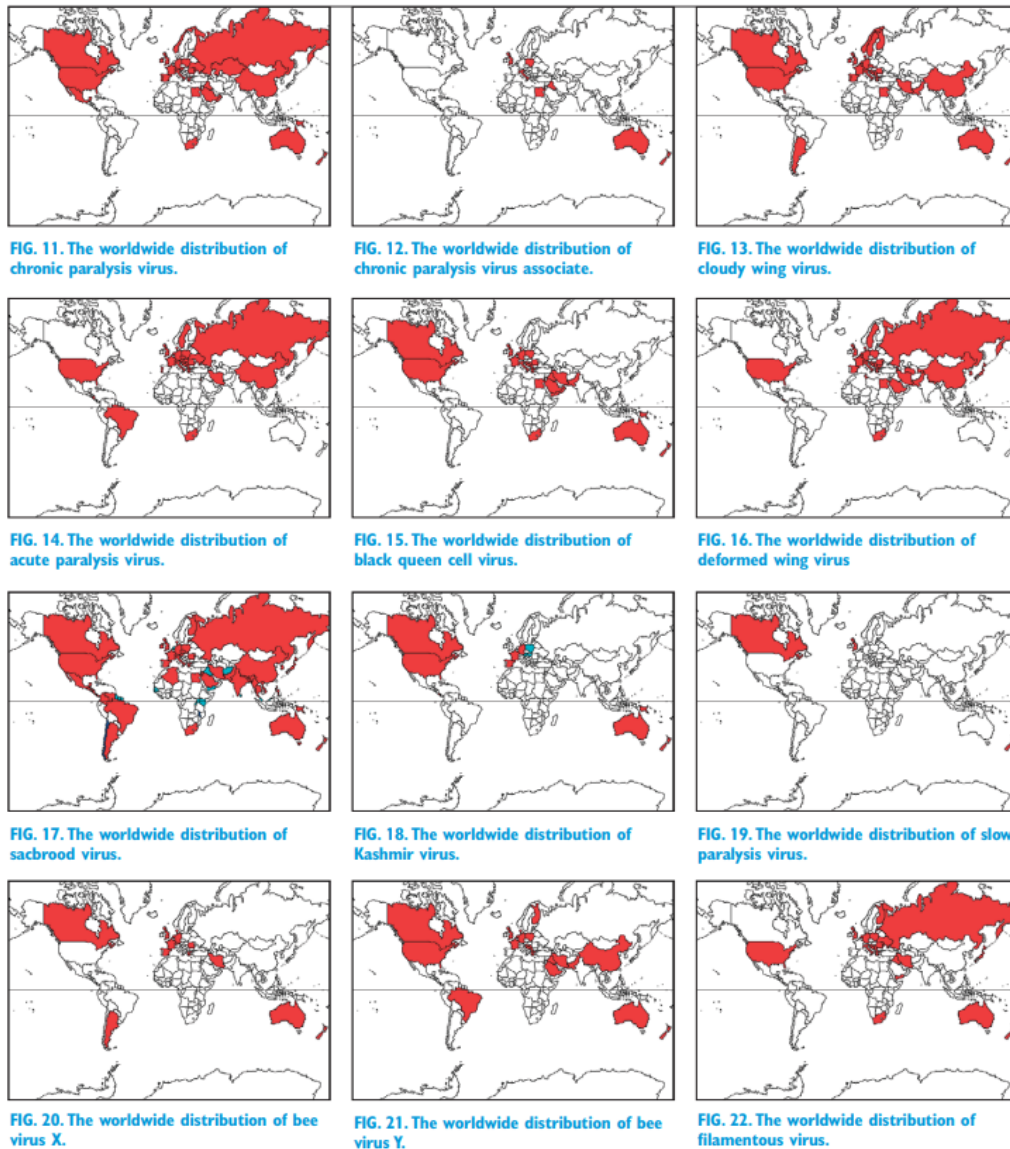


Figure 3: The distribution of different honey bee viruses around the world (Ellis & Munn, 2005)

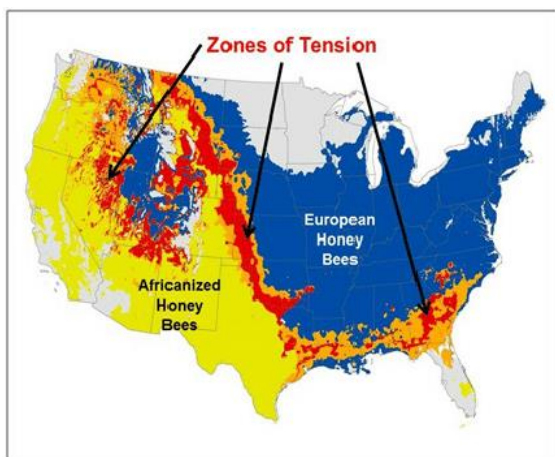


Figure 4: Spread of African killer bee into the U.S.A as of 2014 (Stohlgren et al. 2014)

Many parks are protected due to large amounts of biodiversity and are also used to create eco-tourism that benefit the wildlife as well as the economy (Mayer 2014). However, the notion of calling a pristine environment such is met with some skepticism. Some argue that a truly pristine environment does not exist in the globe at present and with the extensive pollution and other anthropogenic impacts we are seeing, evidence has shown that some of the environments we have called “pristine” in the past and now shown to have been molded by anthropogenic activities. (MacDougall & Turkington 2005).

One example that highlights this contention is the historical land use of the Maasai culture, which has been criticized in the past for overgrazing certain areas of land in Kenya, resulting in the degradation

of many “natural” landscapes (Cronon 1996, Phillips 1998). However, the pristine Serengeti landscapes and other great African wildlife parks that we are used to seeing huge Blue Wildebeest (*Connochaetes taurinus*) migrations in popular wildlife documentaries are said to have been developed through the past use of cattle grazing by the Maasai over hundreds of years and there is evidence to support this (Phillips 1998). There have been rigorous debates for many years about these “pastoral” lands, and whether they should be preserved for wildlife reserves and national parks or whether they should be used by the local Maasai tribes as they historically were. If these lands have a long history of cattle grazing by the Maasai culture, is it fair to say the Serengeti is “pristine”. Throughout time, there are many examples of landscapes that have been molded and manipulated by human kind for more than 500 years and are now considered to be pristine (Newson 1996, Romme et al. 2011).

So, what does pristine really mean? If environments have been shaped by humans and “disturbed” for almost 1000 years, then are they not damaged ecosystems now, and should we be protecting them? With the fact that these areas maybe do not represent thousands of years of natural and undisturbed evolution, should we be redefining our classifications and justifications of a “pristine” environment? Or does the notion of a “pristine” environment even exist?

IV. THE ECOSYSTEM: CONSTANT FLUX OF NATURE

This question about the notion of a pristine environment is tied in with topics around the constant flux of nature. Invasive species are often accused of altering population dynamics, and disrupting the overall stable equilibrium and balance of an ecosystem (McNeely 2001, Dukes & Mooney 2004, Mooney et al. 2005). A large amount of research has been done in the field of trophic ecology, and ideas of top-down and bottom-up effects have helped shape our understanding of ecosystems (Thompson et al. 2007, Bennie et al. 2018). The idea that the removal of an organism in an already-natural environment can lead to a trophic cascade, which has catastrophic impacts on the ecosystem and alters food webs has helped us understand nature in such a way that we view it as a tightly bound system whereby organisms have specific niches and have adapted in a particular way (Estes et al. 2011, Ripple et al. 2016). Invasions are thought of as altering these tightly bound food webs and disrupting the natural balance of an ecosystem by pushing out “native” species, replacing once-existing tightly bound community systems that were

in place and thus reshaping the ecosystem processes (Gallardo et al. 2016).

However, skeptics would argue that invasions do not pose as much of a threat as they are perceived to be and that they do not have the overwhelming negative effects on ecosystems that is represented in the literature. Skeptics feel that this idea of a “perfectly balanced” ecosystem is something that does not exist. Nature is constantly in a flux and changing, and some skeptics believe Darwin’s idea of natural selection and survival of the fittest fit the description of these invasive species - that we should be letting nature decide its own path of evolution and survival, and that species have the ability to adapt to changing environments (Frank 2021). So why then are scientists so concerned about changing environments driven by invasive species?

Hybridization is another issue related to invasive species and the flux of nature (Mallet 2005, Blair & Hufbauer 2010). In general, biologists disapprove of the “intermingling” of genes between invasive and native species (Malukiewicz et al. 2014). A famous controversial example of hybridization is that of the Ruddy duck (*Oxyura jamaicensis*), which is native to North America and was introduced into Europe into a waterfowl reserve in early 1940 (Robertson et al. 2015). After being introduced to the reserve, some of these ducks escaped and soon began breeding in the wild (Smith et al. 2005, Robertson et al. 2015). As the feral populations grew, they began to move south to Spain (Muñoz-Fuentes et al. 2007). The issue here was that the White-Headed duck (*Oxyura leucocephala*), which are native to these areas, began breeding and hybridizing with the Ruddy duck and produced fertile offspring (Muñoz-Fuentes et al. 2007). The White-Headed duck had recently just been brought out of near extinction by Spanish ornithologists who had dedicated many years of research and conservation strategies to protect this species (Muñoz-Fuentes et al. 2006). Conservationists thus saw the hybridization between the two species as a significant threat to the local duck species and began a nation-wide extermination event that would seek to exterminate as many Ruddy ducks as possible (Henderson 2009). The effort was huge with multiple European countries hiring people to shoot the ducks, destroy nesting sites and even shooting the hybrids (figure 6). Studies showed that this reduced the number of hybrids and allowed the recovery of the White-headed duck, with numbers now exceeding over 2500 individuals (Muñoz-Fuentes et al. 2013).

Some skeptics thought of the Ruddy duck example as an extreme strategy and a pointless conservation effort. The concept of trying to protect the “perfect genes” and “pure” genetic line seems to some to be egotistical and contradicts the well-

established notion in ecology that species and genetic diversity are important factors in maintaining healthy ecosystems. So, if the hybridization between two species and the formation of a sub-species is not an addition to diversity, how can it possibly be regarded as a threat if nature is seemingly finding a way to adapt? Some feel that we are in the way of nature's flux and are consistently trying to mold the world and its species in our image.

So, is our view of nature a bit skew? Is there such a thing as a perfect genetic line?

V. HUMAN PERSPECTIVES: CORRELATION VS CAUSATION

The final important topic in the argument around invasion biology is that of correlation versus causation. Biologists often accuse invasions as being one of the leading causes of species extinctions that lead to biodiversity loss (Hulme 2009, McGeoch et al. 2010).

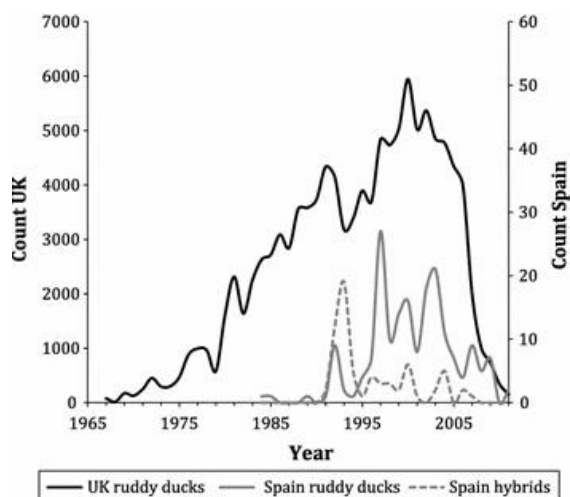


Figure 6: Population trends of ruddy ducks in the UK and Spain (Muñoz-Fuentes et al. 2013).

This is because many studies show that “native” species populations decline simultaneously with the start of an invasion (figure 7) (Mooney et al. 2005, Byrnes et al. 2007, McGeoch et al. 2010). Skeptics argue that scientists sometimes confuse correlation with causation and that just because we are seeing a decline of “native” species at the same time of an invasion, that does not necessarily mean that invasions are the main reason for these declines

Hawaii is often used in the defense of invasion biologists and conservationists to display that invasions can definitely result in the extinction of native species (Wilcove et al. 1998). One study

showed that, of the U.S.A.’s species that are most threatened by invasive species, around 43% of the birds and 39% of the plants occur exclusively in Hawaii (Gurevitch & Padilla 2004). One such example is that of the endemic and endangered plant species, *Xylosma crenatum* (Wood 2012). However, skeptics point out that other issues are often ignored when making this claim, such as the fact that the species has extremely low reproductive rates that could be impacting its population numbers (Gurevitch & Padilla 2004). Skeptics also point out that habitat degradation through the extensive husbandry of domestic livestock such as pigs, goats and cattle on the island is argued by some to be a far larger concern with a much greater effect on the native species (Wilcove et al. 1998).

Scientists have acknowledged this argument and a recently-conducted study experimentally evaluated the difference between correlation and causality within biological systems (MacDougall & Turkington 2005). The study investigated two contrasting ideas called the ‘Driver’ and ‘Passenger’ models (figure 8) (Cassini 2020). The Driver model predicted that invasive species are interactive and competitive with native species, and thus are a leading driver in native population declines and potential extinctions (Cassini 2020). The Passenger model predicted that invasions are rather aided by non-interactive processes (like human disturbance) which allow for exotics to be less constrained and thus dominate the habitat. This study found that a high number of invasive species is not always associated with increased competitive interactions with native species (Cassini 2020). Rather, the study showed that other factors such as dispersal limitation, environmental conditions and disturbance also have a role to play in population declines (MacDougall & Turkington 2005). In other words, the study found support for both models individually and interacting with one-another, and neither was found to work consistently in isolation.

So, is it sufficient to say that invasive species are the only causes of declines in local/pristine biodiversity? Or are their impacts rather correlated with other factors that are equally or more responsible for negative impacts on native species?

This demonstrates that environments should perhaps be looked at in a more context-dependent way whereby multiple drivers are considered altogether.

VI. CONCLUSION

The topic of invasive species is a tricky one to debate and as biologists we cannot ignore some of the controversy and public perception on the work

of invasive species. It is formally our duty to make sure the science and work we do makes sense and that it is as accurate as can be. Here we will conclude some of the thoughts we have on the discussed topics.

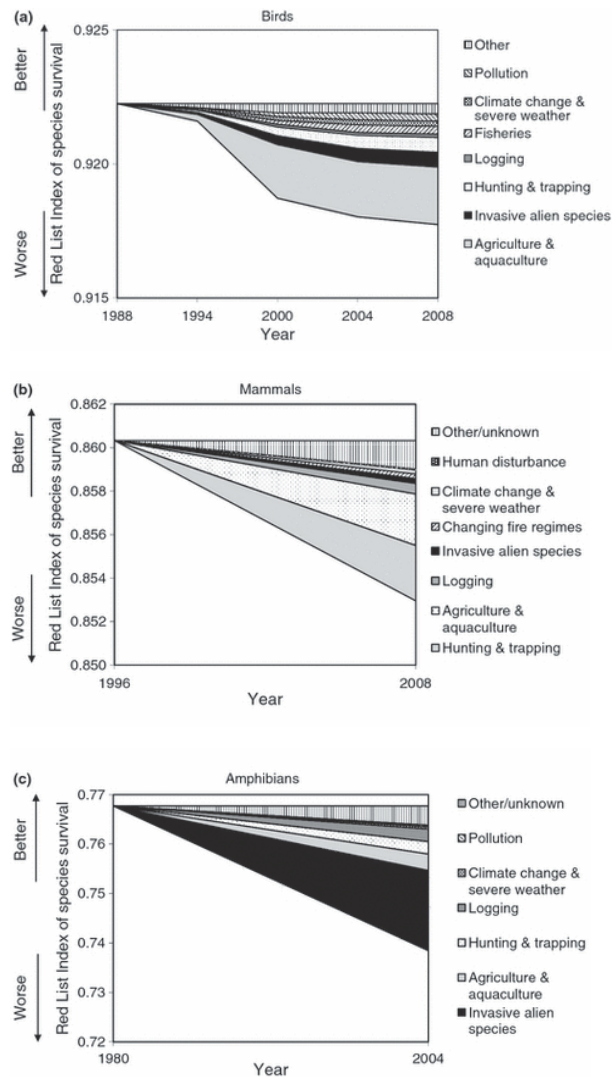


Figure 7: The relevant drivers of biodiversity loss within a.) birds, b.) mammals and c.) amphibians (McGeoch et al. 2010).

It seems there is consistent confusion between the terms “invasive” and “native”. To skeptics, there seems to be exceptions to these definitions, as seen by the EHB example. One would argue that we as humans are more concerned about economics than the environment and that these financial concerns should not be confused with concerns for the greater good of nature. But are the two mutually exclusive? A study showed that in the U.S.A alone, 25% of the global honey production comes from the U.S.A, supplying many jobs and creating a strong economic sector (Apergis 2017). This shows how large the contribution of this natural product is to the economy which should be

considered on both national and local scales. However, there is no doubt that this industry is beginning to have effects on other species and could affect entire ecosystems (Winston 1992, Aizen & Feinsinger 1994, Ellis & Munn 2005, Paine & Roberts 2005, Mutinelli 2011). If the industry is to be maintained, there needs to be more of a push for more sustainable honey production and management practices that reduce the impact on local pollinators and ecosystems. Indeed, it seems we as scientists need to be more honest with our motivations for classifying our “invasive” species as this seems to have sparked confusion and controversy. Both biology and policy are infamous for their complexities, and maybe we should not be focusing on definitions, but rather deal with each “invasion” or “biological issue” on a case-by-case basis, taking all natural and anthropic factors into account. Looking into the definitions of “pristine”, we can see once again that there are possibly some contradictory examples and that maybe our definitions are becoming useless. The argument that there is no such thing a pristine environment is one that I mostly agree with.

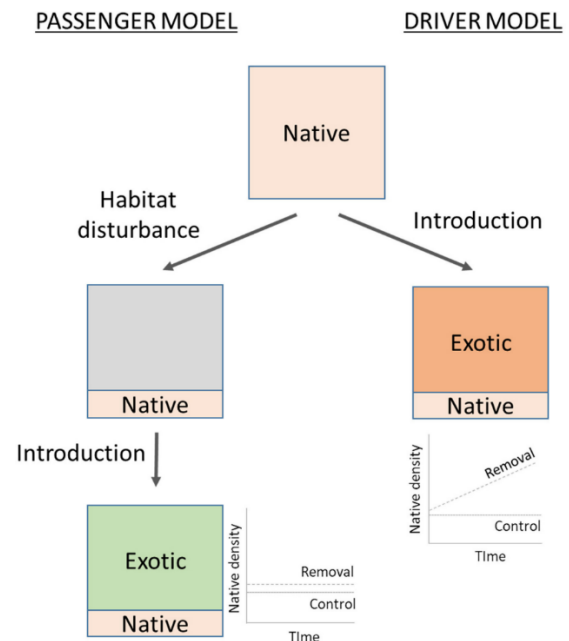


Figure 8: Illustration of the passenger and driver models as used by Cassini, 2020.

I think it is becoming more and more impossible to find environments that are “untouched” by mankind. I would suggest that this is another example of how we should not focus on definitions, but rather use these definitions as a framework and not as a definitive answer to what the ecosystem represents.

Many scientists understand that nature is not as fragile and constant as once thought (Estes et

al. 2011). However, there still exists a need for everyone to think about ecosystems in a more kinetic way. The sudden effects of a new species can alter an environment and is likely to effect community assemblages (Ripple et al. 2016). Based on the data that have been collected on invasions, the risks of this are too unknown and are likely to outweigh the benefits and we need to recognize where our interference has caused an event, like hybridization, that would not naturally occur. If hybridization is left to completely remove a gene pool of a species, we lose that species forever. Once that species is lost, there would undoubtedly be effects to an ecosystem, and one would not know if the hybrid species would fill all the necessary ecosystem function gaps left by the native species. The sentiment behind the result of a hybrid event, that this should be admired and viewed as an increase to biodiversity, is one that has value. However, we as scientists need to be realistic. We as humans are driving too many changes in the environment and although nature is constantly adapting, the rate at which we stress our environments push it to a point whereby there is not enough time to adapt, and we are left with too many consequences. We need to assess the benefits and the risks and there are too many instances where the risks of invasive species do not outweigh the benefits.

It is clear that a more thorough understanding of an ecosystem is needed in order to accurately assess an issue or threat to it. It is definitely true that some environments are affected badly by pollution and this, in turn allows for establishment of invasives and as discussed above, very few “pristine” environments that are free from anthropogenic impacts still exist. Even though invasions might not always be the root cause of biodiversity loss in habitats, it does not mean we should not focus our attention on invasives. In order to deal with the issues of habitat disturbance we also need to remove the invasives, then conservationists and biologists can protect the environment from pollution and prevent the return of invasives as we need to approach. Allow me to explain this using an example. Smoking may cause lung cancer. So, in a person who has the disease, stopping smoking should be the most logical route to recovery. However, once you have lung cancer, the disease itself requires treatment, so focus is required on both the cause and the illness. The same can be said for invasive species. Therefore, I believe that invasions should definitely not be left to continue as the risks could be catastrophic to our communities and our environment, but the other underlying causes (eutrophication, habitat destruction, global climate change etc.) require attention too. This is something biologists need to consider more in the future.

Throughout this review, there is one overwhelming factor we cannot ignore and that most

scientists and skeptics agree with - human disturbance is playing a significant role in ecosystems. As much as there are flaws in invasion biology, the answer seems a lot more grey than both biologists and skeptics would have us believe. Maybe it is true that these constructs of invasion biology are meaningless. Maybe there are no such things as good and bad species. Maybe we as scientists need to accept that there is no right and wrong answer. However, there is one certainty here, the fact that we as humans have had an impact. However, to me it seems easy to say that we do not exist as one with nature, but rather our actions have impeded nature’s survival. Species that enter new ecosystems because of our intervention need to be monitored and we have a responsibility to make the least impact possible on our earth’s systems.

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Why is biology unique?

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ABSTRACT

Is biology a science? There is no argument that the study of biology can be considered a branch of science. However, can biology be compared to the ‘hard’ sciences such as physics, mathematics, or chemistry? The historical-philosophical views of biology are discussed, pertaining to how different the science of biology is from other sciences. Biological sciences, especially evolutionary biology, have had to adapt their own scientific method as the generic one of experimental analysis could not always be applied to this field of science, further highlighting the uniqueness of biology. Complexities and dynamics at every biological level make it nearly impossible to predict outcomes. However, this consensus is questioned. This review emphasizes that some biological processes have some predictability, making biology more comparable to ‘hard’ sciences. This review aims to incorporate both arguments to this question and conclude with a new interpretation.

Keywords: Autonomy; Biological levels; Historical narrative; Predictability.

I. INTRODUCTION

Is Biology a science? Can the study of biology be compared to the so-called ‘hard’ sciences such as mathematics or chemistry? The term science goes beyond natural sciences and now incorporates distant fields such as social science, feminist science and even Creationist Science. On the other end of the extreme, science is described solely on mathematically based physics. The term ‘science’ here is highly misleading. So, where does the line between incontrovertible science and related fields lie? Historically, science was something that could be assessed and known systematically with a high degree of certainty. The main idea of science was that ‘science is physics.’ This perception extended until the late 1900s. As late as the 1970s, multiple philosophies on biology were written, but physical science frameworks and concepts were still applied, which made sense at the time as being a scientist meant obtaining a degree in a ‘hard science’ such as mathematics or chemistry. Using mathematically based frameworks to study the philosophies of biology was not successful as many authors at the time realized that employing standard scientific methods and frameworks in biology was not adequate. Many of the great scientific discoveries in the field of physics in the 1920s made little contributions to the philosophies of biology. Darwin’s book “*On The Origin of Species*” revolutionized biology and contributed to the main philosophies used in this field today (Smocovitis, 2005). Once again, this highlights that there is indeed a line that can be drawn between the soft sciences and the ‘hard sciences.’ This is further emphasized by the fact that some disciplines within humanities employ methods and frameworks that

corresponds to those of the natural sciences. For example, evolutionary biology is analogous to historical science and highly different from sciences, such as mathematics, in conceptualization and methodology (Mayr, 2004). Ultimately, it became clear that employing any standard mathematical methods to describe the philosophies of biology, rather than the unique complexity, would be unsuccessful (Mayr, 2004). Based on the inherent variability within the field of biological science, the consensus is that one cannot accurately predict outcomes. However, several evolutionary biological processes question this and demonstrate that there is a level of predictability within Biology.

II. DIFFERENCES BETWEEN THE SCIENCES

One of the fundamental differences between physical and biological sciences is that the physical sciences use natural laws to explain theories (e.g., Newtonian laws) whilst biological sciences use concepts, such as sexual selection and intraspecific selection, which form prominent bases of all biological theories. Ultimately, the field of physical sciences deals with inanimate objects to which natural laws are applicable. Uniquely, biological systems experience ‘dual causation’ where biological processes are controlled by natural laws and genetic programs (Mayr, 2004). This dual causation not only characterizes the field of biology but applies to both branches of this science (functional and historical biology). There is nothing akin to this in the field of physical science or mathematics (inanimate objects) (Ayala, 2004).

Initially, the main philosophy pertaining to biology was that of 'vitalism'. Vitalism is the philosophical belief that there is a meta-physical force within all living organisms, and regardless of the evidence depicting reason for life, this view insinuates that there still is a 'special force' that drives life (Gatherer, 2010). This philosophy was quickly rejected in the field of biology as it suggested that, like the gravitational pull that drives the movement of planets, a force drives the movements other of organisms. The concept of teleology was subsequently incorporated into the original concept of 'science'. Teleology refers to natural processes with a definite goal or end, such as the development of an embryo to an adult or courtships (Mayr, 2004). However, this insinuates that evolution is goal orientated with a defined end. Teleology would correspond with 'Lamarckism.' Organisms did not acquire characteristics because they 'needed' them to survive. According to Darwin's theory of natural selection, evolution is underpinned by natural selection through the survival of the fittest. For example, the peacock does not need an extravagant tail in order to survive. This ornament displays reproductive fitness and increases the chance of mating (Winters, 2018). Ultimately, the main consensus is that natural selection through variation and selection, which is the critical driver of evolution, is random and by chance, making it impossible to predict (Livnat, 2013). This further emphasizes that biology is unique and not analogous to physical sciences.

III. THE SCIENTIFIC METHOD

The scientific method arose during the 1600s (Betz, 2011). The methodological framework behind the scientific method has evolved and been altered slightly since, but essentially the main steps remain the same (Andersen & Hepburn, 2015). First, a question is asked about an observation that is being investigated. Secondly, research is conducted to understand what is wanted to be investigated. Thirdly, a hypothesis is constructed to predict an outcome. Fourthly, experiments are conducted in order to investigate the hypothesis. Fifthly, the data is analyzed, which allows one to draw conclusions. Lastly, the hypothesis is either accepted or rejected based on the findings of the results (Ritter & Eibl, n.d.).

It is imperative to explain that biology may be viewed as two separate fields: functional and historical biology. Historical Biology, newly termed evolutionary biology, had to obtain its own methodology (one of a historical narrative explanation) as the scientific method of using experiments is not applicable to understanding evolution. 'You can't experiment with two million

years!' (Stebbins in Mayr 2004, 16). Functional biology refers to the physiological processes of organisms. In this aspect of biology, scientific methods can routinely be employed to study cellular processes as the methodological components are mainly comprised of physical and mathematical sciences. However, studying biological organisms and their systems is always accompanied by inherited natural variability. The philosophies of biology try to incorporate the standard scientific method by running numerous different types of tests with multiple replicates to demonstrate a relationship. Innately, biological systems always have inherited variability as they are highly complex and dynamic. This can also be seen on the species level. For example, organisms of the same species, genetically similar, typically display large ranges of variation. The common consensus in behavioural sciences is that the differences in the behaviour of organisms are caused by the differences in genes or the environment (nature vs nurture) (Bierbach et al., 2017). However, a study conducted by Bierbach et al., (2017) questions this consensus. The authors reared clonal fish (genetically identical) in identical conditions but with different durations of social contact. The authors found that individuality was not inhibited by the stable environment nor by the lack of social contact (Bierbach et al., 2017). This variation is not only expressed through the behaviour of species but physiologically too. If you test the photosynthetic rate of two genetically identical plants, the results will not be constant (interspecific variation). This highlights the point that various factors and unknown entities influence biological organisms.

IV. REDUCTIONISM VS HOLISIM

Many scientists within the field of the absolute sciences emphasize that scientific explanations tend to be toward 'reductionism' (Gatherer, 2010). This essentially means that large-scale phenomena can be explained in terms of the small-scale components (Agutter & Wheatley, 2008). Scientists aim to reduce their field of study into smaller units. In the field of physical science, the basic units of analysis are particles or atoms. In biology, reductionism is essential to understanding the basic units of life, such as organelles in cells, but does not contribute to understanding how life works or originated (Bribiescas, 2005). Natural ecosystems are highly complex and dynamic systems which change over space and time. Given the inherent complexity and variability, it is nearly impossible to analyze a system by reducing it to its' basic components as endless interactions and unknown entities influence these systems. How can biological scientists reduce unknown confounding variables? They cannot reduce variability as it is inherent in

biological organisms. Darwin's Theory of Evolution defies reduction to physics and chemistry because of its meta-physical components (reality outside of human sense perception) (Altiero, T., & Zabel, J., n.d.). The concept of holism was then incorporated into the philosophies of Biology, which completely defies reductionism. This concept refers to the idea that biological systems and organisms cannot be reduced to their essential components as there are unknown entities. However, this philosophical view was disproved through accurately predicting hypotheses, although it is incorporated to some degree within studying organisms in their environmental systems (Gatherer, 2010; Trewavas, 2006).

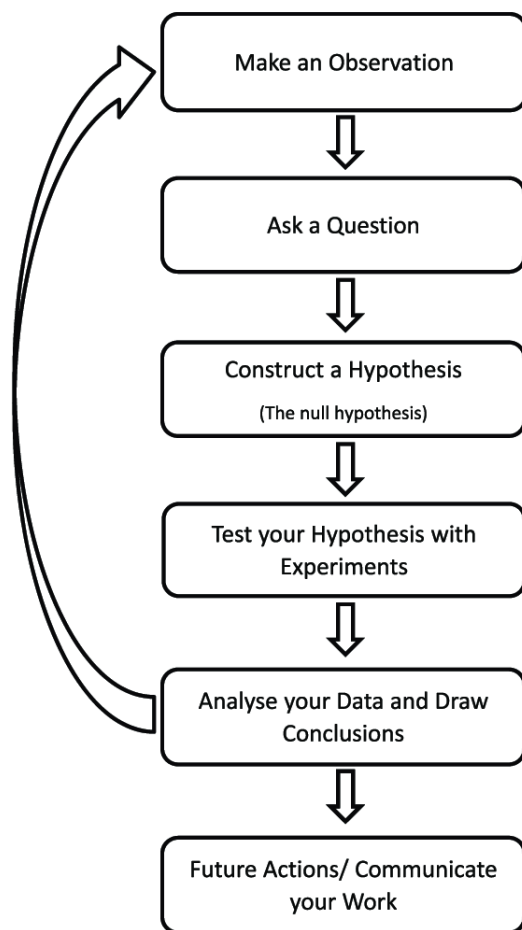


Figure 1: Scientific method (Yeoman et al., 2015)

V. PREDICTABILITY

In the so-called hard sciences, predictability of phenomena is expected as the laws never change. Can biology be compared to these sciences in terms of its predictability? Predicting evolution is complex as it pertains to every biological level. For example, changes in the molecular structure (mutations) will modify the phenotypes and functions of organisms. In turn, community and population structures are altered. Contributing to the complexities, each of these biological levels is intricate, affecting another level as they are all connected (Mas et al., 2020) (Figure 2).

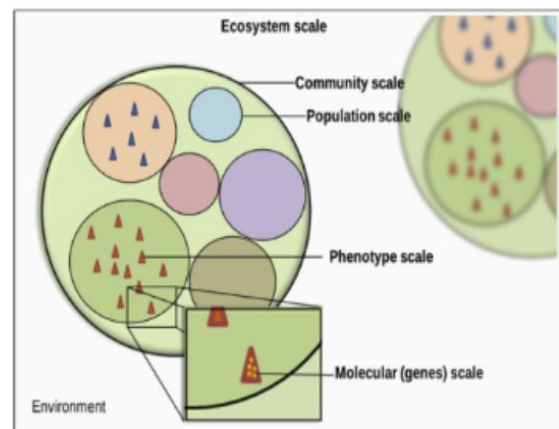


Figure 2: Different levels of ecological hierarchy and the functional level are incorporated as they relate to the dynamics of organisms as they mediate ecological functioning (Mas et al., 2020).

However, if the genetic mutation is favourable or neutral, it is expected to be inherited by the following generation. If we were to rely on the concept of fitness in terms of Darwinism (optimization/maximization), accurately predicting genetic modifications are theoretically possible. To put it simply, if a known mutation benefits the fitness of an organism, one can predict that the mutant will be selected (Figure 3). (Mas et al., 2020). On the phenotypic level, patterns of repeated evolutionary events driven by natural selection are observed (Stern & Orgogozo, 2008). Organisms inhabiting the environments with similarities (in terms of conditions) repeatedly evolve (independently) similar structures and features. This can be seen in fish such as the sticklebacks. Rundle et al., (2000) demonstrated that stickleback populations, that evolved under different ecological conditions, displayed strong reproductive isolation. Populations that evolved independently under similar ecological conditions lacked reproductive isolation. Thus, similar phenotypes emerged based on the similarity of the ecosystems (Rundle et al., 2000). Patterns of repeated convergent evolution are also observed, such as the loss of pigments in

organisms living in caves, which makes evolution somewhat predictable, making it more comparable to the ‘hard sciences.’



Figure 3: Environmental constraints drive selection demonstrated on a genetic level: https://www.cs.mcgill.ca/~rwest/wikispeedia/wpcd/wp/n/Natural_selection.htm

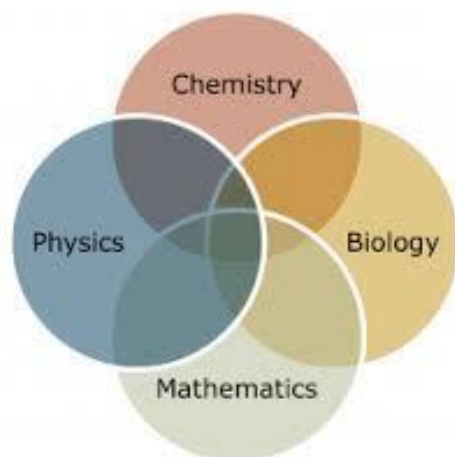


Figure 4: Integration of sciences to assist biological methodologies and frameworks (Hinrichs et al., 2020)

Biology is different from other so-called absolute sciences such as physics and chemistry because the key underlying principle in biology is evolution, more specifically natural selection, which is random and by chance, ultimately making it impossible to predict. Ultimately, this contributes to

the inherent natural variability that we observe in the field of biology. I would say that biological science is a unique science due to the complexities and inherited variability associated with biological systems. However, it employs and integrates methodologies and frameworks from other scientific fields to assist in explaining biological processes with a degree of predictability (Figure 4). The ability to predict phenotypic traits in response to environmental stressors is possible but limited (Pelletier, 2019). Predictive frameworks, such as the model constructed by Blank et al., (2014), demonstrate how organisms will respond to environmental changes, which is significant under the phenomenon of rapid climate change that we are currently experiencing.

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Current theories on evolutionary drivers for high body temperatures in birds and mammals.

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ABSTRACT

Many studies have been conducted on the possible drivers for the evolution of high body temperature in birds and mammals. As a result, theories with evidence have been proposed to better understand and explain what could have driven it. It is predicted that this evolutionary event was not driven by a single cause or selection pressure. The evolution of high internal body temperature is said to have occurred millions of years ago from the Permian through to the Cenozoic periods. The first theory is the evolution of a greater aerobic capacity. It is thought that this could have driven the conquering of land by birds and mammals and at the same time the evolution of endothermy. This theory is supported by the fact that birds and mammals have more enhanced aerobic capacity than ectotherms. Parental care is another theory that is supported by improved fitness and energy transfer in birds and mammals which makes them successful. It has also been proposed that birds and mammals underwent an evolution in body sizes and went from big to small body sizes with fewer energy costs, this is supported by fossil evidence. Insulation is also said to have played a huge role in the evolution of high body temperature as feathers and fur trap heat, increasing the body temperature of endotherms. Other very significant and recognized theories are the evolution of enhanced locomotion modes (Bipedalism, unguligrade, digitigrade, Plantigrades, and Power flapping of wings in birds and mammals). This theory is supported by correlation analyses such as the metatarsal: femur ratio. Lastly, climate change is said to have been a contributor to body reduction evolution and insulation evolution, this is supported by evidence from after the global cooling periods which saw animals living in cold areas maintaining high internal body temperatures. All these theories are recognized but only some of them have strong evidence to support them.

Keywords: Endothermy, Miniaturization, Thermoregulation, Aerobic metabolic scope, Normothermy, Supraendothermy, Basoendothermy.

I. INTRODUCTION

Ectotherms are organisms whose internal body temperature (T_b) varies with the environmental 'exogenous' temperature; they maintain a fairly low internal body temperature (Lovegrove, 2017). Whereas, whereas birds and mammals have the ability to generate internal body heat (Hayes & Garland, 1995; Lovegrove, 2012). These organisms' internal body temperature is independent of the environment, they are known as endotherms (Lovegrove, 2017). The energy cost of being endothermic is much higher than that of being an ectotherm, and it is still a considerable debate why birds and mammals evolved endothermy (Lovegrove, 2012, 2017). The evolution of the ability to maintain a constant body temperature was very important for birds and mammals as it allowed them to be able to live in a wide range of habitats (Gatesy & Middleton, 1997; Grigg, Beard & Augee, 2004; Lovegrove, 2012, 2017; Smit *et al.*, 2013; Nowack *et al.*, 2017; Levesque, Tuen & Lovegrove, 2018). An animal's body temperature has remarkable consequences on its fitness. Many animals have acquired the capacity to regulate their body temperature and keep it at elevated an optimal range (Levesque *et al.*, 2018). Animals maintain

these optimal body temperatures by modifying their behavior and physiology (Clarke & Portner, 2010; Smit *et al.*, 2013).

The exact fitness benefits of endothermy have never been clear and there are many questions about the evolution of high body temperatures in birds and mammals (Koteja, 2004; Kemp, 2006; Lovegrove, 2012, 2017). There are three levels of endothermy based on normothermic body temperature namely; basoendothermy (endotherms with normal body temperatures below 35°C), mesoendothermy (endotherms with normal body temperatures that vary between 35°C and 7.9°C), and supraendothermy which are endotherms that maintain their normal body temperature at or above 37.9°C (Lovegrove, 2012).

Due to uncertainty as to why birds and mammals evolved endothermy many theories have been proposed to try and understand the drivers of this evolution. Some of the theories have evidence to support them. For example, Lovegrove (2012) suggested that meso and supra endothermy levels came about because of the events of global cooling and change in vegetation after the Palaeocene Eocene Thermal Maximum (PETM).

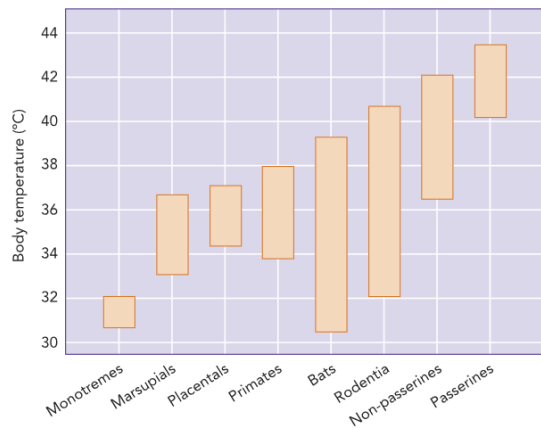


Figure 1: Variation in mammal and bird normal body temperature modified from McKechnie and Wolf (2019)

It has been shown that in extant smaller mammals Basal metabolic rate and body temperature is explained best by their geographic distribution (zoogeography) (Lovegrove, 2012). In big mammals the explanations for body temperature variability remains unclear. In another argument, Lovegrove (2012) proposed that the energies for improved locomotion is the best way to explain mammals evolution of supraendothermy in large cursorial mammals for example orders Lagomorphs and Artiodactyla are supraendothermic (Lovegrove, 2012). While other animals of the same size as monotremes and macropod kangaroos are either meso or basoendothermic (Figure 1). Studies and analysis that have been done to better understand the evolution of endothermy rely on the use of fossils. As stated in Lovegrove (2017), increase in palaeontological data of early birds and Crustaceous archaic mammals was as a result of a large amount of fossils that were well preserved during the Jurassic Cretaceous periods (Lovegrove, 2017). In this review I will be looking at some of the current existing theories on evolutionary drivers of high body temperatures in birds and mammals.

II. PARENTAL CARE

In birds and mammalian lineages the initial pulses or drivers in endothermy and BMR happened during the Triassic era and they were related with maintaining of a fairly high body temperature, taking care of offspring and aerobic capacity (figure 2) (Lovegrove, 2017). These early endothermic pulses helped birds and mammals survive the dry periods of the Triassic era (Lovegrove, 2017). Endothermy is said evolved at the same time as parental care, as endotherms became highly active looking for food for their young and keeping them warm (Farmer, 2000). This

therefore led to an increase in their metabolic rate generating more energy and muscles giving off more heat, as a result more body temperature was generated (figure 2) (Koteja, 2000, 2004; Angilletta, Jr. & Sears, 2003, 2003; Clarke & Portner, 2010). The significance of parental care in endothermy is best shown by Lesser hedgehog tenrec which are known to maintain high body temperature during their reproduction cycles (Oelkrug *et al.*, 2013). This improves their fitness as they can keep their young warm protecting them from dying of cold; therefore, increasing their survival chances (Farmer, 2000). Fitness is measured by how the number of surviving offspring an organism can reproduce in its lifetime. Therefore, organism that can transfer energy to their young increasing their survival chances are rendered more fit.

III. AEROBIC CAPACITY

In endotherms high adaptability in aerobic metabolisms is very crucial for surviving in environment with detrimental temperature conditions (Angilletta, Jr. & Sears, 2003; Clarke & Portner, 2010; Stager *et al.*, 2016). Endothermy is characterized by a more enhanced aerobic capacity to that of ectotherms (Geist, 2000). Selection for enhanced aerobic capacity is a widely accepted theory in the evolution of endotherms (Hayes & Garland, 1995; Angilletta, Jr. & Sears, 2003; Clarke & Portner, 2010), the aerobic scope hypothesis. Another theory presented by Lovegrove (2017) states that for mammals and birds to be able to colonize land they had their aerobic capacity and activity evolve. It is said that mammals went from just sitting and waiting for their food and became more active when foraging (Lovegrove, 2017). This drove them to requiring more energy for sustained activity and an enhanced aerobic capacity (Lovegrove, 2017). Lovegrove (2017) suggested that the change in foraging style is what triggered the evolution of enhanced aerobic capacity. This event was not likely to be the only cause which led to elevated metabolism and enhanced aerobic capacity. In mammals, aerobic metabolic scope is theorised to increase with increasing latitude (Stager *et al.*, 2016). This theory could be supported by the climatic variability hypothesis. The climatic variability hypothesis states that aerobic metabolism variability should increase with normal average environmental temperature (Stager *et al.*, 2016). Another support is the cold adaptation hypothesis which says that cold temperatures are drivers of variation in metabolic scope (Stager *et al.*, 2016). Therefore, according to these hypotheses ideally then endotherms should be well adapted to living in areas with either cold or excruciatingly hot weather due to fairly flexible and high aerobic capacity, which is the case. For birds, the theory says

enhanced activity might be the cause of elevated aerobic capacity, but that it is even more challenging and complicated to predict (Stager *et al.*, 2016; Lovegrove, 2017). A proposed hypothesis states that huge muscles in non-avian dinosaurs came about due to selection for NST in muscles (Lovegrove, 2017). This therefore led to development of a theory that states that enlargement of muscles could have selected for the evolution of high body heat and energy utilisation (Lovegrove, 2017). Better energy utilisation in muscles requires more oxygen. This is because oxygen is required by muscles to break down glucose, giving off energy. This therefore drives the evolution of an enhanced aerobic capacity to support high activity.

IV. BODY SIZE REDUCTION

Another theory that is widely recognised as a potential driver of endothermy is the evolution of extremely small body size in adult birds and mammals. This evolution of smaller bodies is referred to as body miniaturization. It is thought to have occurred at the same time with the evolution of capacity for thermoregulation during the Triassic to the Jurassic era (Lovegrove, 2017). Lovegrove (2017) stated that in mammals, becoming smaller had been related to endothermy and global cooling for a long time. Therefore, evolution of smaller body size in mammals is said to have been driven by cooling climate and changes in vegetation (Lovegrove, 2005; Lovegrove & Mowoe, 2013; Clavel & Morlon, 2017). It is said that if in mammals there was a selection for smaller bodies from homeothermic large mammals (Lovegrove, 2005, 2017). The smaller individuals retained homeothermy by selecting for increased metabolic rate per-gram and a higher mass specific basal metabolic rate (Lovegrove, 2005, 2017). The evolutionary event of body size miniaturization is said to have been displayed by older mammaliaformes before true mammals during the Jurassic era (Lovegrove, 2017). It also seen that throughout the Triassic era there was sustained miniaturization by organisms such as synapsids (Lovegrove, 2017). The evidence presented to support the idea of body-size reduction was that presented by Lovegrove (2017). This evidence shows that *Morgannocodon* an old mammaliaforme weighed approximately 27g, 10 times more than a younger recent mammaliaforme *Hodrocodium* which weighed about 2g (Lovegrove, 2017).

To map out the evidence of body size miniaturization in bird lineages, scientists use methods like the maximum likelihood methods (Lovegrove, 2017). The evidence shows that the body size of birds has reduced greatly from body weights of approximately 163kg during the

tetanuran era, to body weights approximately 1kg in the birds that were first capable of muscle powered flapping flight (Figure 2) (Lovegrove, 2017). This is linked to evolution of high body temperatures (Figure 2) (Lovegrove, 2017). Lovegrove (2017) stated that reduction of body size was very important to how well insulated endotherms evolved (high body temperature). It is said that if it had not occurred during the Jurassic era, we would not be seeing internal thermoregulation in modern birds and mammals.

V. INSULATION: FUR AND FEATHERS

It has also been said that insulation in the form of feathers and fur had a very significant role in evolution of elevated body temperature in birds and mammals (Figure 2). This evolutionary event could have occurred to allow mammals and birds to thermoregulate as the feathers and fur can trap heat and keep the animals warm (Lovegrove, 2017). There is not much fossil evidence to support this theory. In mammals is said that small older lineages had fur throughout their bodies (figure 2) (Lovegrove, 2017). The first credible evidence of fur in mammals is said to be 164-million old, found in an otter like mammaliaform (Lovegrove, 2017). This was discovered through the use of the model of homologous character state reconstruction (Lovegrove, 2017). Using this model, an assumption that fur only evolved once in synapsids was made and it was concluded that ancestral mammalian forms did possess fur (Lovegrove, 2017).

The knowledge of the first appearance of feather like structures in dinosaurs and birds has been highly advanced due to extraordinarily preserved fossils that have been discovered in north-eastern China (Clarke & Portner, 2010; Lovegrove, 2017). Initially the density of filamentous feathers that were discovered during early coelurosaurs era were not enough insulate the birds sufficiently (Clarke, 2013; Lovegrove, 2017). The first animal that would have likely benefitted from feather insulation is said to be *Sciurumimus* specimen which was apparently a post-hatchling dinosaur (Figure 2). Birds feathers have higher insulation compared to fur of mammals as a result feathers trap the air, heat produced by metabolic activity and environmental heat to keep the body temperature of the birds high (figure 3) (Lovegrove, 2017). Feathered birds have fairly high internal body temperatures. Therefore body temperature might be a better indicator of endothermy in birds than basal metabolic rate (Lovegrove, 2017). Feathers being able to trap more air than fur has probably had an impact on birds having a higher normothermic temperature than mammals (Lovegrove, 2012, 2017; Clavel & Morlon, 2017).

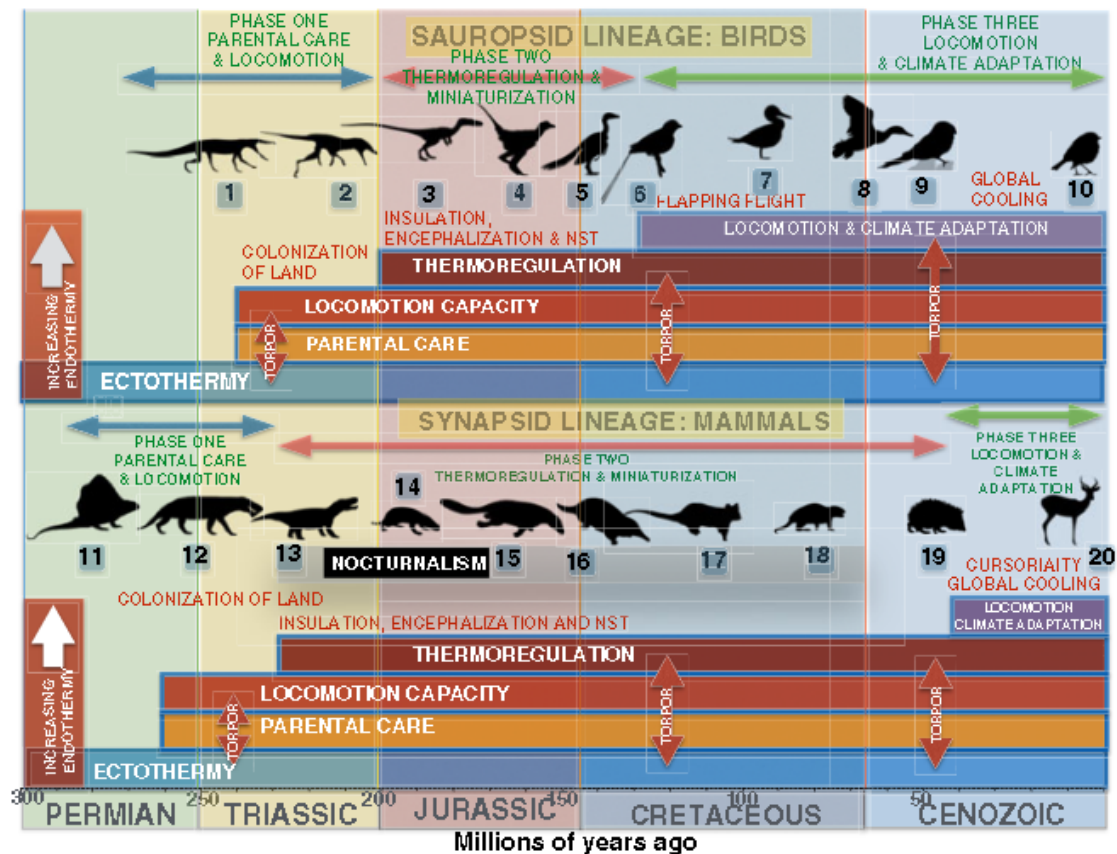


Figure 2: This image from Lovegrove, (2017) shows a Thriphasic model of evolution in birds and mammals.

The idea on evolution of endothermy was very important to knowing how insulation in the form of feathers and enhanced metabolic activity in birds came about (Stager *et al.*, 2016; Lovegrove, 2017).

VI. MODE OF LOCOMOTION

Mammals are a very diverse group of animals and as a result they differ in the ways in which they walk and run. In mammals' cursorial locomotion is said to have been one of the divers of evolution of high body temperature. The hypothesis that explains locomotion capabilities and elevated endothermy in mammals is associated with the different forms of locomotion (Grigg *et al.*, 2004; Lovegrove, 2017). These forms are plantigrade, unguligrade and digitigrade. Mammals had to undergo morphological modification to the limb bones to acquire these forms of locomotion (Gatesy & Middleton, 1997; Lovegrove, 2004, 2017). Plantigrades (rodents and primates) mammals are said to have existed in the Mesozoic era and are slow runners compared to digitigrades (carnivores) and unguligrade (ungulates); as they were not capable of high running speed (figure 2) (Lovegrove, 2017). Due to this a hypothesis was developed that

enhanced cursoriality (fast speed running) in mammals led to or made necessary maintaining of high body temperatures (Lovegrove, 2017). This is due to elevated metabolic rate and body heat production in muscles (Lovegrove, 2017). This hypothesis is said to be supported by correlation analyses. The first correlation analysis support states that basal metabolic rate is correlated positively with increasing speed in mammals which is as a result of a form of locomotion (Lovegrove, 2017). The second correlation analysis support is that the metatarsal: femur ratio which is used as a measurement for cursoriality also has a positive correlation with body temperature (Lovegrove, 2012, 2017). This is seen in cursorial mammals like lagomorphs and carnivores which have the greatest level of endothermy (Lovegrove, 2012, 2017).

Since locomotion depends more on muscles than it does in internal organs (Lovegrove, 2017). In theory cursorial mammals are not expected to show increased basal metabolic rate associated with cursoriality (Lovegrove, 2012, 2017). Therefore, the best way to explain the elevated basal metabolic rate in cursorial mammals is that this generates more body temperature. This in turn maximises metabolic and muscle power as more glucose is broken down to form energy

(Lovegrove, 2017). Elevated body temperature facilitate muscle power which is a requirement for sustaining peak performance in cursorial mammals such as high running speed (Lovegrove, 2017).

In birds the method of locomotion associated with evolution of high body temperature is muscle powered flapping of the wings. For this to occur birds had to undergo morphological modification. The avialan birds inherited many flight characteristics from their ancestors, these include the three fingered hand, the wishbone, s-shaped neck, upright walking, bigger brains, arm folding ability, short tails, knee based locomotion and hollow bones, to mention some (Lovegrove, 2017). However it is said that occurrences that were important for the power flapping of the wings are said to have evolved in early cretaceous era (figure 2) (Lovegrove, 2017). The most important and recognised is the evolution of a keeled sternum which supported by muscles from the pectoral region (Lovegrove, 2017). Another important event was the evolution of a wing pulley system associated with the ability to rotate wings and lastly decreased tail length (figure 2) (Lovegrove, 2017). These morphological modification events were very important in driving the evolution of muscle powered flight in birds as they allowed for more developed pectoral muscles and balance. Pectoral muscles are flight muscles of birds. Flying birds use a lot of energy due to increased BMR and muscle activity which gives off heat keeping their internal body temperatures high (Lovegrove, 2017). In comparison to power flapping flying birds, flightless birds such as kiwis have smaller pectoral muscles and lower BMR (Lovegrove, 2017). Power flying birds maintain high body temperatures which could be a challenge under conditions of high or too low environmental temperatures.

In high temperature extremes birds therefore employ heat dissipation behaviours mostly by evaporative cooling (which is very costly) to keep their body temperatures at an optimal range (Smit *et al.*, 2013; Levesque *et al.*, 2018; Cunningham, Gardner & Martin, 2021), this can also be observed in mammals.. This form of cooling is ultimately likely to have an impact on their foraging efficiency, reproduction and their fitness (Smit *et al.*, 2013; Levesque *et al.*, 2018; Gerson *et al.*, 2019). But if the body temperature of the birds or mammals goes too high it could also lead to lethal hyperthermia (Smit *et al.*, 2013; Levesque *et al.*, 2018; Gerson *et al.*, 2019; McKechnie & Wolf, 2019; Cunningham *et al.*, 2021).

To reduce the cost of evaporative cooling by evaporative water loss endotherms are thought to store heat and maintain a very high temperature when going through a period of high heat stress, this therefore reduces the need for evaporative cooling as

it keeps their body temperatures high enough (figure 3) (Smit *et al.*, 2013; Levesque *et al.*, 2018; Gerson *et al.*, 2019; Cunningham *et al.*, 2021). This is called facultative hyperthermia (Gerson *et al.*, 2019). Gerson *et al.* (2019) has shown that small birds with very low evaporative cooling benefit the most from hyperthermia. They are able to maintain a thermal gradient that enables them to thermoregulate without any evaporative cooling (Gerson *et al.*, 2019). Birds also employ panting to reduce evaporative heat loss as they exchange air via their nasal passages reducing water loss, which saves them energy (Geist, 2000; McKechnie & Wolf, 2019).

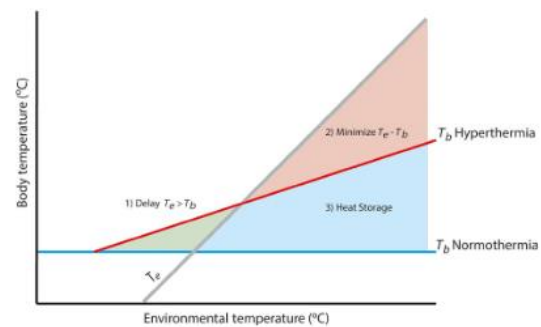


Figure 3: Graphical representation of the benefits of hyperthermia namely 1). Maximising of non-evaporative heat loss, 2). Reduced water loss by evaporation and 3). Heat storage, modified from Gerson *et al.* (2019).

VII. CLIMATE CHANGE

Another theory is that cooling climate influenced by latitude was also a driving force of insulation and high body temperatures in mammals (figure 2) (Lovegrove, 2012, 2017). Small mammals that live in high altitudes are said to have high basal metabolic rate and internal body temperature than tropical small mammals (Lovegrove, 2012, 2017). It is thought that this happens due to colder climates in high latitudes which therefore means that species that live there need high metabolic rate to facilitate heat production in order to elevate body temperatures in response to cold (figure 2) (Lovegrove, 2012, 2017; Stager *et al.*, 2016). The adaptation to cold hypotheses predicts that endotherms will have a variable metabolic scope due to cold temperatures and this was proven to be correct in birds by Stager *et al.* (2016). Also the metatarsal: femur ratio of ungulates is said to have evolved in the time of global cooling following the PETM (Lovegrove, 2012). Another evidence is that of small desert mammals having a lower basal metabolic rate than their mesic counterparts, this is said to have been evolved as a response to low and unpredicted resources and rainfall unpredictability to the species that live in habitats prone to this (Lovegrove, 2012, 2017). Lovegrove proposed a

theory that meso and supraendothermy are highly associated with global cooling and vegetation changes. Low basal metabolic rate therefore allows these mammals to conserve energy. Birds are said to possess similar patterns of climate-related basal metabolic rate to mammals (Lovegrove, Heldmaier & Ruf, 1991; Lovegrove, 2017). The climate related adaptations of elevated body temperatures could have been adapted after the event of global cooling that occurred after the Eocene Thermal Maximum (figure 2) (Lovegrove *et al.*, 1991; Lovegrove, 2012, 2017). Birds and mammals show high standards of endothermy in high latitude cold environments and low endothermy levels in deserts (Lovegrove, 2017).

Another support for this theory was selection for a scrotum in many cursorial mammals which ensures that temperature for spermatogenesis is maintained at an optimal range through stabilizing selection (Lovegrove, 2014, 2017)

VIII. CONCLUSION

The evolution of high body temperature in birds and mammals was a very complex event and still requires a lot of research. There are many proposed theories on the drivers of high body temperature. Although the theories have support, they still require research to gather more credible evidence. These theories include the reduction of body size, Parental care, evolution of locomotion to cursoriality in mammals and muscle-powered flapping of wings in birds, and adaptation to climate change. These influence or have an impact on the animals' metabolic rate and body temperature. This therefore shows that endothermy evolution was not driven by a single cause but many different drivers. Maintaining a high optimal body temperature requires a lot of energy, therefore the cost of endothermy is more than that of ectotherms.

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Are the Shell seismic surveys off South Africa's Wild Coast really as damaging as social media portrays?

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ABSTRACT

Marine seismic surveys are used for underwater exploration to identify offshore reserves of natural gas and oil. If viable deposits of natural resources are found, drilling and blasting processes are used to extract these non-renewable resources and ready them for consumption. The benefits involved with seismic surveys are largely attained by governments in an effort to boost local economic viability. Surveys of this nature have been met with huge controversy from scientists, environmentalists and even the general public. The cause for concern surrounding marine seismic surveys is largely due to the ecological impacts that these surveys may have on marine life, however there have been significant findings to support the claim that seismic surveys may have negative socio-economic impacts as well. There is compelling evidence that shows marine seismic surveys are a sizeable contributor to global climate change, both directly and indirectly. All of the resultant impacts of seismic surveys can in some way be considered to have cascading effects on various stakeholders. South Africa is a country with a huge potential for renewable energy generation, with the two main sources being solar and wind energy. If the country is to take steps towards a more sustainable and safer future, a drastic shift from fossil fuels to green energy, which is implicated in the halting of marine seismic surveying, is likely to be the answer we seek.

Keywords: marine life, ecology, cascades, economic growth, non-renewable resources, green energy, global climate change.

I. INTRODUCTION

There are considerable concerns regarding the possible impacts that marine seismic surveys could have on the environmental and ecological integrity of South Africa's coastline, with a pronounced outcry from the general public and environmental conservation groups opposing the survey entirely (McCauley *et al.*, 2000a; Christiansen, Mecklenburg & Karamushko, 2014). There has been a sizeable amount of research conducted to highlight the potential impacts that seismic surveys could have on ecological systems (McCauley *et al.*, 2000a; Pichegru *et al.*, 2017). These impacts could include potential damage to the environment, possible disturbance and disruption to marine organisms, as well as the overall loss of biodiversity (Nowacek *et al.*, 2013; Christiansen *et al.*, 2014; Guan *et al.*, 2016). Therefore, conducting marine seismic surveys has now been met with huge controversy, as it has become a question of risk versus benefit. Bearing this in mind, the ensuing manuscript will aim to discuss the contentious issues surrounding marine seismic surveys and provide an opinion on whether they should be allowed to continue along the South African coast.

II. WHAT ARE MARINE SEISMIC SURVEYS?

Marine seismic surveys are the primary tool used for the exploration and research of underwater geological features (Pichegru *et al.*, 2017; Przeslawski *et al.*, 2018). The use of sound as a means for geological imaging has been around for more than 50 years, with South Africa's first marine seismic survey dating back to as early as the late 1960's (Srinivasan, Watson & Sumaila, 2012; Gisiner, 2016). The functional principle of seismic surveys is the reflection and refraction of sound waves in the form of compressed air released into the water column (Greene & Richardson, 1988; Gisiner, 2016). Basically, these surveys make use of an array of airguns and acoustic receivers towed behind a large seismic sea vessel (Greene & Richardson, 1988; Gisiner, 2016; Pichegru *et al.*, 2017; Przeslawski *et al.*, 2018). These airguns direct high-pressured air towards the seabed, resulting in the emission of low frequency sound waves (10-100Hz) that can exceed 200dB (Greene & Richardson, 1988; McCauley *et al.*, 2000a; Gisiner, 2016; Pichegru *et al.*, 2017). These sound waves are then reflected off the seabed and underlying rock strata through density and velocity discontinuities, where they are then received by the acoustic receivers (hydrophones and accelerometers) towed behind the seismic vessel (Greene & Richardson,

1988; McCauley *et al.*, 2000a; Gisiner, 2016; Przeslawski *et al.*, 2018). These returned signals can be processed by scientists to obtain detailed profiles of the underlying rock formations and geological features of the sea floor (McCauley *et al.*, 2000a; Gisiner, 2016; Pichegru *et al.*, 2017). This technique is critical for underwater exploration and development, as it is currently the most effective method for identifying potential offshore oil reservoirs and natural gas deposits, which can later be extracted through mining processes (McCauley *et al.*, 2000a; Saenger *et al.*, 2009; Przeslawski *et al.*, 2018).

III. POTENTIAL BENEFITS OF SEISMIC SURVEYS

Marine seismic surveys are in some ways hugely beneficial, with local governments, economies and the industry sector set to be the greatest beneficiaries (Beasley, 2008; Cisneros-Montemayor *et al.*, 2013; Gisiner, 2016). Due to the increasing demand for a stable supply of energy, the extraction and exploitation of non-renewable energy sources, such as that that occurs when seismic surveys locate such resources, appears to be the most viable short-term option (Przeslawski *et al.*, 2018; Singh, le Roux & Naidoo, 2022). Although this review focusses on the aspects of seismic surveying itself, the mining activities and consequent impacts that may potentially take place following the conclusion of the surveys are briefly discussed further below.

With a coastline spanning approximately 3900km, South Africa has access to an extensive Exclusive Economic Zone (EEZ) with a rich potential for non-renewable resources (Przeslawski *et al.*, 2018; Singh *et al.*, 2022). An EEZ is described as an area of the sea off the coastline that a country has exclusive rights to regarding the exploration and extraction of natural resources (Srinivasan *et al.*, 2012; Purdon *et al.*, 2020). Taking this into account, governments may green light marine seismic surveys to explore and potentially find these resource deposits in an effort to boost economic growth, revenue and stability and supplement nationwide energy supply (Cisneros-Montemayor *et al.*, 2013; Merchant, 2019). In 2020, the global seismic survey market recorded roughly 8 billion US dollars in revenue, a figure that is expected to increase to almost 12 billion US dollars by the end of the coming decade (Singh *et al.*, 2022). Over the next ten years, the African seismic survey market is expected to grow exponentially, with South Africa presenting surveyors with key opportunities to facilitate exploration and economic growth (Przeslawski *et al.*, 2018; Singh *et al.*, 2022). As a result of this, the South African government has

openly signalled its intention to promote the surveying and drilling of 30 hydrocarbon wells by 2030 (Singh *et al.*, 2022). Keeping the immense potential for economic growth in mind, it is clear to see why the cessation of marine seismic surveys would significantly undermine the country's potential for global economic growth (Przeslawski *et al.*, 2018; Singh *et al.*, 2022).

In addition to this, marine seismic surveys could also benefit stakeholders through job creation, ultimately helping to reduce the country's unemployment rate (Cisneros-Montemayor *et al.*, 2013). These employment opportunities could include jobs in the mining and processing sectors, as well as in the energy and service delivery sectors (Cisneros-Montemayor *et al.*, 2013; Singh *et al.*, 2022). Moreover, if energy resources are found, it could potentially lead to a more sustained and stable power supply, nationwide energy security and lower energy consumption costs for consumers (Cisneros-Montemayor *et al.*, 2013; Singh *et al.*, 2022). This may be seen as a meritorious benefit given the irregular supply of energy in our country. Barring the aforementioned benefits, marine seismic surveys could also have other uses, namely harbour engineering, geological research, site selection for renewable energy sources (wind, wave or tidal infrastructure) and ocean-floor infrastructure (cables and pipelines), as well as identifying possible expansions to the EEZ (Gisiner, 2016).

IV. NEGATIVE IMPACTS OF SEISMIC SURVEYS

i. Ecological impacts

The marine environment is home to a richly diverse number of organisms including both vertebrate and invertebrate species (Cerchio *et al.*, 2014; Kavanagh *et al.*, 2019). Marine species are known to make full use of their surrounding environment for crucial biological activities and functional processes (Pichegru *et al.*, 2017; Singh *et al.*, 2022). Over the next few paragraphs, the impacts of seismic surveys on marine mammals, reptiles, plankton and fish are discussed.

One example of this is how marine animals use sound and acoustic signalling for a wide range of activities and processes, such as foraging and locating prey, communication, navigation and orientation, predator avoidance, as well as mate selection, breeding and spawning (Cerchio *et al.*, 2014; Nelms *et al.*, 2016; van der Wal *et al.*, 2016; Pichegru *et al.*, 2017). One of the major concerns surrounding marine seismic surveys is the impact these surveys may have on the marine "soundscape", and how this consequently affects the associated marine life (Di Iorio & Clark, 2009; DeRuiter &

Doukara, 2012; Nelms *et al.*, 2016). Sound detectability and energy propagation is a major consideration of seismic survey impacts, as sound waves can travel for great distances underwater (as far as 75km in shallow waters and 4000 km in deep waters) (Gisiner, 2016; Pichegru *et al.*, 2017; Purdon, 2018). There is compelling evidence to support the claim that marine seismic surveying and the significant accumulated exposure to underwater noise pollution may disrupt, harm and potentially threaten marine organisms by hindering their ability to perform basic functions and may induce changes in behaviour, physiology and morphology (Pendoley, 1997; Di Iorio & Clark, 2009; DeRuiter & Doukara, 2012; Nowacek *et al.*, 2013; van der Wal *et al.*, 2016; Pichegru *et al.*, 2017; Kavanagh *et al.*, 2019). Underwater noise pollution can have significant short- or long-term impacts and can affect organisms on both an individual or populational level (Gomez *et al.*, 2016; van der Wal *et al.*, 2016). It is unlikely that loud, low frequency sounds would have significant, lasting implications on marine life if they were brief and only occurred for a short period of time (McCauley *et al.*, 2000a; Nelms *et al.*, 2016; van der Wal *et al.*, 2016; Purdon, 2018). However, seismic surveys are usually run for months on end, with constant, relentless blasting throughout the survey period (McCauley *et al.*, 2000a; Gordon *et al.*, 2004; Nelms *et al.*, 2016; van der Wal *et al.*, 2016).

Marine mammals, such as whales, are likely to be the animals most directly affected due to their heavy dependence on acoustic signalling (McCauley *et al.*, 2000b; Madsen *et al.*, 2002; Gomez *et al.*, 2016). The behavioural implications of seismic surveys on marine mammals could include the disruption of foraging and feeding efforts, altered diving patterns, temporary displacement from suitable habitats, alterations in migratory habits and interference of communication between conspecifics (Gordon *et al.*, 2004; Compton *et al.*, 2008; Gomez *et al.*, 2016; Elliott *et al.*, 2019). There have been numerous studies conducted to determine the influence of seismic surveys on whale abundance in exploration zones with significantly reduced sightings being a consistently recorded trend, implying that whales are travelling out of the affected areas (Mate, Stafford & Ljungblad, 1994; Madsen *et al.*, 2002; Kavanagh *et al.*, 2019). The physiological changes that whales and other marine mammals may be subject to include temporary (short-term) or permanent (long-term) hearing loss, alterations in respiratory patterns, induced stress responses, and even disruptions in reproductive fitness, which could ultimately impact future generations of the respective populations in that region (Gordon *et al.*, 2004; DeRuiter & Doukara, 2012; Gomez *et al.*, 2016; van der Wal *et al.*, 2016; Elliott *et al.*, 2019). In some cases, marine animals

may even experience physical damage to the auditory system and to body tissues (McCauley *et al.*, 2000b; Compton *et al.*, 2008; Purdon, 2018). Damage to tissues can occur as a result of barotrauma, which is any injury or physical trauma that an organism may be subject to due to rapid changes in underwater pressure (McCauley *et al.*, 2000b; Compton *et al.*, 2008; Purdon, 2018). Furthermore, marine animals could be subject to physical harm as they could get entangled in seismic equipment, which could lead to severe injury or even death (Nelms *et al.*, 2016).

As is exemplified above, much work has been done on the impacts of seismic surveys on oceanic mammals. In contrast, there has been limited research conducted on the impact of seismic surveys on marine reptiles, such as free-ranging turtle populations, yet it is a justifiable argument to assume that the implications faced by turtles are similar to that faced by marine mammals (van der Wal *et al.*, 2016; Purdon, 2018). For example, in a controlled study conducted in Angola, turtles were found to increase their swimming speed and erratically change direction in response to airgun releases, indicating active avoidance and alterations to their behaviour (Weir, 2007).

Primary production is arguably the most important ecosystem service provided by the marine environment, with primary producers such as cyanobacteria and phytoplankton acting as the base of all oceanic productivity (McCauley *et al.*, 2017; Fields *et al.*, 2019). There have been documented cases of increased planktonic mortalities as a direct result of seismic surveys, which causes significant cascading effects throughout the associated ecosystem (McCauley *et al.*, 2017; Purdon, 2018). An example of one such cascade could be the reduction in food availability for higher trophic levels that prey on planktonic organisms (Sommer, 2008; McCauley *et al.*, 2017; Purdon, 2018). With this in mind, it is safe to assume that decreases in plankton abundance, as a direct result of seismic noise pollution, will negatively impact ocean functioning and ecosystem health (McCauley *et al.*, 2017; Purdon, 2018). There are various contributing species to plankton communities in the open ocean, with fish larvae and small juveniles making up a crucial part of this community (McCauley *et al.*, 2017; Fields *et al.*, 2019). It is possible to come to the conclusion that healthy and sustainable fish populations (both predatory and prey species) depend heavily on the survival of plankton communities (McCauley *et al.*, 2017; Paxton *et al.*, 2017).

In addition to this, there is extensive literature that attempts to underline the direct effects that seismic surveys have on fish species, both from an individual focus and a population perspective

(Nelms *et al.*, 2016; Carroll *et al.*, 2017; Bruce *et al.*, 2018; Purdon, 2018). Similarly to the effects on other marine animals, seismic surveys may also cause physical, physiological and behavioural changes in fish species (Carroll *et al.*, 2017; Paxton *et al.*, 2017; Bruce *et al.*, 2018). The alterations in underwater air pressure from seismic blasting is likely to lead to barotrauma in fish species, which could have significant lasting effects or even lead to mortality (Carroll *et al.*, 2017; Paxton *et al.*, 2017). The potential physiological implications faced by fish species include decreased natural fitness, altered hearing sensitivity and reduced communication ability (Carroll *et al.*, 2017; Paxton *et al.*, 2017; Bruce *et al.*, 2018). The most commonly observed impacts in fish species have been linked to behaviour, such as changes in swimming and feeding patterns, as well as in population distributions (Slotte *et al.*, 2004; Paxton *et al.*, 2017; Bruce *et al.*, 2018). It has been reported that seismic surveys have a direct negative impact on fish population distribution and abundance, with local populations travelling as far out as 30km from the seismic source (Slotte *et al.*, 2004).

ii. Socio-economic impacts

Marine seismic surveys are often conducted in coastal waters that support abundant pelagic fisheries (Fields *et al.*, 2019). The seismic surveys sometimes overlap with breeding or spawning grounds of commercially important fish species, leading to arising conflicts between various stakeholders (McCauley *et al.*, 2017; Fields *et al.*, 2019). Local fisherman are insistent that seismic surveys negatively affect catchability rates, which ultimately undermine their livelihoods (Slotte *et al.*, 2004; Pichegru *et al.*, 2017; Bruce *et al.*, 2018). The findings of a study conducted by Paxton *et al.* (2017) off the coast of North Carolina in 2014 supports the claim that seismic surveys do indeed reduce the rate of catch, with low catch rates still being recorded for several days after seismic activity ends. During periods of seismic activity, commercial fisheries experience low catch rates, which could have negative consequences for the associated fishermen (Cisneros-Montemayor *et al.*, 2013; McCauley *et al.*, 2017). If catch rates are poor, the supply of fish is limited and fisheries have to endure decreased revenue, which could potentially lead to job losses and an increased rate of unemployment (Cisneros-Montemayor *et al.*, 2013; Bruce *et al.*, 2018). Similarly, if catchability rates are reduced, local villages and subsistence fishermen who depend on the nearby fisheries for sustenance can be faced with dire circumstances (Cisneros-Montemayor *et al.*, 2013). In these cases where both subsistence and commercial fisheries are negatively affected, local and national food security may come under threat

(Cisneros-Montemayor *et al.*, 2013; Pichegru *et al.*, 2017).

A further sector that may be negatively impacted by seismic surveys is marine ecotourism (Cisneros-Montemayor *et al.*, 2013; Ressurreição *et al.*, 2022). This sector is ultimately fuelled by a healthy marine environment and essentially promotes ecological conservation (Cisneros-Montemayor *et al.*, 2013; Cusack *et al.*, 2021). The income from marine ecotourism is beneficial to both the national economy and local communities, through generated revenue and employment opportunities (Cisneros-Montemayor *et al.*, 2013; Cusack *et al.*, 2021; Ressurreição *et al.*, 2022). Ecotourism sites also have recreational value, as they serve as a place of natural beauty and provide leisure (Cisneros-Montemayor *et al.*, 2013; Cusack *et al.*, 2021; Ressurreição *et al.*, 2022). With these points in mind, seismic surveys should not be conducted in waters that are of great ecological, commercial or recreational importance.

iii. What if the surveys yield significant results?

The goal of marine seismic surveys is to identify viable natural resource deposits within the Earth's crust (Gisiner, 2016; Przeslawski *et al.*, 2018). If the surveys are considered a success, the next step would be to extract and exploit these resources through means of drilling and blasting (Gisiner, 2016; Przeslawski *et al.*, 2018; Singh *et al.*, 2022). The risks involved with the extraction of marine resources are considered to be immeasurable and way beyond those associated with the surveying itself, as it may lead to environmental degradation and ecological devastation (Singh *et al.*, 2022). Seismic drilling and blasting would have a direct impact on habitat use and overall integrity through fragmentation and destruction (Putman *et al.*, 2015; Paxton *et al.*, 2017). For example, coral reefs are crucial habitats for a wide range of marine animals, as it provides them with an abundance of resources and accommodates various biological activities (eg. courtship rituals, refuge from predators, spawning grounds, etc.) (McCauley *et al.*, 2000a; Paxton *et al.*, 2017). The drilling into and blasting of the Earth's crust would then directly influence the integrity of once-favourable habitats, forcing the displacement and mortality of affected species (Nowacek *et al.*, 2015; Putman *et al.*, 2015; Paxton *et al.*, 2017). There has been research conducted to investigate the direct effects of drilling and blasting on species, with injuries and mortalities being observed in animals in close proximity to drill sites (Putman *et al.*, 2015; Gomez *et al.*, 2016).

Though thorough risk assessments and mitigation strategies are put in place prior to the

commencement of drilling, the possibility of devastating oil spills cannot be ignored (Cisneros-Montemayor *et al.*, 2013; Singh *et al.*, 2022). Oil spills are arguably the worst case scenario, as they have cascading effects that can impact multiple stakeholders (Cisneros-Montemayor *et al.*, 2013; Singh *et al.*, 2022). In the event of an oil spill, toxin and chemical pollution becomes a concerning health issue for both humans and marine wildlife, as these chemicals and toxins can pollute the environment and associated food webs, as well food sources that impact the food market (Cisneros-Montemayor *et al.*, 2013). An oil spill is not an accident South Africa can afford to endure, as it would likely lead to the devastation of our coastline and associated ecosystems in a very short period of time (Singh *et al.*, 2022). This would have cascading impacts on fisheries, livelihoods and the ecotourism sector, with the local economy set to take a substantial loss of revenue in an attempt to restore ecological health (Cisneros-Montemayor *et al.*, 2013; Singh *et al.*, 2022).

iv. Seismic surveys and global climate change

The impacts of seismic surveying and the associated mining thereafter extend beyond local ones. The continued reliance, use and burning of fossil fuels, such as those that seismic surveying both use and seeks to find, is only further exacerbating global climate change, which affects the entire globe as a whole (Seaborne, 2002; Singh *et al.*, 2022). All of the processes involved with seismic activity, from exploration and drilling, to transport and energy consumption, are major contributors to greenhouse gas (GHG) emissions (Seaborne, 2002; Singh *et al.*, 2022). The release and accumulation of these GHG's in the Earth's atmosphere leads to the absorption, entrapment and reemission of radiation and heat, which ultimately causes a rise in the average global temperature (Seaborne, 2002; Singh *et al.*, 2022). The increase of global temperatures can have ramifications across the globe; for example, and in the context of the marine ecosystems that are discussed in this review, extensive research has been carried out in the past few decades to investigate the complex interactions between the drivers and consequences of climate change, and how these impact polar ice sheet dynamics and global sea level fluctuations (Heide-Jørgensen *et al.*, 2013; Cotterill *et al.*, 2017). The rise in atmospheric carbon dioxide, which results from the burning of fossil fuels, is the primary driver of ocean acidification, defined as the reduction in oceanic pH as a direct result of extensive carbon dioxide absorption (Seaborne, 2002; Singh *et al.*, 2022). These alterations in temperature and oceanic chemistry may further add to the impacts faced by

marine life (Seaborne, 2002; Singh *et al.*, 2022). These impacts could include induced sensitivity or physiological stress, and may result in hindered biological functioning (Seaborne, 2002; Singh *et al.*, 2022). It can be concluded that seismic activity and the consequent exploitation of non-renewable resources is a core contributor to global climate change and that the potential for green and renewable energy is the only sustainable way forward (Przeslawski *et al.*, 2018; Singh *et al.*, 2022).

V. A MORE ENVIRONMENTALLY CONSCIOUS WAY FORWARD

As the human population continues to expand, the demand for resources such as food and energy sources will increase exponentially (Nowacek *et al.*, 2015; Singh *et al.*, 2022). It is for this exact reason that environmental sustainability is of utmost importance, especially when the welfare of future generations is considered (Przeslawski *et al.*, 2018; Singh *et al.*, 2022). The reliance on fossil fuels within the energy sector is responsible for approximately three quarters of all global greenhouse gas emissions (Singh *et al.*, 2022). This indicates that the first (and possibly most significant) step that we can take is to rapidly reduce the exploitation and consumption of non-renewable resources in an effort to strive towards environmental sustainability (Merchant, 2019; Singh *et al.*, 2022). South Africa has tremendous potential for green and renewable energy (Jain & Jain, 2017; Singh *et al.*, 2022). However, the procurement of infrastructure and implementation of such programmes are hindered by monetary constraints and may even be impeded by stakeholders who have vested interests in non-renewable energy like those linked to seismic surveying and who stand to lose out if there is a sudden shift to renewable energy (Jain & Jain, 2017; Singh *et al.*, 2022). The country has large potential to generate both solar and wind energy due to the high levels of solar radiation and fast mean annual wind speeds (Jain & Jain, 2017). On a smaller scale, the country also has the potential capacity to facilitate biofuels and hydropower (Jain & Jain, 2017). The long-term benefits of sustainable and renewable energy significantly outweigh that of non-renewable energy sources, especially when the potential short- and long-term risks involved with the use of fossil fuels are taken into account (Cisneros-Montemayor *et al.*, 2013). In light of the points mentioned above, I believe that the country should rather be investing money into and developing infrastructure for renewable energy sources.

VI. CONCLUSION

If the demand for non-renewable energy continues to increase, the use of marine seismic surveys still remains the most effective technique for identifying potential offshore resource deposits (Przeslawski *et al.*, 2018). This is primarily due to the lack of advancements in alternative techniques for marine geological surveying (Przeslawski *et al.*, 2018). Although numerous studies have been conducted to investigate the impacts of seismic surveys on the marine environment and associated ecosystems, there is a lack of convincing evidence to conclude that the direct impacts of the surveys will cause permanent damage to marine life (Pendoley, 1997; Slotte *et al.*, 2004; Carroll *et al.*, 2017; Fields *et al.*, 2019). Further research is required to address the question of lasting marine impacts. That being said, there is still an abundance of literature that unfolds the complexities and negative implications of marine seismic surveys, both directly and indirectly and on short- and long-term scales (McCauley *et al.*, 2000a; Cisneros-Montemayor *et al.*, 2013; Singh *et al.*, 2022). These negative impacts are in conflict with the economic and socio-economic benefits for the country. I believe that the significant short-term economic benefits that governments stand to accrue should not overshadow the serious (and inevitable) long-term ecological and socio-economic impacts that we, as a country, will be faced with if we continue on the path of non-renewable energy (Cisneros-Montemayor *et al.*, 2013). Thus, the use of alternative, renewable energy is the best way forward as it is a sure-fire way of ensuring a more sustainable, safer and greener future (Jain & Jain, 2017; Singh *et al.*, 2022).

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Future Imperfect: The Ethics of Human Gene Editing with CRISPR-Cas9

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ABSTRACT

The gene editing tool CRISPR-Cas9 is part of a series of technologies making it increasingly possible for humans to modify their own genomes and those of future generations. This has raised, and continues to raise, moral and ethical problems which are of great concern to all sectors of society. This review begins with a brief history and explanation of the technology and the early medical applications of gene therapy. We then discuss behavioural genetics, recent controversies around the use of CRISPR for human germline engineering, the battle to regulate genetic technologies, and the risks and potential of current and future germline engineering, including genetic discrimination, eugenics as a tool of oppression, and human enhancement. At several points we refer to cautionary tales of human modification and eugenics from 20th- and 21st-century fiction.

Keywords: CRISPR-Cas9, genetics, ethics, genetic engineering, gene editing, eugenics

I. INTRODUCTION

Gene editing has been portrayed in popular culture as both a potent cure for all human ailments and an instrument of oppression and destruction (Teo, 2009). In the 2007 videogame *BioShock*, a utopian underwater society's discovery of a gene-altering substance in a deep-sea slug opens the door to virtually unlimited genetic modification – and the society's downfall. In some ways the plot was almost prophetic. In 2012, a team of molecular biologists at the University of California, Berkeley published their discovery that a DNA-protein complex first studied in a real-life extremophile – salt-loving archaeobacteria dwelling in inhospitable solar salterns in Spain (Mojica & Rodriguez-Valera, 2016) – could be used as a programmable gene-editing tool (Jinek et al., 2012). The complex was dubbed CRISPR-Cas9: CRISPR standing for Clustered, Regular Interspaced Short Palindromic Repeats, and Cas9 for CRISPR-associated protein number 9. Although a mouthful, it is changing the face of science and medicine forever in ways that we cannot predict.

II. HOW CRISPR WAS DISCOVERED

At the time that the UC Berkeley study was published, it had already been known for many years that these palindromic repeats existed in a diversity of prokaryotes (archaea and bacteria) including the *E. coli* in our intestines (Mojica & Rodriguez-Valera, 2016). How did scientists fail to realize that they were sitting (quite literally) on a powerful gene editing tool at a time when genetic technologies were enjoying a surge in popularity among both academics and the public? The field of molecular microbiology was even smaller in the 1990s than it is today, and genetic sequencing was much slower

and more expensive. Initially, no one noticed that the palindromic repeats being discovered in prokaryotes through gel electrophoresis (a technique of sorting DNA fragments based on their size) and early sequencing were exact matches for the genetic sequences of certain bacteriophages; the sequencing simply had not been done yet. Even once a handful of scientists were able to connect the dots, the original research finding that palindromic repeats were part of an adaptive genetic immune system in prokaryotes was rejected by sceptical editors of several scientific journals, languishing in obscurity until other studies emerged in support of it (Mojica & Rodriguez-Valera, 2016). In the preceding years, CRISPR had been known by several amusing acronyms, from TREPS to SRSR to SPIDR, but it was only the discovery of its potential for gene editing – years after the first publication on its role in prokaryote immunity – that made it into the buzzword it is today (Mojica & Rodriguez-Valera, 2016).

III. HOW CRISPR WORKS

In nature, the function of CRISPR can be likened to that of a genetic criminal database, with Cas proteins functioning as the genetic police. Each time a new species of bacteriophage inserts its deadly genetic code into the cell of a prokaryote, assuming that the prokaryote can survive the attack, Cas proteins guided by special RNAs integrate the viral genome into the prokaryote's own genome at a very specific location within the CRISPR library (Doudna, 2015). The genetic memory of past viral invaders can therefore be retained indefinitely and passed down from generation to generation. From these stored viral genomes, RNAs can be synthesized that guide other types of Cas proteins to identify the same foreign genomes the next time they

are encountered and suppress them or even remove them, without removing anything important (Figure 1) (Doudna, 2015). CRISPR-Cas9 tools simply exploit this natural system by “programming” specific Cas proteins (Cas9) with custom, or chimeric, RNAs (cRNAs) that direct them to “snip” segments of DNA at any location the genetic engineer chooses. Because DNA has a natural self-repairing mechanism, it can be induced to “splice” a desired sequence into the newly created gap. This snip-splice process is how most gene editing is accomplished.

Bear in mind that gene editing did not begin with CRISPR-Cas9. Other nucleases (DNA-snipping proteins) had already been in use for years, namely the Zinc-Finger Nucleases (ZFNs) and TAL Effector Nucleases (TALENs) (Uddin et al., 2020). Yet, CRISPR-Cas9 changed the game by providing a highly accurate tool that was also faster, easier, and cheaper to use (Doudna & Charpentier, 2014). Gene editing had become more feasible for someone with a sufficient background in molecular biology, but limited funding, to perform in his or her garage – in fact, among the first generation of “biohackers”, a DIY bioengineering movement was already well underway (Wohlsen, 2011). Before we see just how much of a game-changer CRISPR has been in the field of human gene editing, we need to see how far human gene editing has come.

IV. GENE THERAPY

Gene therapy and other genetic services have been greatly oversold in the past by the media and by some geneticists. When the US National Center for Genome Resources conducted a survey in

1997, 36% of respondents who were aware gene therapy had been applied to humans thought it was completely successful. In reality, no human disease had been cured with gene therapy until years later (Andrews, 2001). The first successful gene therapies were designed to treat Severe Combined Immunodeficiency (SCID), a disease caused by a congenital defect in the immune system's production of white blood T-cells that normally kills affected children before their first birthday (Hacein-Bey-Abina et al., 2002). These early gene therapies, first tested on rats, worked by removing T-cell-producing bone marrow cells from the infant patient, treating these cells with retroviruses modified to splice working copies of the defective gene into the patient's genome, and then transfusing them back into the patient (Hacein-Bey-Abina et al., 2002). A pair of infant girls were the first to receive this gene therapy in 1990. Only one patient saw any improvement, but her immunodeficiency was not cured (Uddin et al., 2020). In a larger trial that ran from 1999 to 2002, four of eight children to receive the gene therapy contracted leukaemia, a common side-effect of retroviral vectors, and one child died as a result (Hacein-Bey-Abina et al., 2010). Three of the children did not improve significantly and had to receive traditional transfusion therapy. Of the four children who received gene therapy and survived to early adolescence, their immune function recovered almost to normal capacity; they had been cured, it seems permanently (Hacein-Bey-Abina et al., 2010).

And so, some of the earliest gene therapy trials turned out to be a perfect mixed bag, with outcomes ranging from life-saving for half of the patients to useless or tragic for the other half.

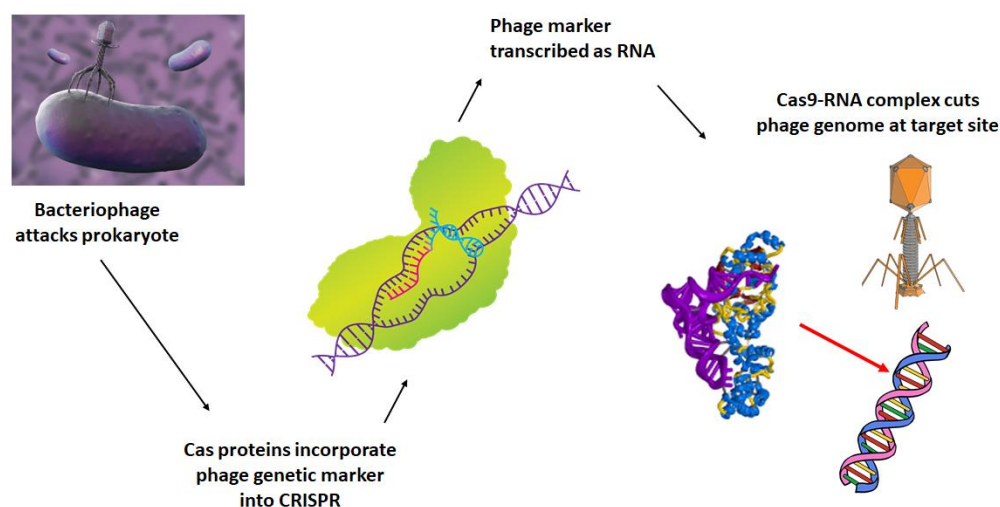


Figure 1. Simplified description of the CRISPR-Cas system in prokaryotes.

The failure of the therapy to correct the disease in some cases probably resulted from the insertion of the healthy gene into a part of the genome where it could not be properly expressed by the nanoscopic machinery of the cell. Similarly, the induction of cancer is a result of gene insertion into exactly the wrong place, disrupting parts of the genome that regulate cell division. Because it is difficult to control where viral vectors insert the new genes, the risk is far too high to make this kind of gene therapy a viable option except in desperate cases such as SCID. A much higher degree of accuracy for genetic edits was required for gene therapy to play anything more than a marginal role in medicine. The future looked promising with the development of nucleases that could make more accurate edits than viral vectors, but translating these to new medical interventions was a slow process. The effect of the appearance of CRISPR-Cas9 gene editing of human cells in 2012, on the other hand, has been sudden and astonishing. In 2014, monkeys with CRISPR-edited genes were born, showing that multiple genes could be edited successfully in primates (Niu et al., 2014). In 2017, 2018 and 2019, clinical trials began for human gene therapies to treat HIV/AIDS, certain cancers, and congenital blindness. Now, other clinical trials using the technology are working to correct heritable diseases like sickle cell anaemia, beta-thalassemia, and hypertrophic cardiomyopathy (Uddin et al. 2020). Although CRISPR is not the only gene editing tool being used today, it has helped to make this rapid advancement possible.

By now we are quite accustomed to the idea that our genes determine how our bodies look and function and our risk for certain diseases. We accept that some of these diseases can be corrected with gene editing. But our bodies include our brains, and the brain's most important and mysterious function – something we call “the mind” – also has a relationship with our DNA.

V. BEHAVIOURAL GENETICS AND THE GENE POLICE

In an issue of the *Judge Dredd* comic, a man called “Ruff” stands accused before a panel of judges for protecting his female partner from the city's abortion mandate and its enforcers (“the Gene Police”) when it is found that her unborn son carries a gene predisposing him to antisocial behaviour. After hearing the judges' case, Ruff argues that the same gene “*also turns out geniuses, the kind of people who change society*” – to which the judges reply “*We'd rather not take that chance. We have enough criminals as it is*” (Grant & Davis, 1999).

Objectionable though it may be to some, there is increasing evidence that antisocial

behaviour, or any other kind of behaviour for that matter, has a strong genetic basis and is heritable. According to Robert Plomin, an expert in the field of behavioural genetics, there is not a single human psychological trait studied to date that does not show some degree of genetic influence (Plomin, 2018). The claims of Plomin and his peers are backed up by solid evidence, in the form of almost 80 years of research on what Plomin calls “nature's gift to science” – twins. Identical and fraternal twins share the same environment for at least the first nine months of their lives, and usually much longer. But identical twins also share the same DNA, down to the last nucleotide. By comparing identical twins raised in different families, it is possible to see how big of a role genetics and the environment each play in how each twin turns out. The results are surprising. The influence of genetics on school achievement and reading ability – a trait we usually attribute to good parenting or good schooling – is about 60%. For general intelligence, it's 50%. Even general personality traits like neatness, aggressiveness, or joyfulness, have a genetic influence of about 40% on average. This is nowhere near the 95% genetic influence on a trait like eye colour, but it does show that much of who we are is determined before we are born and moulded by the world (Plomin, 2018). With this in mind, it is not surprising that Ruff was a revolutionary, willing to break the law to protect his partner and their baby, and to stand up for his beliefs. It may have been in his genes.

There is no single gene that determines something as complex as intelligence or personality, but thousands of small genetic differences add up to have a big effect. To add to the complexity, our environment and our experiences can have significant effects on how our genes are expressed, leading to very visible physical and personality differences even between genetically identical individuals (Spector, 2012). We are far from being able to use CRISPR to edit embryos so that they will reliably grow into children more intelligent and outgoing. We have identified not even 20% of the genetic factors that determine how tall a person will be as an adult – and that is considered a well-mapped trait (Plomin, 2018). But with research advancing at a steady pace, the gaps in our knowledge of which parts of the genome do what become ever narrower. Human psychological traits are complex – but it would be naive to think that we are so complex that we will never reach a point where we can influence such traits. What happens when we do?

VI. CAUTIONARY TALES

In the 1997 film *Gattaca*, humanity is divided into two classes: those who are genetically

acceptable and have socioeconomic freedom, and the genetically inferior, who are marginalized. Babies are screened at birth for undesirable traits, such as susceptibility to heart disease or violent tendencies. Their estimated lifespan is calculated, and they are assigned the genetic status that will define them for the rest of their lives. For couples who can afford artificial fertilization, only those embryos destined to become the healthiest, most intelligent and most attractive adults are chosen (Niccol, 1997). The film was released at a time when the international project to sequence the entire human genome was making rapid progress, genetic screening at birth for mental disabilities was standard practice in some parts of the United States, and artificial fertilization had become a relatively commonplace procedure (Andrews, 2001). That genetic technology would soon radically change society, for better or worse, seemed a sure thing. One writer in 1996 predicted that by 2020, everyone would carry around their genomes, documenting all their genetic characteristics, on a tiny plastic card in their pockets – and suggested that genetic discrimination, already a real danger in the U.S. health care insurance system, could become the new peril of the job market (Garrett, 1996; Andrews, 2001). This prediction came true a few years later when, in Germany, several young police cadets were denied long-term employment in the police force on the basis of their genetic risk of developing Huntington's disease (Lemke, 2013).

Some thinkers had been contemplating the consequences of human modification long before the genetic revolution of the 1990s. Aldous Huxley's 1932 novel *Brave New World* predates the discovery of the double helix, but also envisions a future where humans are divided into castes – in this case through a combination of cloning and physical conditioning of the developing embryo (which, although Huxley could not have known it at the time, is a form of epigenetic modification). In Huxley's world, humans are mass produced and engineered for specific roles. Boredom and aggression are suppressed with games, drugs, and casual sex. Those endowed with the intelligence to do so are sent into exile if they question the way their society is run (Huxley, 1932).

Both of these fictional worlds reflect real concerns about what should become of humanity should it begin to tamper with the biological basis of individuality, viewing human life in purely materialistic and utilitarian terms. In both worlds, humans are dehumanized, either because they fail to meet their society's standards or because these standards are forced upon them. Both paint an unsettling future where one idea in particular, which is still fostered in most societies today, has been either rejected or forgotten. That is the peculiar idea that every human being comes into the world as they are for a purpose that was not determined by any

other human being, and therefore has a right to be in the world, as they are. This is obviously a completely anti-utilitarian idea. It is also anti-materialistic, because it is a religious idea, and assumes that another type of reality exists beyond physical reality and takes priority over physical reality. This idea has survived many thousands of years of competition with many forms of prejudice that do not favour the right of certain individuals to exist as they are. This is a good indicator that there is something important about this idea, and it would be unwise for humanity to discard it.

Brave New World and *Gattaca* should not be seen as mere sci-fi, but as social commentary and cautionary tales. The deliberate alteration of the human gene pool is a reality that we now have to contend with. The SCID trials mentioned earlier are an example of a kind of gene therapy called somatic cell editing. That is, the gene edits that the children received were isolated in specific, non-reproductive tissues and could not be passed on to their own children. Germline editing, which changes all the cells in the body, including the reproductive cells, is the Pandora's box of genetic engineering – and it has already been opened.

VII. THE SAGA OF HE JIANKUI AND THE CRISPR BABIES

In November 2018, He Jiankui of Shenzhen University in China announced that he and his colleagues had edited the genes of two embryos before implanting them in their mother's womb, leading to the birth of twin girls. "Nana" and "Lulu", dubbed the "CRISPR babies" by the press, had each received edits to the CCR5 gene, which is linked to HIV resistance (Uddin et al. 2020). The announcement was an international scandal. He's actions were widely condemned by the scientific community, although the story was suppressed in the Chinese media (Cyranoski, 2019). Two months later, He was fired from his university, and his team was banned from conducting any work in human reproduction or receiving research funding by the ministry of health. In December 2019, courts sentenced He, already under house arrest, to three years in prison and a three million yuan fine. In April 2022, He was released from prison (Regalado, 2022).

The scientific community was not convinced by He's claimed solution to the AIDS epidemic. The twins were not at high risk for contracting HIV – it was their fathers, not their mothers, who were HIV-positive. Nor was it clear that editing their CCR5 genes would really grant them HIV-resistance, or any medical benefit at all. In fact, their risk of contracting another fatal disease, cancer, may have been increased. Why, then, did He

decide to edit their genes? And why twins with HIV-positive fathers, when there are many more single births to HIV-positive mothers, whose children are legitimately at risk of contracting the same illness? There are two other important details to take note of. First of these is the fact that one of the two twin embryos received edits to both copies of the CCR5 gene, while the other had only one copy altered successfully – but He decided to implant both embryos anyway (Raposo, 2019). Like the behavioural geneticists, He may have chosen twins as his test subjects intentionally in order to control for genetic differences that occur between non-identical siblings. The claimed purpose of medical intervention may merely have been a pretence for testing the effects of his gene editing technique. The second important detail is that the CCR5 gene is not only implicated in HIV resistance: it plays an important role in brain function, and suppressing the gene improves learning and memory in mice (Zhou et al. 2016). Scientists seeking to alter the CCR5 gene may have less noble goals than preventing infection with HIV – they may have dreams of human genetic enhancement.

VIII. WHO REGULATES THE REGULATORS?

In January 2015, in response to rumours of gene editing being applied to human embryos in China, eighteen academics convened in Napa, California to discuss amongst themselves what was to be done. The new development could panic lawmakers and the public in countries with tighter restrictions, leading to a clamp-down on researchers working with gene editing technology. Two months later, the Napa eighteen published a paper in the prestigious *Science* journal calling for a worldwide moratorium on human germline editing until a much larger group could be convened to decide how the technology should be regulated (Baltimore et al. 2015; Hurlbut 2015). Forty years earlier in another Californian city, Asilomar, another group (including two of the same academics that later met in Napa) had called for a wholesale moratorium on genetic engineering. They feared that if tight regulation was not applied immediately, destructive new pathogens could be created and escape the laboratory, destroying the public's, and the government's, confidence in their field (Hurlbut 2015). Whereas Asilomar was successful in putting the brakes on gene editing and getting the desired regulations put in place, Napa was not. Calls for a moratorium continue, and continue to be argued against (Lander et al. 2019; Kaan et al. 2021).

The U.S. bioethicist J.B. Hurlbut points out that the Napa conference can be rightly criticized for making the same mistakes as Asilomar. The

problem, he says, is not in deciding whether there should be a moratorium on gene editing or no – it is in genetics researchers taking it upon themselves to decide when the world should discuss how their own research should be regulated, and along what lines (Hurlbut, 2015). The Asilomar conference opened by defining the limits of discussion in the narrowest terms: this was to be a discussion about how gene editing technology could be used “safely” in a biosecurity sense, but not a moral discussion about whether or not it should be used at all, or what the eventual, non-medical implications of “safe” uses would be for society (Hurlbut, 2015).

Forty years later, there is reason to worry that we are repeating history. When 26 geneticists and bioethicists responded to requests for comment from *Nature Biotechnology* regarding the 2015 controversy, all conceded that human germline editing was ethically challenging – but many framed this as a technical safety issue, suggesting that so long as the unintended impacts of heritable modification on human bodily health can be kept within reasonable bounds, then it is acceptable. The idea that the technology could be abused, with far-reaching social consequences, was dismissed by one respondent as a “science-fiction scenario” (Bosley et al. 2015). This tunnel vision is endemic to the field; a later review of regulatory perspectives on gene therapy by a group of Korean researchers emphasized ensuring the safety of clinical trials, but contained not a single mention of the ethical problems that gene therapy raises (Shim et al., 2017). Fewer respondents were willing to broach the topic of human enhancement and the consequences this could have on humanity's values. Primate gene editing pioneer Weizhi Ji states that gene editing of human cells will have social consequences – particularly if only the wealthy can afford to have healthier, “more intelligent and beautiful babies”. Stem cell expert Martin Pera noted that people with deafness or dwarfism do not necessarily consider themselves unwell or defective (although he calls them “patient groups”). These people may lead slightly different but nonetheless happy lives, and people of average height and hearing might equally consider them to be part of the normal range of human variation. Pera describes germline correction of these conditions as “genetic cleansing” of deaf and short people from the human population. In societies where genetic correction is normalized, even individuals with less visible disabilities – such as cystic fibrosis or schizophrenia – may face even greater discrimination if it is perceived that they are accidental births, having fallen through the cracks of a system designed to prevent the existence of burdensome people (Andrews, 2001). Annelien Bredenoord, a bioethicist who later became a Dutch senator, expressed concern that editing the “essential characteristics” of unborn children would set them

along a path predetermined by their parents or by social needs, denying them the “right to an open future”. This, she says, should not be allowed, but she would allow “modifications that we can assume give children traits that are useful for all conceptions of a good life... being healthy should clearly be included” (Bosley et al. 2015). On one hand, Bredenoord’s voice sounds like the voice of conscience, or at least reasonable. But is a deaf person healthy? Is a person with Asperger’s healthy? Who defines what true health looks like? Should this be the responsibility of a panel of genetic experts, or a panel of medical experts, or a panel of lawmakers? Or should the public, including deaf and autistic people, put it to a vote? One of the Napa eighteen, Jacob Corn, stated some years ago that “a wider discussion among representative stakeholders... is underway” (Bosley et al., 2015). This discussion almost certainly did not include anyone without a doctorate. Yet who are the “stakeholders” if not the people who are most directly affected – like the hundreds of millions of young women, most of whom are not genetics experts, who may or may not want to give birth to gene-edited children in the future? Clearly decisions of enormous consequence are being made, but by a very small number of people, non-transparently, without widely agreed-upon definitions (Hurlbut, 2015). With most of the public under-educated and under-engaged about gene editing technology, the societal response has for decades been almost purely reactive (Andrews, 2001; Hurlbut, 2015). Shocking developments take the world by surprise, causing momentary outrage; regulations are put in place only after the fact, but it is not clear who makes them and what informs them; and those who might object, but are not allowed to participate in the decision-making process, are left impotent. This weak system of regulation may be exactly what some users of gene editing technology want.

Proponents of lax regulations on human gene editing have many advocates. One of these is Vera Lucia Raposo, an academic associated with the law faculty of the University of Macao, China. Raposo characterizes He’s gene editing not as a crime but as “a leap of faith in science” in a paper that lauds China’s supremacy in the field of genetics (Raposo, 2019). It is disquieting to read such praise from a legal expert who also freely admits that He’s actions constitute an illegal experiment in human genetic modification, carried out without the informed consent of the families involved (Raposo, 2019). In an earlier paper, Raposo makes the case that gene editing should not be restrained by concern for human dignity or human nature, which she argues are not affected by gene editing, if these concepts mean anything at all (Raposo, 2018). Human dignity is the freedom to make our own choices, she says, and we exercise this freedom

when we choose to change our genes. The paper’s title, “*Gene Editing, the Mystic Threat to Human Dignity*”, is probably an allusion to the 1997 Oviedo Convention – otherwise known as the Convention for the Protection of Human Rights and Dignity of the Human Being – which banned genetic discrimination, germline editing, and genetic engineering for non-medical purposes. The Convention has been ratified by 29 European states, including Raposo’s native Portugal, but, notably, not the United Kingdom. Article 13 of the Convention cites “*the ultimate fear*”: “*intentional modification of the human genome so as to produce individuals or entire groups endowed with particular characteristics and required qualities*” (Council of Europe, 1997). The ultimate fear, in other words, is eugenics – the pursuit of genetic enhancement and superiority, which has come to be associated with some of the darkest events in humanity’s recent history. “*We should not be afraid of eugenics*”, Raposo argues – emphasizing its use to spare present and future generations from disease and pain, rather than as a justification for racial hierarchies and murder. But what are we to make of her support for He Jiankui’s gene editing, an intervention with no likely medical benefit? Was the freedom and dignity of the twin girls affirmed when someone else chose to change their genes for them, even though they would have been healthy otherwise? Will their quality of life be improved by having the eyes of the world upon them, perhaps with a mind to “safeguard” them, but nonetheless eager to see the outcome of the experiment (Mallapaty, 2022)?

Like it or not, there are more gene-edited humans on the way. Not even a year after He’s announcement, a Russian scientist, Denis Rebrikov, declared that he also plans to create a CRISPR-edited baby – as soon as he can get approval from parents and regulators (Cyranoski, 2019). His stated goal is to cure a form of congenital deafness. “*Gene editing... does not change human nature*”, claims Raposo (2018). This is evading the truth. Our personalities – which most of us would agree is the foundation of our nature as individuals – are in part genetically determined, and therefore modifiable by gene editing. It is entirely possible that CRISPR and other gene editing tools could, in the near future, be used to decide the future personality traits of the unborn. However, these are tools that we have had at our disposal for decades only. There is another genetic tool that humans have been using for far longer, and with great success – selective breeding.

IX. BREEDING HUMANITY – EUGENICS AND GENETIC DISCRIMINATION

“*Through recorded history, animal breeders knew that you could breed for behaviour*”,

says Robert Plomin (2016). The results of this ancient practice are very apparent to us today. Livestock have been bred for a degree of docility that never occurs in their wild relatives. We know well that some dog breeds are smarter, or more aggressive, or more friendly with children; after all, they have been created with these exact qualities in mind. But when it comes to humans ourselves, we assume that we are too complex to be manipulated in this way, or at least that we ought to be. Throughout history, military force and belief systems have been used by states to influence the behaviour of their citizens. The possibility has scarcely been raised that modern states could control the behaviour of its citizens by influencing their genetic makeup. If this seems far-fetched, consider that genetic manipulation of populations has been practiced for a long time. One recent manifestation occurred during the era of European colonialism of the past few hundred years. During the colonization of South and Central America, Portugal and Spain promoted intermarriage between their settlers and native Americans, creating a mestizo population with blood ties to both Europe and the Americas – a population more likely to be sympathetic to the colonial endeavour (Raby, 2002). Following Australian government policy in the early to mid-20th century, aboriginal Australian children were taken from their families and assimilated culturally and genetically into white society, and similar policies were in practice in North America (Muller, 2011; Keating, 2020). Portions of indigenous populations were subjugated and partially integrated in this way. But to effectively mould the personality of whole populations through selective breeding has never been feasible. That, however, could soon change.

Over the last few years, China has been in the process of implementing a social credit system, which scores the “trustworthiness” of individuals based on copious amounts of data obtained from government records, surveillance cameras, and smartphones that track everything about their users: where they go and when, who they interact with, and their internet browsing activities (Deutsche Welle, 2021). Citizens are assigned a score that can determine what services they can access, such as public transportation, and what privileges they are entitled to (Lee, 2020). Following the state's standards of good behaviour improves your score and might even improve your lot in life, but dissenting individuals are punished (Lee, 2020). Even businesses are assigned credit scores, and it matters who your partners are – those that do business with low-scoring partners might put their own credit scores at risk (Donnelly, 2022). It is not much of a stretch to imagine that in the near future, this could also apply to citizens' reproductive choices. Imagine that choosing to marry someone

with a high social credit score increases your own score, while choosing a partner with a lower score causes you to lose social credit. With the cost of marriage and childrearing going up in China, it is conceivable that couples lower on the socioeconomic ladder may choose to have fewer children (Koetse, 2019). Compliance and obedience, like all personality traits studied so far, are likely to be heritable to some degree. They are also the very currency of the social credit system. This means that there is, to however small an extent, selective pressure being exerted against rebelliousness and nonconformity – a more obedient generation in the making. Artificial selection is, of course, also being applied by more brutal means, such as the forced sterilization of men and women in minority communities that refuse to integrate with the prevailing culture (USCIRF, 2020). Perhaps only those who by nature tend to comply will be afforded even the chance to reproduce: a nauseating mixture of genocide and eugenics.

Although such practices are deserving of our revulsion, modern China is not the first state to practice eugenics on its citizens. In the 1800s and early 1900s, some U.S. states sought to improve the gene pool by sterilizing individuals with undesirable personality characteristics – “*idiots, imbeciles, rapists, or habitual criminals*” (Andrews, 2001). The mentally disabled were clearly targeted for elimination, and the racial inferiority of blacks was considered by some to be scientifically proven. Americans were also warned against breeding with European immigrants, as this would make the population “*more mercurial... more given to crimes of larceny, kidnapping, assault, murder, rape and sex-immorality*”. Although laden with erroneous stereotypes, the eugenics laws were clearly directed at controlling human behaviour, including reproduction – and as a consequence, the impacts fell disproportionately on women. Consider the case of the orphan Carrie Buck, who after being raped by a relative of her foster parents and conceiving a daughter, was committed to an institution as “feeble-minded”, deemed an “imbecile” and a “menace” by a Supreme Court judge on the basis of her supposed genetic characteristics, and was forcibly sterilized – not because she was violent or mentally deficient (she and her daughter did well in school) but because she had had a child out of wedlock (Andrews, 2001). Nazi Germany later imitated and outdid the U.S. in the formulation of its own, particularly vicious eugenics program (Andrews, 2001).

Societies that allow such treatment of human beings in the name of the greater good are not likely to think twice about using genetic technologies for similar purposes. In matters of ethics and morality, the technology is almost irrelevant; rather, it is the values of the society that wields that technology. Even before CRISPR,

genetic modification to promote desirable traits in humans and eliminate undesirable ones was possible. Abortion and embryonic selection on the basis of genetic traits was already being practiced in wealthy nations, and was being promoted – either subtly or blatantly – by geneticists and by physicians with strong views about which combinations of physical and mental traits constituted a life worth living (Andrews, 2001; Lemke, 2013). In many cases these instances of genetic discrimination and reproductive coercion were only brought to light years after the fact, and similar cases very likely continue in some form today (Lemke, 2013).

Some modern philosophers suggest that the concern people have about governments assuming more control of reproductive genetics is overblown – rather, we should care more about whether or not we as individuals will take responsibility for the outcomes of our reproductive choices, which we are becomingly increasingly able to influence with genetic technologies (Anomaly, 2020).

X. THINKING TWICE

The path ahead is daunting. All parents want their children to have the best possible chance of a good life, but what constitutes a good life is difficult to define and differs between people. Even if we have clear standards of success, our children will have to cooperate – or compete – with other parents' children. There is always the possibility that people will try to imitate and outdo each other in their choice of genetic traits, leading us into the trap of using CRISPR and genetic profiling to create always “more beautiful and intelligent” children, ideal and homogenous, regardless of whether or not that is in the best interests of our children and the society that they will live in (Bosley et al., 2015; Anomaly, 2021). We will have to decide whether enhanced people are actually better than others, not just in terms of health or appearance or mental ability, but morally as well. And we will have to decide, collectively and hopefully individually, whether we have a social responsibility to better humanity by directly modifying and selecting all possible traits of our children, and to avoid producing people with traits that are likely to make them, in one way or another, a burden on others (Anomaly, 2020).

We should not rush to embrace or reject any technology for which both the benefits and risks are not known. Should we use CRISPR and other gene-editing tools to try to save lives? Most would say absolutely, we should. Should we alter the genome of the next generation, and the generations to follow, to make our children smarter, healthier, and more likely to succeed – by our own standards, that is? Would this make them happier, or would it

only constrain the range of possibilities for what a human being can be?

In *BioShock*, the main character discovers that he is genetically engineered and psychologically conditioned to be a perfect, unwitting slave. He is ordered to sit down, stand up, run, and kill on command by the founder of the fictional city of Rapture, Andrew Ryan, whose lofty vision is presented in the game's opening minutes: “A city where the artist would not fear the censor. Where the scientist would not be bound by petty morality.” Ryan makes his vision a reality, but it proves to be a dehumanizing and ultimately a doomed one, a mistake not worth repeating. Lest this plot prove to be even more prophetic than we supposed, let us be unwilling to abandon what seem to be petty ethical concerns, and let us be willing to restrain our ambitions, our prejudices, and our desire for perfection, which may destroy the good that we already have.

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Rewilding; are we doing it the right way and does it work?

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Human existence depends entirely on our biosphere and the state at which it is in (Díaz et al., 2019). The biosphere provides us with important physical elements; ecosystem services including resources and water purification as well as metaphysical elements like spiritual connections, both of which directly influence the quality of human life (Díaz et al., 2019; Donadio et al., 2022). The functionality and abundance of these services have been significantly transformed by human induced pressures (Díaz et al., 2019), such as rising temperatures, increasing greenhouse gas emissions (e.g. methane and CO₂), pollution of water through agriculture, mining and forestry (Sassi 2006). These changes in the environment exert multiple pressures on food security and human health through hazards like extreme temperatures and their influence on disease outbreaks like cholera, salmonella and mosquito borne infections (Arnell, 2022; McMichael et al., 2006). Furthermore, many organisms have been adversely affected and as a result we are currently in a biodiversity crisis, with around one million species predicted to become extinct in the decades to come (Díaz et al., 2019; Sassi 2006; Quintero & Wiens 2013). However, societal norms and urban settings have restricted citizens from living and interacting with an environment that is coexistent with “wild” nature, making many unaware of this crisis (Perino et al., 2019; Sassi 2006). Sustainable restructuring of urban environments, industries and conservation methods need to be implemented to improve long-term conservation efforts and enhance accessibility to natural spaces across socio economic groups (Danford et al., 2018; Sassi 2006). Over the years, conservation efforts have taken several different approaches with diverse positive and negative outcomes (Perino et al., 2019). An example of one of these conservational approaches is rewilding, and studies have reported mixed conclusions surrounding it, with the lack of social incorporation and the threat of predator reintroduction to livestock being common criticism (Johnson & Wallach 2016; Perino et al., 2019). Despite the cases of successful environmental benefits from rewilding, like most conservation plans, the realistic thresholds of society are often disregarded and must therefore be combined with the social and economic sciences to make use of land and resources effective, ethical and sustainable (Cvitanovic et al., 2015).

Rewilding is a broad term that has been used in restoration ecology with the limiting of current and historical human alterations to the environment being the common backbone for most projects (Lorimer et al., 2015; Sandom et al., 2013). However, it is objectively difficult to do a comparative analysis of rewilding as it lacks a fixed definition, with a wide range of strategies used across different projects (Lorimer et al., 2015; Sandom et al., 2013; Sweeney et al., 2019) (Fig 1). At its core, the term appears obvious, however since its initial usage in 1990 as returning land to a wilder and more natural state, its definition has taken many different forms (Fig 1) (Jørgensen 2015). A common misconception of rewilding is the nostalgic idea that it will restore our landscapes to a pre-existing age of a “pristine ecosystem” (Jørgensen 2015). Instead, it is about facilitating the spontaneous growth of natural systems back into modern times by incorporating rewilding strategies into our urban and conserved settings (Danford et al., 2018; Jørgensen 2015). However, rewilding often focuses on ecosystems prior to human establishment and can result in conservational methods that essentially exclude humans from having access to them (Danford et al., 2018; Jørgensen 2015). Initially, rewilding focused on what was known as the three *cs*: *cores*, *corridors* and *carnivores*, the focus was mainly on the reintroduction and protection of large extinct predators (Jørgensen 2015; Kopnina et al., 2019). The introduced animals resemble extinct Pleistocene megafauna (such as mammoths, giant sloths and three-ton wombats) in their functional roles as grazers, browsers or predators, to prevent the further development of weeds and pests (Jørgensen 2015; Marris 2009). This method of rewilding was conducted in places such as Oostvaardersplassen (OVP) in the Netherlands where non native species like Scottish Red deer (*Cervus elaphus*), Polish Konik horses (*Equus ferus*) and German Heck cattle (*Bos taurus*) were incorporated into OVP due to their morphological and functional resemblances to extinct Dutch megafauna (Kopnina et al., 2019). Rewilding has diversified into a much broader term in recent years and includes other methods such as *Trophic rewilding*, which is the restoration of landscapes through reintroducing certain native keystone species that have top-down or bottom up effects on their environment, and example of this being the well known reintroduction of the grey wolf (*Canis*

lupis), an apex predator, back into Yellowstone National Park in the United States of America and the impact on elk and tree populations that followed (Bakker & Svenning 2018; Jørgensen 2015; Sweeney et al., 2019). Lastly, rewilding had also been practiced in the form of *productive land abandonment*, where previously cultivated areas are left to develop without human interference (Gellrich et al., 2007; Jørgensen 2015). In the Swiss mountains, productive land abandonment, particularly of agricultural areas, resulted in the reestablishment of natural forests which are important support systems for a wide variety of species (Gellrich et al., 2007). Although the definition of rewilding infers limited human involvement, in most cases it has not been achieved as a passive act (except for productive landscape restoration), but rather involves direct action that usually involves four steps (Sandom et al., 2013). These steps include; defining the conservational issue at hand; the recognition of a certain ecological processes that is missing; the detection of important functional characteristics that are needed in the ecosystem; and lastly choosing a species that is applicable for reintroduction in a given area (Sandom et al., 2013).

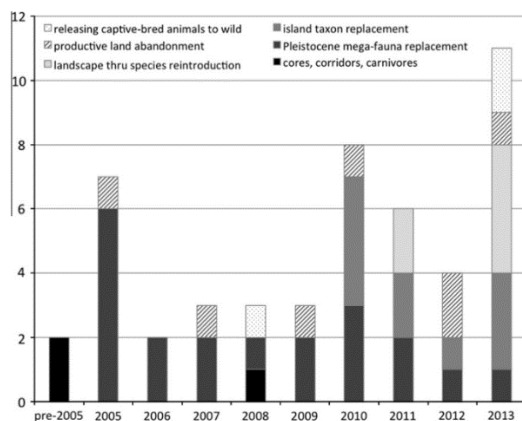


Figure 1: Fluctuations in the definition of rewilding from 1999-2013 in different scientific literature.

From a conservation perspective, rewilding has been an effective approach in some parts of the world and to date, it has been introduced in over 400 projects globally (Donadio et al., 2022). Successful *trophic rewilding* was conducted in the Pacific Ocean, where sea otters (*Enhydra lutris*) were returned to the ocean to control urchin populations, which consequently improved kelp forests as this was the primary food for urchins (Donadio et al., 2022; Estes & Duggins 1995). Apex-predators can have profound ecological influences across the trophic levels and their densities are therefore

fundamental to some areas. In the Yellowstone National Park in the United States of America, the grey wolf (*Canis lupis*) an apex predator, was reintroduced into the park in 1995 (Bakker & Svenning 2018; Ripple et al., 2001). This previously abundant predator, was eradicated from the majority of North America and Europe during the 1920s and as a result herbivores like elk (*Cervus elaphus*) increased substantially and an increased feeding decreased overall vegetation biomass in a trophic cascade (Bakker & Svenning 2018; Beschta et al., 2015). The reintroduction of the grey wolf resulted in a shift back to an ecological stable environment by cascading effects through the different trophic levels by decreasing prey foraging patterns and subsequently improving previous vegetative states such as riparian cotton wood which in turn prevented erosion and improving channel conditions (Beschta et al., 2015; Lorimer 2017; Ripple et al., 2001) (Fig 2).

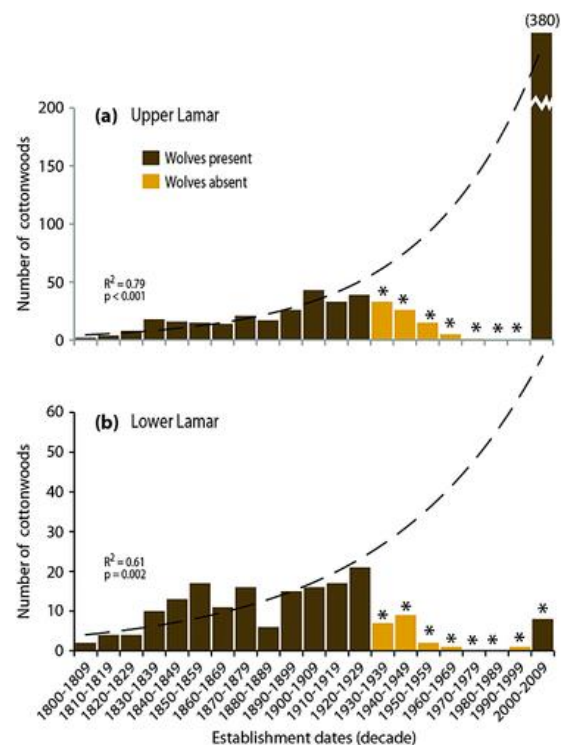


Figure 2: The influence of the presence of wolves on the frequency of riparian cottonwoods over several decades for two different study sites, the upper and lower Lamar.

Trophic rewilding was also initiated in several areas of Australian as a novel approach to improve high biodiversity losses, particularly through the reintroduction of apex-predators such as dingoes (*Canis dingo*) (Sweeney et al., 2019). Dingoes in particular are important top-down regulators of herbivores such as Kangaroos and non-

native invasive mesopredators such as the feral cat (*Felis catus*) and red fox (*Vulpes vulpes*) (Johnson & Wallach 2016; Sweeney et al., 2019). Dingoes influence the behavioral patterns (such as feeding) of large herbivores along with mesopredators and subsequently the diversity, and distribution of smaller mammals and vegetation is greater in areas with high dingo populations (Johnson & Wallach 2016). However, citizens are often reluctant to agree with reintroducing predators due to the risk of damage to live stock and human safety, and the lack of knowledge on their ecological importance (Johnson & Wallach 2016). This resulted in the common attempt to control dingo populations by lethal means, an approach that has proven to be ineffective in controlling dingo numbers and their behaviour towards livestock. The use of predator-friendly farming through secure fencing could protect both the livelihoods of farmers and promote conservation of biodiversity (Johnson & Wallach 2016) (Fig 3). The rewilding of dingoes has therefore been proposed as a low-cost solution to subdue the feeding behaviours of non-native invasive mesopredators that feed on native species (Sweeney et al., 2019). Rewilding in these cases has resulted in the successful ecological reestablishment of species that have been in disproportionately low densities for decades.

Despite this form of reintroduction being successful in a multitude of cases, there are several environmental concerns that can be associated with the moving of both native and non-native animals from a different geographical area (Caro & Sherman 2009). For example, the animals used for the Pleistocene reintroductions are translocated from another area often spanning over large geographical ranges. Each animal that is introduced is associated with a microbiome that can lead to the transfer of disease and viruses and have negative effects on local fauna. For instance, the amphibian chytrid *Batrachochytrium dendrobatidis* is a fungus that had detrimental affects on an extremely wide range of global amphibian populations as well as some bat species and was initially transferred across countries through human movement of the common platanna (*Xenopus laevis*) (Weldone et al., 2007). This species was transported due to its use in pregnancy testing in the 1950s, as the frog ovulated after injection of the urine from a pregnant human female (Weldone et al., 2007). Similar to this, species that are translocated from one area to the other for rewilding purposes also have the risk of being vectors of a multitude of diseases. They might migrate further than their intended niche and lead to hybridization and predators can pose a threat to livestock if the correct parameters are not put in place (Caro & Sherman 2009). Furthermore, the importation of animals that were not previously

found in an area might lead to reductions in their native ranges and consequential negative effects on that environment when they are captured for rewilding and moved outside of their distribution (Caro & Sherman 2009). Lastly, rewilding is difficult by nature due to the complexity of organisms and how their coevolved genomes influence an ecosystem (Wade 2007). Species interactions may be misunderstood because they are complex and tightly woven, when reintroduction is considered, its effects must take into consideration the effects at the community or population level.

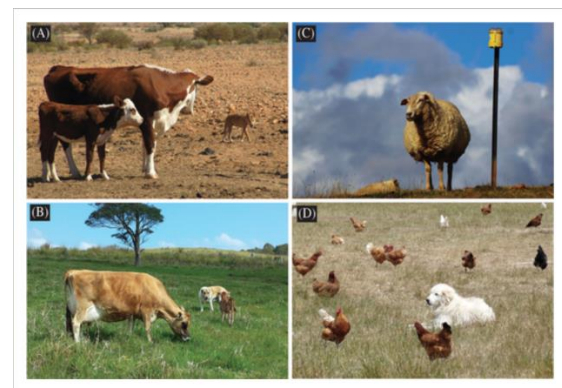


Figure 3: Different methods of predator-friendly agriculture in Australia that uses non-lethal practices as a means of protecting livestock from reintroduction of dingoes. This includes; taking care of proper feeding of livestock in order for animals such as cattle to defend themselves and their calves against dingoes (A), the use of guard dogs to defend cattle on smaller farms (B), Technology such as “foxlights” that influence the behaviour of animals and scares them off (C) and the use of proper fencing and guard dogs particularly for smaller livestock such as chickens.

Additionally, conservation that is based primarily on scientific evidence regularly disregards realistic thresholds of society and must therefore be managed in conjunction with the social and economic sciences to make effective, ethical and sustainable decisions of resource and land usage (Cvitanovic et al., 2015). Many scientists and practitioners in this field have not involved a wider community of citizens in such efforts, although there has been an increased pressure for improved integration of the social sciences in conservation decision making (Bennett et al., 2017; Campbell 2005). Involving citizens in rewilding is important to ensure that decisions are considerate of local people and planned with cultural norms, values and belief systems being considered to increase their success in a community (Cvitanovic et al., 2015). The limited outreach of researchers to local

communities is sometimes based on the belief that rural communities in particular are not interested in participating or being involved with conservation (Infield 1988). However, in 1988, a study conducted in South Africa focused on the “Attitudes of a rural community towards conservation and a local conservation area in Natal, South Africa” and challenged this view by showing that conserving wildlife, specifically through protected areas of land was well supported amongst rural communities (Infield 1988). Despite unfair Apartheid policies, land shortage, low income and other hardships that directly relate to the establishment of conserved areas, many participants strongly agreed with wildlife protection (Infield 1988). However, many people benefit from ecological services and limiting accessibility to conserved areas may exclude certain groups of people from accessing such benefits (Egoh et al., 2007). Ecosystem services are indirect and direct human benefits derived from ecosystems and include pollination, fuel wood, water purification and nutrient cycling (Egoh et al., 2007). The inclusion of these services into conservation plans would likely aid effective conservation action by benefitting and consequently gaining support from surrounding communities (Egoh et al., 2007). However, in countries such as South Africa, historical divisions still influence conservation efforts and are often a barrier to people wanting to access such benefits (Kockott, 2022). Vusi Tshabalala, manager of an environmental monitoring program in the Kruger National Park, Blyde River Canyon and Wolkberg region (a total of 2.5 million hectares) comments on the conflict of interest often present in hunting and conservation: “It is only allowed to certain people, those who have money, who are educated enough to go to the department to get a permit and to go through all those processes. And then there is the villager, a hunter, who has not been taught about all these processes, and who, anyway, does not have money to pay for a permit...So, there is a big war, a big fence, between who can and who can't,” (Kockott 2022). The conflict between hunting and conservation brings up the bigger question of what purpose these conserved areas have and for whom. If it is built primarily on the ideas of re-establishing the environment but excludes people from interacting with the nature, it is ineffective as there will be a lack of community interest in the overall importance of protecting the environment. In South Africa, conservation still has its roots in Western scientific standards and limited focus on the metaphysical meanings of nature to local communities, this can be improved by conservationists, social scientists and communities in close proximity to an area being directly involved in the management and planning (Cocks et al., 2012).

Rewilding can help improve the divide between social sciences and conservation efforts if it is planned in accordance with the needs of local people due to its ecological and spiritual importance (Cocks et al., 2012). Argentina is an example of a country that has involved some social strategies into its rewilding program, particularly through ecotourism. Since many of the species introduced are unique to that landscape, they have a great appeal to tourists (Donadio et al., 2022). The result is an economic incentive that allows for more local people to be involved in workshops based around conservation where they are trained as nature guides, artisan craftsmen and cooks amongst others (Donadio et al., 2022). Rewilding can be done on a smaller scaled urban setting. Marginalized groups are often limited from participating in outdoor recreation due to physical barriers of how cities have been built, rewilding in a smaller scale (urban spaces) is a possible solution to physical and mental benefits for a wider range of citizens (Danford et al., 2018). Many studies have shown that exposure to nature, particularly in cities can help alleviate mental stresses and aid community building (Danford et al., 2018). Parks and other green spaces are often located in areas that are inaccessible to poorer communities (Danford et al., 2018). Governments could easily introduce community gardens of native and noninvasive plants to people as a means to improve this unfair disconnect with nature. Other places such as Kruger National Park have achieved significantly decreasing numbers of poaching by involving the nearby community in their conservation plans in an all-female anti-poaching unit (Schmidt 2020). This method of conservation has had many beneficial effects on the community such as women empowerment, facilitation of education to nearby communities as most women on the team are from nearby villages and an 80% reduction in poaching (Schmidt 2020).

Ultimately, rewilding is a term that is for the most part loosely defined and is varied in all its different practices, making it difficult for researchers to objectively make decisions about its effectiveness. However, trophic rewilding has been successful in several countries, when native species, particularly predators are reintroduced (Sweeney et al., 2019). Conservation as a whole can benefit greatly from an increased inclusion of the community of non-scientists to understand whether different strategies are liable and will realistically be supported by its surrounding population of people.

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Can informally protected areas provide the safe haven required for South African reptiles? An assessment of the total conservation estate in South Africa and its potential for reptile conservation

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ABSTRACT

Protected areas are crucial towards reaching global conservation targets. At present in South Africa, the protected area estate constitutes ~ 8.69% of terrestrial land. The total conservation estate consists of protected areas and conservation areas. Areas that do not meet the International Union for Conservation of Nature (IUCN) criteria of a protected area can be referred to as “other effective area-based conservation measures” (OECMs). The number and extent of OECMs in South Africa is currently unknown. Furthermore, many protected areas are not fulfilling their conservation goals, and are rather protected only on paper. South Africa is currently not meeting its biodiversity stewardship requirements for reptiles through formally protected areas (both current and planned) and should seek alternative solutions to address this concern. This review uses the South African total conservation estate to provide a framework to measure where alternative means of protection can be applied. Methods have been formed to assess the degree of protection provided by the informally protected areas in South Africa.

Keywords: Biodiversity; demography; global conservation; IUCN; red list

I. INTRODUCTION

Protected areas are critical to reaching conservation targets (Watson et al. 2014). The IUCN defines a protected area as “A clearly defined geographical space, recognised, dedicated and managed, through legal or other effective means, to achieve the long-term conservation of nature with associated ecosystem services and cultural values.” (Dudley 2008; IUCN WCPA 2018). In South Africa, there are formally protected areas (i.e. National Parks, private and state owned Nature Reserves, Protected Environments, Forest Nature Reserves/ Wilderness Areas, Mountain Catchment Areas and World Heritage Sites) and informally protected areas (i.e. Conservancies, Non-binding stewardship sites, Heritage sites, Biodiversity reserves, and several more see OECMs below). The formally protected areas (PAs) in South Africa are legally proclaimed under the National Environmental Management: Protected Areas Act of 2003, and subsequently form part of the PA estate (Government of South Africa 2003). At present, the total protected area estate (<https://www.protectedplanet.net/country/ZAF>) constitutes about 8.69% of terrestrial land in South Africa.

Formal PAs in the protected area estate can be privately owned (Cumming & Daniels 2014). Privately owned PAs (hereafter PPAs) gazetted under the Protected Areas Act (Government of South Africa 2003) are thus included in the PA estate (De Vos et al. 2019). PPAs are formed through the voluntary relinquishment of private property rights.

At present, about 60% of South African mainland protected areas are state-owned, with PPA's contributing approximately 35% and an additional 5% being community-owned land (Mitchell et al. 2018). Since gazetted PPAs in South Africa fit the IUCN Protected Area definition (Stolton et al. 2014; Bingham et al. 2017), they are included when assessing the protection status of a species. Areas privately managed by either non-governmental organisations or corporations can be expressed as a PPA (Stolton et al. 2014). Not all areas in the total conservation estate of South Africa fit the IUCN definition of a protected area (Dudley 2008). Rather, some should be referred to as informally protected areas, otherwise known as other effective area-based conservation measures (OECMs) (Marnewick et al. 2020).

PPAs should not be mistaken for OECMs, which could be considered less resilient than formally PAs, as OECMs lack long-term security and permanence through legal status (Mitchell et al. 2018). An area cannot fall under both the PPA and OECM categories; it should be listed as one or the other (Mitchell et al. 2018). The number and extent of OECMs in South Africa is currently unknown and these are under review (Mitchell et al. 2018; Marnewick et al. 2020). Many of South Africa's informally protected areas that likely qualify as OECMs are listed under the South African Conservation Area Dataset (SACAD), and the informally private protected areas (such as conservancies and conservation areas) listed under the private land conservation areas (PLCA) dataset

(Shumba et al. 2020). OECMs include, but are not restricted to, biosphere reserves, conservation areas, conservancies, Ramsar sites, and botanical gardens. These areas are the global standard for potential OECMs, however within South Africa, several Ramsar sites have been proclaimed and received formal protection status, and are thus included in the formal protected area estate. Areas in South Africa that can be classified as OECMs according to the IUCN criteria (IUCN WCPA 2019) do not register as a formally protected areas, however, still contribute towards the total conservation estate (Mitchell et al. 2018). The criteria set out by the IUCN is the key separation between a PPA and an OECM (Mitchell et al. 2018). An OECM is defined by the IUCN as “*A geographically defined space that is not recognised as a protected area but it is governed and managed over the long-term in ways that deliver the effective in-situ conservation of biodiversity with associated ecosystem services and cultural and spiritual values*” (IUCN WCPA, 2019). Approximately 48% of the Key Biodiversity Areas (currently being identified, see Marnewick et al. 2020; Marnewick et al. 2021) in South Africa’s could meet the definition of an OECM (Donald et al. 2019). Key Biodiversity areas are geographical sites that are identified based on globally standardised criteria, but are essentially regions that improve the global perseverance of biodiversity (Knight et al. 2007; IUCN 2016). Although not considered when assessing the degree of protection for a species when following IUCN criteria (e.g. Tolley et al. 2019), OECMs contribute to the Convention on Biological Diversity’s (CBD) Aichi Target 11 (Laffoley et al. 2017; Wright 2019). The Aichi Target 11 is focused on the importance of biodiversity and ecosystem services through the use of formal protected areas and OECM’s (Zafra-Calvo et al. 2019).

Concerns have been raised regarding the brushstroke approach applied when assigning protection status to a species. These concerns are largely based on the species occurring within several formally protected area (thus, well protected), while these PAs are not fulfilling their conservation goals they were established for (Craigie et al. 2010), and are rather protected only on paper (Rife et al. 2013). The Management Effectiveness Tracking Tool or METT framework (current version: METT 3 (Stolton et al. 2007)) is a management assessment that can be applied to the total conservation estate, following standardised criteria (Stolton et al. 2007). Currently, only state-owned protected areas in the SAPAD have been assessed (Cowan et al. 2010), of which 275 have received a METT score (<https://www.protectedplanet.net/country/ZAF>). As formally protected areas with low METT scores are more likely to be degazetted than those with higher scores (Cowan et al. 2010), long-term protection of

threatened species occupying areas of poorly managed areas is unreliable.

South African reptiles are among the most well studied on throughout Africa (Tolley et al. 2016). As a result, approximately 25% of mainland African reptiles have been recorded within the national borders (Tolley et al. 2019). Tolley et al. (2019) noted that threatened South African reptiles were poorly represented in the protected area network. For many of these threatened reptile species, neither the current extent of the South African Protected Area Database (SAPAD) nor the national protected area expansion strategy (NPAES) for South Africa mitigates the extinction risk for these species. Consequently, South Africa is currently not meeting the biodiversity stewardship requirements for reptiles through formally protected areas (both current and planned) and should seek alternative solutions to address this concern. At present, a total 34 of the 414 terrestrial and freshwater reptile species in South Africa were given threatened category in the updated IUCN Red Listing (Table 1). Due to a lack of information, an additional 12 species could not be assessed, and have subsequently been listed as Data Deficient (Tolley et al. 2019). Two species were listed as Critically Endangered (Table 2), both not being protected at all, while 50% of species with an endangered listing were either poorly protected or not protected (Tolley et al. 2019).

To begin addressing the documented lack of protection for reptile species of conservation concern, I aim to use the review of the current conservation estate, both informal and formal protected areas, to provide a framework where conservation measures can be applied through alternative means. To do this, I will first reproduce the assessment by Tolley et al. (2019) using newly updated interpreted geographic distributions of near-threatened and threatened South African reptiles, and the updated formally protected areas in South Africa (SAPAD 2021). Thereafter, the degree of protection provided by the informally protected areas in South Africa, represented by the South African Conservation Area Database (SACAD 2021) and additional informally protected areas such as those defined by Shumba et al. (2020) will be assessed for the same reptile species. A final step will involve identifying any important informally protected areas, which provide protection for reptile species which are poorly represented in formal PAs, indicating suggested steps towards formal inclusion of these areas into the conservation strategy for reptiles in South Africa.

This review will allow one to investigate near-threatened and threatened reptiles within informally protected areas across South Africa and identify specific areas which are contributing

towards conservation. The informal protected dataset and the conservation area dataset will act as a comprehensive (yet incomplete) baseline to address this issue until more comprehensive datasets are made available (e.g. Key Biodiversity Areas and OECM's). The proposed methods will allow a systematic and strategic approach to identify key areas of conservation priority for near-threatened and threatened reptiles in South Africa. The reasoning behind the review is that despite the extensive existing protected area network and the NPAES in South Africa, near-threatened and threatened reptiles are and will remain poorly protected or not protected within the country. Conservation actions need to be put in place and implemented to reduce the extinction risk of poorly protected species, and identifying key areas for the conservation of poorly protected species is crucial to achieving this goal. I question how much protection do informally protected areas offer near-threatened and threatened reptiles in South Africa, and how does this differ from formally protected areas. Through this review, I aim to quantify the conservation importance of informal protected areas for South Africa's near-threatened and threatened reptiles. Along with assessing which informal protected areas have important conservation value for poorly protected near-threatened and threatened reptiles and how these can be integrated into a more effective conservation strategy for reptiles in South Africa. This review provides an opportunity to develop a list of near-threatened and threatened South African reptile species and indicate their protection statuses in both formally and informally protected areas. Furthermore, highlighting those species which are poorly protected by formally protected areas but have the opportunity to receive significant protection from informally protected areas. Develop a list of informally protected areas that are capable of significantly contributing to the protection of near-threatened and threatened South African reptile species. Identify the minimum number of informally protected areas that should be included in the national conservation strategy to afford necessary protection to fully protect near-threatened and threatened South African reptiles. Herein I present the prescribed methods to address total conservation estate in South Africa and its potential for reptile conservation.

¹ Total conservation estate: All Protected Areas (legally declared) and Conservation Areas (areas responsibly managed for biodiversity conservation but not legally declared as Protected Areas).

II. METHODS

- These analyses require QGIS¹ version 3.22, SAGA² version 7.6.3 and Rstudio³ version 1.2.5042.
- Acquire the following ESRI shapefiles and incorporate into GIS:
 - South African border;
 - South African protected area database (SAPAD);
 - South African conservation area database (SACAD);
 - South African informally protected areas (Shumba et al. 2020);
 - Interpreted geographic distributions (Tolley et al. In press) of all near-threatened (NT) and threatened (CR, EN & VU) South African reptile species (SANBI). Data Deficient (DD) species are excluded since by definition, their geographic distributions are not well known;
- Reproject the coordinate reference system (CRS) of all shapefiles to Africa Albers Equal Area Conic (ESRI: 102022).
- Clip all shapefiles to the South African border.
- Thereafter clean and prepare shapefiles for analysis.

i. Preparation of shapefiles

1. Apply the *multipart to single parts* function separately for the SAPAD, SACAD and Shumba et al. (2020) shapefiles. This ensures that polygons combined under a single name yet do not share a border are separated (e.g. single protected area made up of several separate polygons, such as the Mountain Zebra National Park).
2. Apply the *Buffer* function of 5 m to each of the shapefiles. This allows protected areas that share a border to act as a single protected area network and prevents the exclusion of properties that are too small (< 10 km²) to be included in the analysis as separate entities, as performed by (Tolley et al. 2019). This process ensures that the total connected protected area is evaluated rather than the discrete properties. While this process results in a more realistic and accurate calculation of the total protection area, it does underestimate the total number of discrete entities providing protection. This limitation is not of concern, as my focus is the

² <https://www.qgis.org>

³ System for Automated Geoscientific Analyses (<http://www.saga-gis.org>)

⁴ <https://www.rstudio.com>

total protection provided to a species. However, for species that have < 5 distinct protected areas, I will interrogate them further manually, thus being able to mention the true number of PA's providing protection. This is an important step as it allows for additional analysis of the management effectiveness of the PA using the METT 3 scoring system (Table 3). The degree of management effectiveness can further influence the stability of the protected areas that moderately or poorly protected species occur in.

- The interpreted distributions of all reptile species of conservation concern will be edited to excise all transformed habitats (Table 4), resulting in a more realistic interpretation of how much land area a species of conservation concern currently occupies.
- Intersect the interpreted geographic distributions of each threatened or near-threatened reptile species to the manicured formal and informal protected area network shapefiles.
- The surface area of each species' geographic distribution falling within a protected area will be determined and compared between the informally and formally protected areas.
- Informally protected areas that can contribute significantly to the protection of near-threatened and threatened South African reptiles will be identified as priority areas. Significance of protection contribution is evaluated by the follow methods.

ii. *Identifying priority informal areas*

1. Informally protected areas that provide connectivity between other protected areas that offer a safe-haven for reptiles of conservation concern. This will be performed through manually inspecting the intersects between the interpreted geographic distributions of threatened and near-threatened reptiles and the informally and formally protected estates.
2. Informally protected areas that provide sufficient "protection status" for near-threatened and threatened South African reptiles. This will be done using the methods of Tolley et al. (2019) to identify species that are Well protected, Moderately protected, Poorly protected, Not protected. These methods will be applied for the formally protected area estate in isolation, and again but including the informally protected area estate also. If the latter evaluation improves the "protection status" of the species, then identify the key informally protected areas that increased its protection.

iii. *Data analysis and presentation*

1. Visual comparison between the proportions of each species of conservation concern's geographic distribution that falls within the formally and informally protected areas (using similar graphics to Tolley et al. (2019)).
2. Statistical comparison (Wilcoxon Test for paired samples) between the proportions of each species of conservation concern's geographic distribution that falls within the formally protected areas and the summed total of the formally and informally protected areas.
3. A table capturing a comprehensive list of threatened and near-threatened species and the total surface area of protection provided by formally and informally protected areas and the total conservation estate as a proportion of the geographic distribution of the species.
4. List of informal protected areas with the proportion of additional protection they provide and connectivity to the formal PA estate.

III. CONCLUSION

Informal protected areas have shown to be important conservation areas globally (Bhagwat et al. 2005; Waithaka & Njoroge 2018; Alves-Pinto et al. 2021). The South African informal protected area network is a potential framework that allows for identifying priority conservation areas and offering alternative means of protection. Several limitations were identified from this review. The protected area network as represented in the SAPAD needs major modification to perform any analysis. While the potential informally protected area network also had major challenges, mostly to do with the lack of transparency in areas that classify, as there are currently no comprehensive special layers that can be used. The potential OECM sites appear to be best represented in the SACAD special layer. The reporting and sharing of data, specifically of Private Protected Areas or informal protected areas into public databases is lacking (Clements et al. 2018). Furthermore, using interpreted distributions would likely still require presence/absence confirmation at proposed priority areas. Using the METT scoring system is a poor system to measure conservation success (Hockings 2006). Scores could be biased based on methods applying the METT criteria. Thus, METT scores should be logically applied when addressing the overall protection of a species. Although several limitations to the project have been identified, the methods outlined from this review allow for a baseline investigation and provide a framework to begin addressing the lack of protection for reptiles of conservation concern.

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Supplementary information

Table 1. A review of the conservation status of all listed reptiles found in South Africa. Least Concern (LC), Near Threatened (NT), Vulnerable (VU), Endangered (EN), Critically Endangered (CR), Extinct (EX), Data Deficient (DD), Not Evaluated (NE).

	LC	NT	VU	EN	CR	EX	DD	NE	Total
Chelonians									
Cheloniidae								4	4
Dermochelyidae								1	1
Pelomedusidae	4		2						6
Testudinidae	6	1	2	2	1				12
Crocodylians									
Crocodylidae			1						1
Squamates									
Agamidae	6								6
Amphisbaenidae	9							1	10
Chamaeleonidae	10	6		3					19
Cordylidae	39	2	2						42
Gekkonidae	77	1					5		82
Gerrhosauridae	10		1			1		1	13
Lacertidae	26								26
Scincidae	52	3	1		1	1	5		63
Varanidae	2								2
Colubridae	15								15
Elapidae	11		1					1	13
Lamprophiidae	55						1		66
Leptotyphlopidae	9							1	10
Natricidae	1							1	2
Pythonidae	1								1
Typhlopidae	6							1	7
Viperidae	9	1	1	1			1		13
Total	348	14	11	6	2	2	12	11	414

Table 2. South Africa reptiles that fall under a Threatened or Near-threatened category in the IUCN red-list. Near Threatened (NT), Vulnerable (VU), Endangered (EN), Critically Endangered (CR), Extinct (EX).

Chelonians (Tortoises, Terrapins)			
Pelomedusidae	<i>Pelusios castanoides</i>	VU	A4ac
Pelomedusidae	<i>Pelusios rhodesianus</i>	VU	A4ace
Testudinidae	<i>Chersobius boulengeri</i>	EN	A4ace
Testudinidae	<i>Chersobius signatus</i>	EN	A4ace
Testudinidae	<i>Kinixys lobatsiana</i>	VU	A4cde
Testudinidae	<i>Kinixys natalensis</i>	VU	A4c
Testudinidae	<i>Psammobates geometricus</i>	CR	A4ace
Testudinidae	<i>Psammobates tentorius</i>	NT	A4ce
Crocodylians (Crocodyles)			
Crocodylidae	<i>Crocodylus niloticus</i>	VU	A2ac
Squamates (Lizards)			
Chamaeleonidae	<i>Bradypodion caeruleogula</i>	EN	B1ab(i,ii,iii)+2ab(i,ii,iii)
Chamaeleonidae	<i>Bradypodion caffer</i>	EN	B1ab(i,ii,iii,iv,v)+2ab(i,ii,iii,iv,v)
Chamaeleonidae	<i>Bradypodion dracomontanum</i>	NT	B1b(iii)
Chamaeleonidae	<i>Bradypodion kentanicum</i>	NT	B1b(i,ii,iii)

Chamaeleonidae	<i>Bradypodion melanocephalum</i>	NT	B1b(iii)
Chamaeleonidae	<i>Bradypodion nemorale</i>	NT	B2b(iii)
Chamaeleonidae	<i>Bradypodion ngomeense</i>	NT	B1b(iii)+2b(iii)
Chamaeleonidae	<i>Bradypodion pumilum</i>	NT	B2b(iii)
Chamaeleonidae	<i>Bradypodion thamnobates</i>	EN	B1ab(iii,v)
Cordylidae	<i>Hemicordylus nebulosus</i>	VU	D2
Chamaeleonidae	<i>Platysaurus monotropis</i>	NT	B1b(iii)
Chamaeleonidae	<i>Smaug giganteus</i>	VU	A2bcd+4bcd
Gekkonidae	<i>Cryptactites peringueyi</i>	NT	B1a
Gerrhosauridae	<i>Tetradactylus eastwoodae</i>	EX	
Gerrhosauridae	<i>Tetradactylus fitzsionsi</i>	VU	B1ab(i,iii)
Scincidae	<i>Acontias rieppeli</i>	NT	B1b(iii)
Scincidae	<i>Scelotes borquini</i>	VU	B1ab(i,iii,v)
Scincidae	<i>Scelotes guntheri</i>	EX	
Scincidae	<i>Scelotes inornatus</i>	CR	B1ab(i,ii,iii)
Scincidae	<i>Scelotes kasneri</i>	NT	B1b(iii)
Scincidae	<i>Scelotes montispectus</i>	NT	B1a
Squamates (snakes)			
Elapidae	<i>Dendroaspis angusticeps</i>	VU	B2ab(ii,iii,iv,v)
Viperidae	<i>Bitis albanica</i>	EN	B1ab(i,iii,iv,v)
Viperidae	<i>Bitis armata</i>	VU	B1ab(i,ii,iii,iv,v)
Viperidae	<i>Bitis schneideri</i>	NT	B2ab(ii,iii)

Table 3. The Management Effectiveness Tracking Tool (METT scores) of state-owned protected areas in South Africa (Cowan et al. 2010)

Management status	Authority		Total percentage score
National	SANParks	Mokala	55
National	SANParks	Tankwa	56
National	SANParks	Namaqua	58
National	SANParks	West Coast	59
National	SANParks	Bontebok	63
National	SANParks	Augrabies Falls	64
National	SANParks	Kalahari Gemsbok	66
National	SANParks	Agulhas	67
National	SANParks	Camdeboo	67
National	SANParks	Golden Gate Highlands	69
National	SANParks	Garden Route	71
National	SANParks	Richtersveld	72
National	SANParks	Mountain Zebra	73
National	SANParks	Table Mountain	75
National	SANParks	Addo	76
National	SANParks	Karoo	78
National	SANParks	Marakele	80
National	SANParks	Kruger	83
National	E C Parks	East London Coast	28
Nature reserve	E C Parks	Tomas Baines -Waters Meeting	28
Nature reserve	E C Parks	Nduli-Luchaba	29
Nature reserve	E C Parks	Great Fish River	32
Nature reserve	E C Parks	Tsolwana-Commando Drift	33
Nature reserve	E C Parks	Mpofu-Fort Fordyce	34
Nature reserve	E C Parks	Formosa	36
Nature reserve	E C Parks	Mkambati	36

Management status	Authority		Total percentage score
Nature reserve	E C Parks	Ongeluksnek	36
Nature reserve	E C Parks	Oviston	36
Nature reserve	E C Parks	Silaka	36
Nature reserve	E C Parks	Dwesa-Cwebe	37
Nature reserve	E C Parks	Hluleka	37
Nature reserve	EKZNW	Soada forest	25
Nature reserve	EKZNW	Kwa Yili	26
Nature reserve	EKZNW	iGxaingenwa	26
Nature reserve	EKZNW	INkelabantwana	26
Nature reserve	EKZNW	Marwaqa	28
Nature reserve	EKZNW	Erf 179, 180 & 181	29
Nature reserve	EKZNW	Ndhloveni	29
Nature reserve	EKZNW	Xotsheyake	29
Nature reserve	EKZNW	Lake Eteza	29
Nature reserve	EKZNW	Marutswa	31
Nature reserve	EKZNW	Tugela Drift	36
Nature reserve	EKZNW	The Swamp	38
Nature reserve	EKZNW	Manguzi	41
Nature reserve	EKZNW	Umvoti vlei	43
Nature reserve	EKZNW	Blinkwater	45
Nature reserve	EKZNW	Emakhosini	45
Nature reserve	EKZNW	Fort Nottingham	45
Nature reserve	EKZNW	Himeville	45
Nature reserve	EKZNW	Doreen Clark	48
Nature reserve	EKZNW	Ntinini	49
Nature reserve	EKZNW	Coleford	49
Nature reserve	EKZNW	Umhlanga lagoon	49
Nature reserve	EKZNW	Isandlwana	51
Nature reserve	EKZNW	Richards Bay	51
Nature reserve	EKZNW	Midmar	52
Nature reserve	EKZNW	Karkloof	53
Nature reserve	EKZNW	Mbumbazi	53
Nature reserve	EKZNW	Opathe	53
Nature reserve	EKZNW	Spioenkop	54
Nature reserve	EKZNW	Mount Currie	57
Nature reserve	EKZNW	Weenen	57
Nature reserve	EKZNW	Dlinza	58
Nature reserve	EKZNW	Impendle	58
Nature reserve	EKZNW	Queen Elizabeth Park	58
Nature reserve	EKZNW	Pongola Bush	58
Nature reserve	EKZNW	Ongoye	58
Nature reserve	EKZNW	Wagendrift dam	59
Nature reserve	EKZNW	Entumeni	59
Nature reserve	EKZNW	Vernon Crookes	59
Nature reserve	EKZNW	Oribi Gorge	60
Nature reserve	EKZNW	Umtamvuna	60
Nature reserve	EKZNW	Ntsikeni	60
Nature reserve	EKZNW	Amatigulu	61
Nature reserve	EKZNW	Sileza	61
Nature reserve	EKZNW	Beachwood Mangroves	62
Nature reserve	EKZNW	Skyline	62
Nature reserve	EKZNW	Chelmsford	62
Nature reserve	EKZNW	Lebombo Mountain	63
Nature reserve	EKZNW	Mpenjati	63
Nature reserve	EKZNW	North Park	63
Nature reserve	EKZNW	Bluff	64
Nature reserve	EKZNW	Umlalazi	64
Nature reserve	EKZNW	Krantzkloof	67
Nature reserve	EKZNW	Hhluhluwe Imfolosi Park	69
Nature reserve	EKZNW	Phongola	69
Nature reserve	EKZNW	Vryheid Hill	6
Nature reserve	EKZNW	Enseleni	69
Nature reserve	EKZNW	Harold Johnson	69
Nature reserve	EKZNW	Ndumo	70
Nature reserve	EKZNW	Kenneth Stainbank	70

Management status	Authority		Total percentage score
Nature reserve	EKZNW	Tembe	78
Nature reserve	EKZNW	Ithala	79
Nature reserve	Free State	Rustfontein Dam	48
Nature reserve	Free State	Sterkfontein Dam	50
Nature reserve	Free State	Soetdoring	52
Nature reserve	Free State	Bathurst	53
Nature reserve	Free State	Erfenis Dam	53
Nature reserve	Free State	Sandveld	57
Nature reserve	Free State	Caledon	57
Nature reserve	Free State	Gariiep	58
Nature reserve	Free State	Maria Moroka	58
Nature reserve	Free State	Kalkfontein Dam	59
Nature reserve	Free State	Willem Pretorius	59
Nature reserve	Free State	Koppies Dam	59
Nature reserve	Free State	Tussen die Riviere	61
Nature reserve	Free State	Seekoeivlei	65
Nature reserve	Gauteng	Abe Bailey	46
Nature reserve	Gauteng	Roodeplaatdam	46
Nature reserve	Gauteng	Leeufontein	52
Nature reserve	Gauteng	Suikerbosrand	61
Nature reserve	Limpopo	Lillie	12
Nature reserve	Limpopo	Kalkbank	16
Nature reserve	Limpopo	Machaka	16
Nature reserve	Limpopo	Leswena	19
Nature reserve	Limpopo	Mogol Dam	22
Nature reserve	Limpopo	Bewaarkloof	22
Nature reserve	Limpopo	Manombe	23
Nature reserve	Limpopo	Stellenbosch	24
Nature reserve	Limpopo	Moletjie	26
Nature reserve	Limpopo	D'Nyala	27
Nature reserve	Limpopo	Manthrombi	28
Nature reserve	Limpopo	Brackenridge	28
Nature reserve	Limpopo	Tzaneen Dam	31
Nature reserve	Limpopo	Nzhelele	32
Nature reserve	Limpopo	Doorndraai Dam	34
Nature reserve	Limpopo	Letaba Ranch	34
Nature reserve	Limpopo	Wonderkop	34
Nature reserve	Limpopo	Mphaphuli	34
Nature reserve	Limpopo	Hans Merensky	36
Nature reserve	Limpopo	Turfloop	36
Nature reserve	Limpopo	Modjadji	36
Nature reserve	Limpopo	Wolkberg Caves	36
Nature reserve	Limpopo	Schuinsdraai	37
Nature reserve	Limpopo	Malebocho	37
Nature reserve	Limpopo	Nwanedi	38
Nature reserve	Limpopo	Nylsvley	39
Nature reserve	Limpopo	Langjan	39
Nature reserve	Limpopo	Blouberg	41
Nature reserve	Limpopo	Lekgalameetse	41
Nature reserve	Limpopo	Percy Fyfe	43
Nature reserve	Limpopo	Potlake	45
Nature reserve	Limpopo	Makhuya	51
Nature reserve	Limpopo	Atherstone	57
Nature reserve	MTPA	Bushbuckridge	9
Nature reserve	MTPA	Mkhombo	30
Nature reserve	MTPA	Barberton Mountainlands	41
Nature reserve	MTPA	Andover	46
Nature reserve	MTPA	Mabusa	50
Nature reserve	MTPA	SS Sokosana	51
Nature reserve	MTPA	Mthethomusho	53
Nature reserve	MTPA	Nooitgedacht	55
Nature reserve	MTPA	Verloren Valei	58
Nature reserve	MTPA	Mdala	59
Nature reserve	MTPA	Sterkspruit	60
Nature reserve	MTPA	Loskop dam	61

Management status	Authority		Total percentage score
Nature reserve	MTPA	Barberton	62
Nature reserve	MTPA	Blyde River Canyon	63
Nature reserve	MTPA	Mahushe Shongwe	65
Nature reserve	MTPA	Manyleti	67
Nature reserve	MTPA	Songenvilo	67
Nature reserve	MTPA	Ohrigstad	68
Nature reserve	N Cape	Nababiep	37
Nature reserve	N Cape	Doornkloof	41
Nature reserve	N Cape	Oorlogskloof	41
Nature reserve	N Cape	Rolfontein	42
Nature reserve	N Cape	Goeap	45
Nature reserve	N Cape	Witsand	46
Nature reserve	NWPTB	Kgaswane	46
Nature reserve	NWPTB	Vaalkop	47
Nature reserve	NWPTB	Barberspan	51
Nature reserve	NWPTB	Molopo	56
Nature reserve	NWPTB	Madikwe	56
Nature reserve	NWPTB	Boskop	58
Nature reserve	NWPTB	Borakalalo	60
Nature reserve	NWPTB	Botsalano	62
Nature reserve	NWPTB	Wolwespruit	65
Nature reserve	NWPTB	Molemane Eye	68
Nature reserve	NWPTB	Pilanesberg	70
Nature reserve	CapeNature	Soetendalsvlei	21
Nature reserve	CapeNature	Verlorenvlei	26
Nature reserve	CapeNature	Waterval	27
Nature reserve	CapeNature	Kleinjongensfontein	36
Nature reserve	CapeNature	Blomboschfontein	37
Nature reserve	CapeNature	Limietberg	39
Nature reserve	CapeNature	Waenhuiskrans	39
Nature reserve	CapeNature	Knersvlakte	39
Nature reserve	CapeNature	Jonkershoek	40
Nature reserve	CapeNature	Matjiesrivier	40
Nature reserve	CapeNature	Vrolijkheid MCA	40
Nature reserve	CapeNature	Walker Bay	41
Nature reserve	CapeNature	Harmony Flats	41
Nature reserve	CapeNature	Outeniqua	42
Nature reserve	CapeNature	Kruisrivier	44
Nature reserve	CapeNature	Riverlands	45
Nature reserve	CapeNature	Rocherpan	47
Nature reserve	CapeNature	Kammanassie	48
Nature reserve	CapeNature	Driftsands	48
Nature reserve	CapeNature	Dassen Island	52
Nature reserve	CapeNature	Salmondsdam	53
Nature reserve	CapeNature	Keurbooms	53
Nature reserve	CapeNature	De Hoop	54
Nature reserve	CapeNature	Kogelberg	55
Nature reserve	CapeNature	Goukamma	55
Nature reserve	CapeNature	Swartberg	55
Nature reserve	CapeNature	Grootvadersbosch	56
Nature reserve	CapeNature	Vrolijkheid	57
Nature reserve	CapeNature	Robberg	63
Nature reserve	CapeNature	Dyer Island	63
Nature reserve	CapeNature	Bird Island	66
Nature reserve	CapeNature	Gamkaberg	69
World Heritage sites	EKZNW	uKhahlamba Drakensberg	73
World Heritage sites	SANParks	Mapungubwe	75
World Heritage sites	iSimangaliso	iSimangaliso Wetland Park	86
Special Nature Reserves	CapeNature	Brenton Blue Butterfly	60

Table 4. Land use cover categories used to remove or keep based on individual interpreted distributions of Threatened and Near-threatened reptiles in South Africa.

ID	SA landcover classes	Natural (1) and transformed (0) habitats
1	Water seasonal	1
2	Water permanent	0
3	Wetlands	1
4	Indigenous Forest	1
5	Thicket/Dense bush	1
6	Woodland/Open bush	1
7	Grassland	1
8	Shrubland fynbos	1
9	Low shrubland	1
10	Cultivated comm fields (high)	0
11	Cultivated comm fields (med)	0
12	Cultivated comm fields (low)	0
13	Cultivated comm pivots (high)	0
14	Cultivated comm pivots (med)	0
15	Cultivated comm pivots (low)	0
16	Cultivated orchards (high)	0
17	Cultivated orchards (med)	0
18	Cultivated orchards (low)	0
19	Cultivated vines (high)	0
20	Cultivated vines (med)	0
21	Cultivated vines (low)	0
22	Cultivated permanent pineapple	0
23	Cultivated subsistence (high)	0
24	Cultivated subsistence (med)	0
25	Cultivated subsistence (low)	0
26	Cultivated cane pivot - crop	0
27	Cultivated cane pivot - fallow	0
28	Cultivated cane commercial - crop	0
29	Cultivated cane commercial - fallow	0
30	Cultivated cane emerging - crop	0
31	Cultivated cane emerging - fallow	0
32	Plantations / Woodlots mature	0
33	Plantation / Woodlots young	0
34	Plantation / Woodlots clearfelled	0
35	Mines 1 bare	0
36	Mines 2 semi-bare	0
37	Mines water seasonal	0
38	Mines water permanent	0
39	Mine buildings	0
40	Erosion (donga)	0
41	Bare none vegetated	0
42	Urban commercial	0
43	Urban industrial	0
44	Urban informal (dense trees/bush)	0
45	Urban informal (open trees/bush)	0
46	Urban informal (low veg/grass)	0
47	Urban informal (bare)	0
48	Urban residential (dense trees/bush)	0
49	Urban residential (open trees/bush)	0
50	Urban residential (low veg/grass)	0
51	Urban residential (bare)	0
52	Urban school and sports ground	0
53	Urban smallholding (dense trees/bush)	0
54	Urban smallholding (open trees/bush)	0
55	Urban smallholding (low veg/grass)	0
56	Urban smallholding (bare)	0
57	Urban sports and golf (dense trees/bush)	0
58	Urban sports and golf (open trees/bush)	0
59	Urban sports and golf (low veg/grass)	0
60	Urban sports and golf (bare)	0
61	Urban township (dense trees/bush)	0
62	Urban township (open trees/bush)	0
63	Urban township (low veg/grass)	0

ID	SA landcover classes	Natural (1) and transformed (0) habitats
64	Urban township (bare)	0
65	Urban village (dense trees/bush)	0
66	Urban village (open trees/bush)	0
67	Urban village (low veg/grass)	0
68	Urban village (bare)	0
69	Urban built-up(dense trees/bush)	0
70	Urban built-up (open trees/bush)	0
71	Urban built-up (low veg/grass)	0
72	Urban built-up (bare)	0

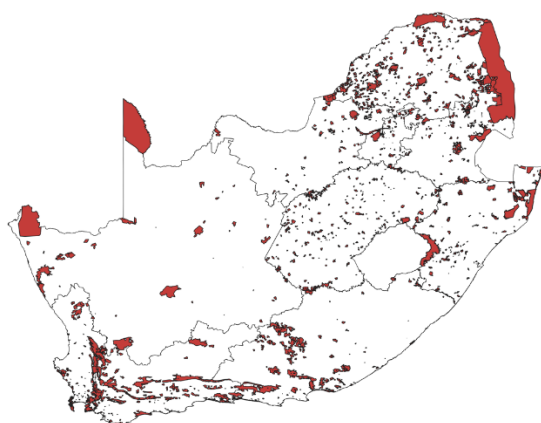


Figure 1. Total South African Protected Area Database (SAPAD) used to estimate protection status of reptiles of conservation concern.

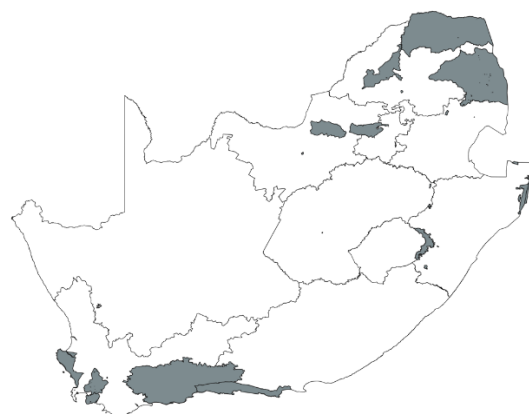


Figure 2. Total South African Conservation Area Database (SACAD) used to estimate informally protected areas and priority conservation areas for reptiles of conservation concern (The dataset is currently incomplete, as areas in the PLCA (Shumba et al. 2020) still need to be identified).

Why do bats carry so many zoonotic viruses? In the height of the current Covid-19 pandemic, this has been a hot topic

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ABSTRACT

Bats are receiving considerable attention for being reservoirs of several viruses that cause dreadful infectious diseases. These include three coronaviruses, one of which is responsible for the current Covid-19 global pandemic (the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)), Middle East respiratory syndrome (MERS), severe acute diarrhea syndrome (SADS), Ebola and Hendra viruses. Interestingly, bats do not show clinical symptoms when infected with the viruses, even if the infection is constant. Many studies have demonstrated that human activity contributes largely to the great variety of emerging diseases. Practices such as deforestation, human encroachment, farming and bushmeat trade, bring wildlife in close contact to humans or intermediate hosts such as domestic animals, facilitating disease transfer from animals to humans. Even so, there is enough evidence to support that bats in particular carry a lot of viruses. Understanding the biology and ecology of reservoir species is vital for the mitigation of future emerging diseases. Therefore, in this review we consolidate information about bats traits that are associated with pathogens. The discussion will mainly focus on the recognised hypotheses that attempt to explain why bats are special diseases carriers. We hope that this will aid bring light to medical research.

Keywords: Bat-borne viruses, bat immunology, Chiroptera, spillover, zoonoses

I. INTRODUCTION

The emergence of unpredictable and unavoidable infectious diseases has been happening for thousands of years, way before the inventions of advanced technologies for discovering the pathogens causing the infections (Morens & Fauci, 2013). However, at the present time, even with the advanced scientific knowledge and curative measures such as diagnostics, effective therapies, and vaccines, infectious diseases are still emerging, at an alarming rate, and they are likely to continue in this fashion (Wang & Crameri, 2014). This does not only pose threat to human health but also to the economic health of communities and global biodiversity (Luis *et al.*, 2013; Wang & Crameri, 2014). Moreover, the rising human population is not helping in any manner. Most of these diseases are the result of zoonoses (Jones *et al.*, 2008), referring to diseases that are transferred from a non-human vertebrate animal to humans and vice versa. Diseases transmitted from animals to humans are not just a present day problem. They have existed side-by-side with the vast diversity of diseases that affect humans, such as tuberculosis (TB), influenza, and yellow fever, but most were involved with domestic animals (Wang & Crameri, 2014). Scientists estimate that 60-75% of the emerging diseases are from animals and over 70% are coming from wildlife (Jones *et al.*, 2008; Smith & Wang, 2013; Wang & Crameri, 2014). Additionally, this is only

expected to increase gradually (Figure 1) (Luis *et al.*, 2013, Smith *et al.*, 2014).

One fifth of the world's extant mammals are bats. Belonging to the order Chiroptera (Moratelli & Calisher, 2015), they comprise of more than 1300 species (Banerjee *et al.*, 2019) and are found in all parts of the world, except Antarctica (Simmons, 2005; Moratelli & Calisher, 2015). Their species-richness is of great importance as bats are great pollinating agents. Bats also disperse a wide variety of seeds, helping to restore and regrow forests, and they aid to suppress arthropod populations (Schneeberger & Voigt, 2016). In addition, bats have evolved adaptations, such as hibernation, echolocation, and powered flight (Wynne & Wang, 2013). However, despite the ecosystem services and traits that bats have, they are amongst the most feared animals today. Bats are notorious for their association with zoonotic infectious diseases. As a result, great scientific research efforts have been made to mainly answer the question "are bats special zoonotic viruses carriers?". In this mini-review we attempt to show that indeed bats may be special harbours of many viruses and to also answer the question that follows, "why bats carry so many viruses?" This will be done by integrating the available theories about bats as important hosts of viruses. Knowing what makes bats special virus reservoirs without being clinically bothered may help us in a wide variety of conditions such as cancer. This will also allow us to apply the

same concept to modify humans immune responses. Lastly, effective countermeasure to prevent future transmissions of zoonotic diseases will be developed

and both scientists and medical personnel's will be able deal with future outbreaks in a more efficient, prepared and organised way.

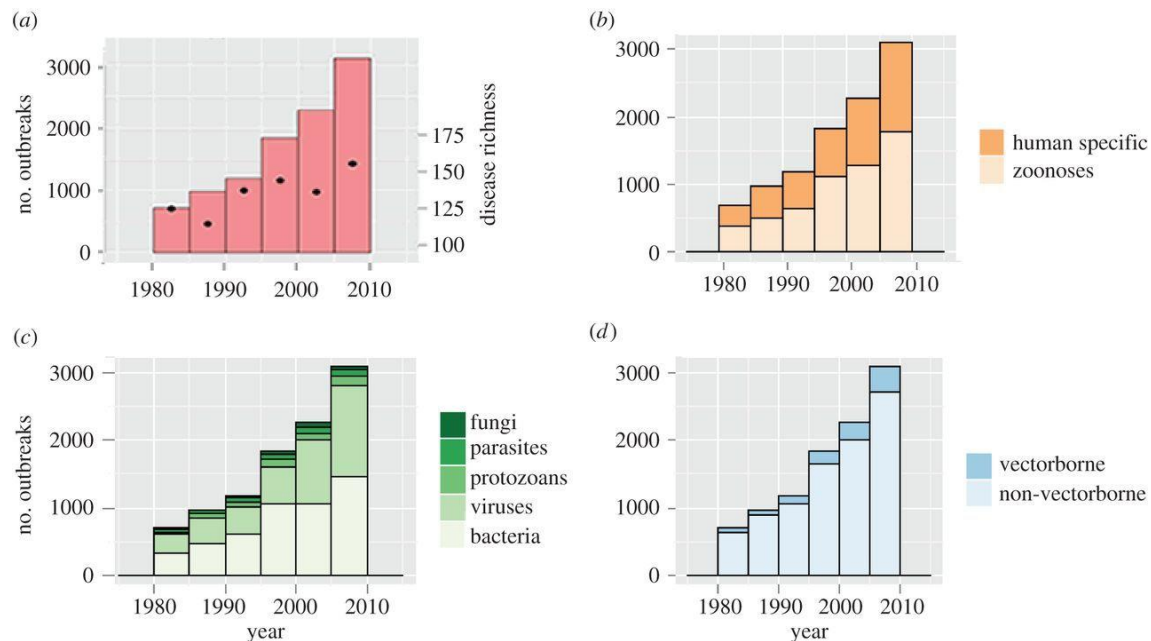


Figure 1: The number and causes of human diseases epidemics worldwide between the years 1980-1990. Graph (a) indicates the number of disease outbreaks with respect to disease richness , (b) host type, (c) pathogen taxonomy and (d) transmission mode. (obtained from Smith *et al.*, 2014)

II. DRIVERS OF ZONONOSES

First and foremost, it is important to understand the drivers and effects of zoonotic diseases because it allows for the early development of effective countermeasures and prompt response. There are many factors that cause zoonoses, and for the most part, humans are blamed. Scientific research shows that anthropogenic practices such as habitat destruction, human encroachment, agriculture, as well as global climate change, are the main drivers of emerging high virulent zoonoses (Wang & Cramer, 2014). The rising human population demands for more resources and therefore agricultural practices have been modernised and require more land, which leads to the destruction of natural habitats (Valitutti *et al.*, 2020). Being a very important traditional food source of many cultures globally, bushmeat is also a significant driver of appearing wildlife diseases. The growth of bushmeat trade can increase the danger of spillover if living animals are transferred to centralised markets where a great diversity of animals is forced to be in close contact in stressful and crowded conditions. This has been the case for outbreaks of some coronaviruses (Wang *et al.*, 2006). Lastly, global climate change plays a key role

in the alteration of vectors distribution. With the warming of temperatures, warmer seasons are lengthened, and this is favourable for disease carrying insects. This will allow them to thrive and expand their geographic range, allowing mixing with already know vectors and introducing the agents to potential new vectors.

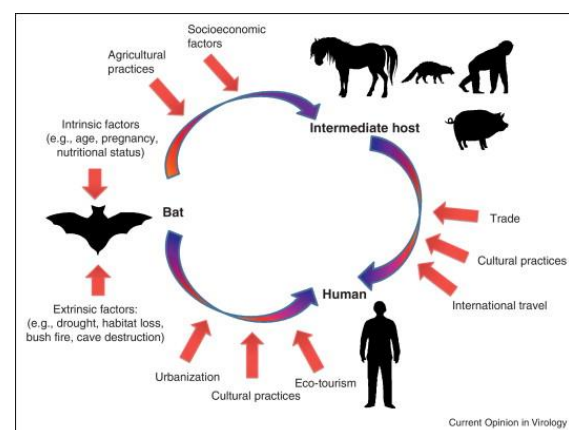


Figure 1: Examples of zoonotic diseases drivers (obtained from Smith & Wang , 2013)

III. EVIDENCE OF BATS AS VIRUS CARRIERS

i. *Are bats special virus carriers?*

In spite of the human drivers of zoonotic infections, bats do carry hundreds of viruses, of which they do not seem show clinical symptoms for diseases, except for rabies (Wang *et al.*, 2011). Molecular studies have demonstrated that over two hundred viruses have been identified in different families of bats globally (Wang *et al.*, 2011; O'shea *et al.*, 2014). The viruses identified are members of over twenty-six families, representing a great diversification, strongly suggesting that the association of bats and viruses is not by accident (O'shea *et al.*, 2014). Most research studies were centred around the question, “are bats special carriers of viruses?” (Wang *et al.*, 2011; Luis *et al.*, 2013). In a comparative study by Luis *et al.*, (2013), they show bats host more zoonotic viruses per species compared to rodents even with rodents species richness being twice that of bats. Certainly, when significant efforts were made to investigate certain virus species, including herpesviruses and coronaviruses, considerable number of viruses were isolated or detected from bats, and most of which were previously unrecognised viruses. Moreover, many researchers only focus on specific viruses, or specific viral families, thus, it is in no doubt that a many viruses must have been missed (Moratelli & Calisher, 2015). Bats host about three of ten viruses of epidemic concerns (Ruiz-Aravena *et al.*, 2022) and below we briefly go over one or two diseases that are caused by the viruses. These are the henipaviruses, coronaviruses and filoviruses. Further information of bats host species and the viruses that each species harbour are well presented by Moratelli and Calisher *et al.*, (2015)

ii. *Rabies and respiratory diseases*

Rabies virus causes a disease also known as rabies, and it is the very first virus that was recognised in bats (Moratelli & Calisher, 2015). Belonging to the genus *Lyssavirus* and family *Rhabdoviridae* (Kuzmin *et al.*, 2011), rabies is one of the most lethal diseases known. Humans and animals that are infected with this virus usually die shortly after infection if they do not receive speedy medical attention. Rabies are thought to have originated from wolves, and it now affects dogs worldwide (Moratelli & Calisher, 2015). Currently, bats are reservoir hosts of almost all (10 of 11) the known *lyssavirus* species (Kuzmin *et al.*, 2011). In some parts of the world, rabies are still a problem (Africa and Asia) and hundreds of people and livestock die every year from bat bites, however, many deaths are caused by dogs (Begeman *et al.*, 2018). Bats are also associated with respiratory

diseases such as coronaviruses and the Middle East respiratory syndrome (MERS). The first outbreak of coronavirus was in 2002 and it infected over 8000 humans leading to more than 800 deaths (Wang *et al.*, 2011). Ever since then, over 4 800 viruses sequences have been identified in 14 families of bats (Ruiz-Aravena *et al.*, 2022). The latest coronavirus outbreak was caused by SARS CoV-2 and it caused the devastating Covid-19. Covid-19 was first recorded in late 2019 and it has caused hundred thousands of human death in a very short period of time. Many countries suffered the loss of frontline health care workers and economies of many populations were affected leaving many people in poverty.

iii. *Henipaviruses and filoviruses*

Hendra and Nipah viruses emerged in 1994 and late 1990s, respectively. The Hendra viruses mostly infected horses and the Nipah infected pigs (Smith & Wang, 2013). However, in both events the viruses were transmitted from the domesticated animals to humans and resulted in mortality of both the animals and infected humans (Moratelli & Calisher, 2015). The encouraging news about these cases is that scientists have found the bats reservoir host species of these viruses and they are species of the flying fox bat (Smith & Wang, 2013). Filoviruses are also one of the most pathogenic viruses ever recorded. Ebola and Marburg viruses are examples of the filoviruses. The Ebola virus caused mortality of many people and great apes in Africa (Wang & Crameri, 2014). This led to considerable scientific effort to be dedicated in the investigations of the host of the virus. After many years of research, scientist recently suggested that the natural reservoir host of these viruses are bats (Wang & Crameri, 2014).

IV. BATS TRAITS ASSOCIATED WITH PATHOGENS

i. *Lifespan and body size*

Bats have a wide range of unique life history characteristics that may contribute to them being special reservoirs of pathogens. These include their longevity. Bats have the benefit of long life span for their body size. Longevity in mammals is usually inversely proportional to body size, therefore, larger animals last longer than smaller animals (Wilkinson & South, 2002). However, this is not the case with bats. Bats body mass range from 2g (bumblebee bat) to 1 kg in some species (flying-fox). Some bat species are known to live for more than 30 years, while rodents of similar size only live for one to two years (Moratelli & Calisher, 2015). In actual fact, when looking at body size, bats outlive all extant mammals. As a result, researchers strongly

believe that this may enable viruses to persist for chronic infections (Wang *et al.*, 2011). Luis *et al.*, (2013) demonstrated that viral richness increases with life span. Moreover, they further showed that viral species richness was higher in bats with a larger body mass, small litters and more litters annually.

ii. Aggregations and hibernation

Substantial bat species are gregarious and some species usually form the most extensive aggregations of the mammalian group worldwide (Moratelli & Calisher, 2015). The Geoffrey's Rousette fruit bats are known to form aggregations of over 2 million individuals in the Monfort bat cave annually (Figure 3). The Mexican free-tailed bats on the other hand can form a cloud of over 20 million individuals in Bracken Cave of central Texas (Moratelli & Calisher, 2015). These remarkably large aggregations can allow for quick transmissions of viral infections in the population and may allow the persistence of immune infections. The first identified zoonotic virus in bats, rabies, was documented in aggregations of bats (Calisher *et al.*, 2006). Bats also have the ability to enter into prolonged daily torpor and seasonal hibernation (in some species), which allows them to save energy. Although there are not many studies on the effect of torpor and hibernation on bats' ability to harbour many viruses, pathogens may hibernate together with bats and bats may consistently spill the virus without detection (Calisher *et al.*, 2006).

iii. Powered flight and sympatry

The trait that receives the most focus of all the traits that bats have evolved is flight. Bats are the only mammals that evolved powered flight. Unlike soaring, flapping requires a lot of energy, therefore, it is speculated that the evolution of flight had some impact on the metabolism and immune system of bats (Brook & Dobson, 2015). In contrast to the seven fold increase in the metabolic rate of rodents when they run until the point of exhaustion, or the two fold of birds when they are flying (Bundle *et al.*, 2007), the metabolic rate of bats is believed to increase by fifteen to sixteen fold when flying (Speakman *et al.*, 2005). Scientists suggest that the increase in both temperature and metabolic rate during flight may select for pathogens tolerant to those conditions and this promotes the number of virus species that many find refuge in bats (O'shea *et al.*, 2014). In many animals including humans, body temperature increases as a defence mechanism which fights pathogens, thereby decrease the period

of the disease. In other words, when we get infected by a pathogen, the fever we get improves the effectiveness of our immune response making us heal faster (Canale & Henry, 2011; O'Shea *et al.*, 2014). Since in bats, flying is an everyday behaviour (unless when hibernating), it is thought that the daily increase in their body temperature may also facilitate heat resistant pathogens in some bats (O'Shea *et al.*, 2014). Bats are the second most widely distributed mammals in the world. Many species are known to migrate long distances to different parts of the world, including isolated oceanic islands where they may be the only mammals native to the area. Their ability to disperse to different places, means that they carry viruses with them to new habitats. This also allows for bats of different species to be found in the same place, known as sympatry. Luis *et al.*, (2013) demonstrated that sympatry is an important characteristic that correlates with viral species richness in bats. This has been the case for the transmission of the rabies virus amidst bats. Sympatry also correlates with viral richness in other animals such as rodents and primates (Davies & Pedersen, 2008).

V. CONCLUSION

This review demonstrated that indeed bats are special virus carriers. Over twenty five families of viruses were detected in bats and bats do carry more viruses per species than rodents. Moreover, we presented a few theories that attempt to answer the question of why bats carry so many zoonotic diseases. Bats life history traits that may promote pathogen harbouring are demonstrated. Most of these are speculative hypothesis, and no evidence is available. However, even though there is not much evidence, we hope that the integration of the different hypothesis will lead to many new ideas and help bring light to medical research. In this study we also show that humans contribute to the rising emergence of infectious zoonotic diseases and we hope that this will also allow us humans to introspect on our ways of living and hopefully change for the better of us and also many wildlife species that are affected by our lifestyles. Future research that will mainly focus of field work for gathering more evidence on the association of bats and viruses vital. Moreover, recent hypothesis such as that of sympatry need to be carefully studies. Lastly, pathogens that are either isolated or detected from bats need to also be studies. Their characteristics and similarities, perhaps a trend could be found that would explain why bats can shelter the pathogens.



Figure 3: A colony of fruit bats on the Monfort bat cave (Photo credit: Gregorio Dantes Jr.)

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Do Animals Have Rights?

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ABSTRACT

Animals and human have lived in proximity for centuries; humans have used them for transport food and in experiments as surrogates. Animals have been bred to have traits that suit human needs; this is done by selectively breeding those with desired traits. They have some characteristics that are similar to humans, like parental care and altruistic behaviours, but they lack the moral character of choosing between wrong and right. Here, we discussed if animals have rights. Rights are legal principles that govern both the claimant and the offender, every right goes with responsibilities. Humans hold the highest moral stature compared to animals, as they know how to differentiate between right and wrong. While animals cannot make calculated decisions and therefore, are seen as amoral and they cannot have rights. The concept of rights is a human construct that can not apply to animals. Although they do not have rights, they still need to be treated fairly or with care as they feel pain and have emotions. Humans have tried to achieve this by having legislations that protect animals from cruelty, organisations, and animal activists around the world report animal abuse to the authorities and try to ensure that people are punished for inflicting pain on animals. In Higher Education institutions, animal ethics committees make sure that animals used in research are treated well by requiring researchers to apply for ethics clearance. Do animals have rights? No animals do not have rights due being amoral, they can only have rights if the definition of rights can be modified to fit beings that hold no moral stature.

Keywords: ethics, morals, experiments, cruelty, rights, nonhuman animals

I. WHAT ARE RIGHTS?

Rights are defined as legal, social and ethical principles that govern both the claimant and the target of the claim (Cohen, 1997). They are fundamental in any civilization and their violation can be punished by the criminal system or court of law (Nobis, 2016). Human rights cover all beings regardless of gender, race, nationality, and any other status. In South Africa, rights include: right to education, freedom of expression, privacy, life, and many others (South African Government 1996). Rights are different from demands as they are justified by pre-existing legal rules (Renteln, 1988). Humans have rights all over the world; they are important for protection of all, especially those who may face abuse they give us the power to speak up and challenge poor treatment, they promote peace in the society (Schaffner, 2010). Rights are created by the legislative body of a country, and they can be held by individuals or corporations. They are the highest form of legal protection that sometimes can be overruled by the government; for instance, during the start of the Covid-19 pandemic in 2020, the right of movement was taken away from citizens (Franjić, 2022). The concept of rights has been successful in humans as they are moral beings and they are notably different from animals in that they are able to categorize, think in an ABSTRACT way, and form situations that enable them to predict or plan

for the future most importantly, humans are able to reason (Ayala, 2010). Compared to other existing living organisms, humans are moral beings: they have beliefs that consider the impact of their decisions to others, this is like altruism, although this may not come at a cost for the benefit of others (Ayala, 2010). Humans can anticipate consequences, make valued judgement, and make alternative decisions of action, their moral behaviour is biological, and their moral norms are sometimes a result of collective human experiences and cultures (Ayala, 2010).

II. ANIMAL HUMAN INTERACTIONS

Animals have lived with or in close proximity to humans for centuries, they are our source of food, companionship and they are used in as human surrogates in scientific experiments (Figure 1; Nacone, 2012). Production of an effective Polio virus vaccine was achieved by several trials done on mice and monkeys, and today the virus has been completely eradicated in the Western hemisphere (Cohen, 1997). Blind mole rats are used to investigate how we can cure cancers and they have been shown to have some form of resistance (Gorbunova *et al.* 2012). We have altered animal genes to suit our lifestyle and domestic needs, by selectively breeding them; for example,

dog handlers and breeders breed dogs that have the most desirable characteristics, which are related to morphology and behaviour of dogs (Arnott *et al.* 2015). German Shepherds were bred for herding sheep and protecting flocks from predators, cows are bred to produce milk, chicken for producing eggs, and others. Animals are beings that have different species and are capable of breathing, hearing, communicate, and have sense of smell. This includes humans, however, when we refer to “animals” we generally refer to all nonhuman species (Franjić, 2022). Different species of animals have different capabilities that humans do not possess some are able to fly, run faster, change colour, and even breath underwater; these characteristics are amongst many that separate the two.

III. ANIMAL ABUSE

Do animals have rights? If so, are their rights being respected? Humans are seen as world builders and meaning makers: early Greek philosopher Aristotle argued that animals occupied a lower natural level compared to humans as they lacked reasoning, and others believed that animals existed to be used (Naconecy, 2012). The use of animals includes insects for waking up humans, cocks to catch scorpions, donkeys for transport (Naconecy, 2012).

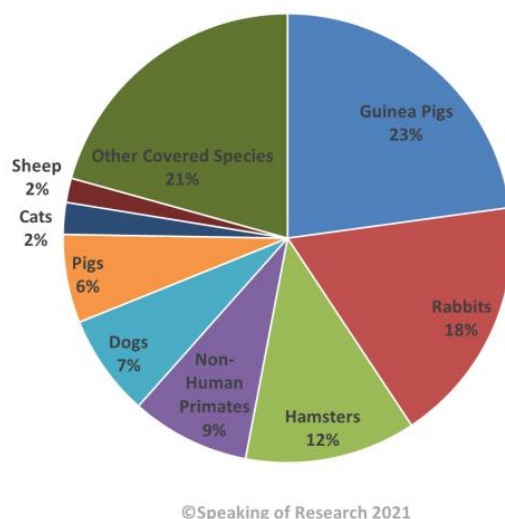


Figure 1: Different animals used in scientific research in the US in 2019.

Even in early theologies of Christians, animals were seen as different from humans as they lack souls, and other philosophers believed that animals were here to satisfy human needs (Naconecy, 2012). The view of animals in this way has brought different forms of abuse to animals

around the world, with abuse being defined as the act that may cause pain, death or compromises their welfare, it can be physical, mental, and passive (Agnew, 1998). Pet.com classified the different types of animal abuse into hording, mutilation, trading, beating, stabbing, and fighting Some forms of abuse have been widely accepted or condemned in our society. Millions of animals are slaughtered for meat production annually and the numbers are increasing with meat demand from population growth. These animals live in the most hostile conditions and they are deprived of sunlight; movement, as they live in small spaces that are expose them to diseases as the conditions allow for an easy spread (Sollund, 2013). Animal rights activists have gathered evidence around the world to show how animals are being treated in farm factories by going undercover as farm workers; one activist showed how calves were being lifted by their tails and ears and being kicked or thrown into transport trailers, which resulted in injuries or death (Figure 2; Fiber-Ostrow & Lovell, 2016). The abuse is not only in animals in close reach to humans, but many marine animals suffer through being tangled by fish nets, exposure to chemicals during mining in the ocean, etc. (Sollund, 2013). Here we show why animals do not have rights as humans do, and what is being, or can be done, differently by humans to make sure that animals are treated fairly, and they are not abused (Sollund, 2013).



Figure 2: Animal abuse in farms: pigs living in small, confined spaces (left) and calves being handled inhumanely (right). (Image source: New York post; <https://nypost.com/2019/06/06/chilling-footage-shows-brutal-abuse-of-calves-at-dairy-farm/>)

IV. WHY ANIMALS DO NOT HAVE RIGHTS

As stated in the rights' definitions, rights go hand in hand with responsibilities. A person has the right to freedom of speech and with that, the community the individual is part of is obliged to respect their rights to do so, but for animal's humans owe more obligations to them than rights (Cohen, 1977). This is because the concept of rights is a human construct that animals will never be obligated to follow. Cohen (1977) made an example: imagine you are on a game drive, and you see a lion preying on a baby zebra; would you intervene and stop the lion as it is violating the right of life of the baby zebra? Surely, you would not, but if it was preying on a human baby, our instincts would not allow us to observe and take pictures; we would make sure we try to save the baby. Why does the human baby have the right of life, but it is okay for the baby zebra to be eaten by the lion? The answer is simple humans have a higher moral status than animals and in the case of lion and the zebra encounter they are displaying a natural behaviour (Cohen, 1997). Another example that shows the different moral grounds that humans have as opposed to animals is in medicinal studies. During the Nazi era, doctors did research on hypothermia by soaking Jewish people in cold water and putting them in refrigerators; this advanced their knowledge of how hypothermia works beyond reasonable limits, but those humans had the right to life. It is not surprising that currently, in medicine and in biology, animals are used to advance our knowledge and in the process most animals lose their life, but since they are seen amoral, this is accepted (Cohen, 1997).

It is beyond reasonable doubt that humans occupy a higher moral stature than any other known species on Earth according to humans, but certain moral traits are seen in other animals or at least one may be observed in animals. Moral emotions like kindness, anger, grief, patience, spite, and being able to make sense of what is fair or unfair is seen in animals (Rowlands, 2012). Evidence from some studies shows that animals can behave morally; for instance, primates show altruistic behaviours to others in the group by grooming them, and elephants have been observed consoling individuals that are in distress by making sympathetic noises and placing their trunk on the distressed individual (Monsó *et al.* 2018). Dogs, rats, and chimpanzees have been found to exhibit hatred after being treated unfairly in groups; they may show this by some kind of irritation (Monsó *et al.* 2018). Although studies and observations show that some animals may have some morality they all lack the ability to formulate many moral principles, they cannot do what is right or wrong and they cannot be held accountable for any wrong doing like jail time or any form of punishment this makes the concepts of rights unapplicable to them (Cohen, 1997). Since they

cannot have any rights does that mean all animals must be treated as stones? No! animals may be amoral, but they are like humans, they are living beings that have "families" within their niche, and some are part of our families it is not surprising that animals can feel pain. Pain is defined as an uncomfortable emotional or sensory experience that is associated with some form of tissue damage (Underwood, 2002). For research experiments, animal pain has been categorized into five categories A-E: A is the category where animals do not experience any pain, they are observed for the study, while E category inflicts pain on the subject through surgery or test that results in discomfort, where anaesthetics are not used as they may affect the results of the experiment with in between categories B-C pain moderately increase. Animals react to pain differently compared to humans: cows grunt, and change their facial expression, pigs squeal and become aggressive, and other taxonomic groups have other behavioural indicators of pain (Underwood, 2002).

V. SOUTH AFRICAN ANIMAL WELFARE

Since animals do not have rights, it is important that humans use their higher moral ability to ensure animals are treated with respect and humility. This can be achieved through well-structured animal welfare laws (set of principles that control human behaviour) and ethics (Franjić, 2022). Animal welfare laws are in place to ensure the wellbeing of animals; this includes nutrition, space, and the physiological and social needs of the animal (Franjić 2022). In South Africa, animal welfare is achieved through the Animal Protection Act No 71 of 1962, which seeks to prevent animal cruelty and it covers all animals except insects and marine species that are under control of humans. According to the south African legislation "animal refers to any equine, bovine, sheep, goat, pig, fowl, ostrich, dog, cat, or other domestic animal or bird, or any wild animal, wild bird or reptile which is in captivity or under control of any person" (South African Government 1996). It is good to note that the definition of animal in the SA legislation it does not include arthropods, although they are part of the kingdom animalia and they are important for ecosystems services like pollination, food, biodiversity indicators and biological control of invasive weeds (Rosenberg *et al.* 1986). From the Act, people who are cruel to animals get punished through monetary fines or light imprisonment if they fail to pay.

VI. ORGANISATIONS THAT ENSURE FAIR ANIMAL TREATMENT

The National Society for the Prevention of Cruelty to Animals (NSPCA) and other animal activist organisations like Animal Activist Africa (AAA), animal Voice South Africa (AVSA) and others ensure that animals are treated fairly, and they report those who abuse animals to the authorities. The South African National Standards for care of animals for scientific purposes (SANS10386:2008) is another document detailing how animals used in experiments must be treated. Universities, like Rhodes, Stellenbosch, and others, have animal ethics committees that ensure that animals used in research and teaching are treated humanely and a few individuals are used for studies. Around the world, some countries have laws that protect animals: Egypt has a penal code, Zimbabwe has the Prevention of Cruelty to Animal Act, Germany has the Welfare Act etc, while countries like China, Vietnam, Iran, and others have no or very weak animal protection laws (Global animal law association). According to a database of the Global Animal Law Association, out of 50 countries only Germany ranked the highest in terms of animal laws. They ranked them according to eight cases: Case 1, countries where no animal welfare legislation was found; Case 2, countries having basic national laws, anti-cruelty laws (or penal code provisions) and new legislation on animal welfare; Case 3, countries with a national civil code provision giving a new status to animals; Case 4, countries with a basic national law and a provincial civil code provision giving a new status to animals; Case 5, countries with a basic national law and a national civil code provision giving a new status to animals; Case 6, countries with a basic national law and a provincial or local constitutional principle; Case 7, countries with a basic national law and a national constitutional principle; and Case 8, countries with a basic national law, a national civil code provision giving a new status to animals and a national constitutional principle (Figure 3). Majority of the countries that rank low are developing countries or countries that consume various exotic animals as delicacy food which results in animals in those countries to suffer greatly as there is a demand for them in the food market this means that there should be a push towards the protection of animal protection worldwide.

Another organisation that tracks the animal welfare performance of countries is the voiceless animal protection institute (VACI) which evaluates

50 countries that produce life stock based on nature and extent of animal cruelty considering the animal welfare laws of each country, ranking them in form 1-50 divided in three categories. First category is producing cruelty which ranks the total number of animals farmed for food, second category is the consumption of animal products, and the third category is the sanctioning of cruelty based on countries social and cultural attitudes towards animals. The top country ranked by the VACI is India although it has low animal welfare laws, they consume the least animals and most of the population are vegetarians and their religion Hinduism, Jainism, Buddhism limits the consumption of animals. While Australia ranks the worst as it slaughters triple the global average of 10 animals, and it export many animals for breeding.

A special case of the first country to give rights to individuals animals in the world is Ecuador it ruled that animals possess legal rights, which include the right to exist, flourish and evolve termed the “right of nature”, which does not equate them to human beings, but it allows them to be free in their environment (Koumoundouros, 2022). The “right of nature” is an umbrella term for several rights which includes the right to free development of their animal behaviour, right to physical, mental, and sexual integrity, right to health etc (Koumoundouros, 2022).

VII. CONCLUSION

Since animals do not have rights humans need to make sure that they are treated fairly, the current laws available for animals need to have consequences that are harsh for animal abusers. The legislative bodies also need to make sure that the laws put in place are followed and they try to put systems or even worldwide committees that promote fair treatment of animals. The kingdom animalia has many species of animals some are yet to be discovered and having laws that cover a fraction of them need to be looked at as every animal including insects play an important role in the ecosystem. The conservation of all species starts with fair treatment of species and with our knowledge about animals and earth in general developing we need all species as some are very important in explaining or being indicators of change, we are experiencing on earth example climate change. Humans define the law, and we can use our moral advantage to redefine what a right is and how it functions in the society for the benefit of animals.

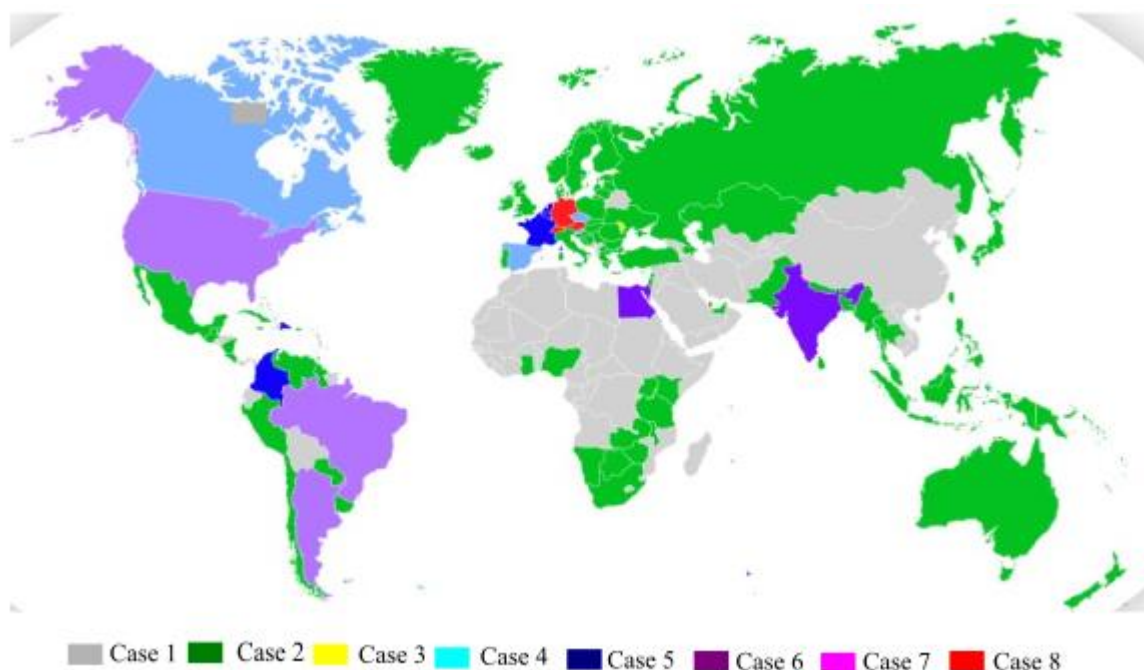


Figure 3: Data base legislation rank in different countries. Colours show different cases

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A review of decapod biological invasions of polar waters

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ABSTRACT

Over recent decades, polar waters have experienced an unprecedented amount of anthropogenic activity due to climate change melting sea-ice and creating new pathways through previously inaccessible waters. As anthropogenic activity increased in these regions, shipping has become more frequent and thus the movement of organisms through ballast water and hull fouling has also increased. As such, polar waters have become particularly susceptible to bio-invasions. Notably, decapod bio-invasions in both the Arctic and the Antarctic have been rising due to increased transoceanic movement through polar waters as well as the overall warming of polar waters. This is particularly interesting as decapods face a range of biogeographical and physiological barriers (such as below freezing temperatures and high magnesium [Mg²⁺] concentrations) that limits successful introductions, establishment, and geographic spread. Available literature suggests that as climate change drives disproportionate warming of polar waters, geographic and physiological barriers are being broken down both directly and indirectly via anthropogenic activity thus driving an increase in observed decapod invasions. This review assessed decapod bio-invasions in both the Arctic and the Antarctic with regard to key vectors of invasion, and physiological determinants to create an overview of decapod invasions in polar waters as whole as well as to create a preliminary list of invasive decapod species in polar waters.

Keywords: Arctic, Antarctic, marine, crabs, climate change, ballast water, hull fouling

I. INTRODUCTION

Over the last century, as the world has changed rapidly with the emergence of mass transoceanic transport, climate change, and industrialisation, many ecosystems have experienced critical ecological shifts. Polar waters, in particular, have been made vulnerable to bio-invasions due to increasingly warmer waters and increased human activity in the region. Arctic waters have experienced a significant decrease in sea ice, increase in water temperatures, and increased anthropogenic activity (e.g., tourism (increasingly), research, and aquaculture practices; Goldsmit et al, 2018; Legezyńska et al., 2020). Antarctic waters, in contrast, has seen an overall increase in sea ice (mainly in Autumn) while also experiencing an overall increase in water temperature as well (Parkinson and Cavalieri, 2012; Maksym et al, 2012). Interestingly, the increase in sea ice in the Antarctic has been observed in only some areas, while others have experienced a decrease or total loss of sea thus creating new pathways for anthropogenic activity and consequent bio-invasions (Maksym et al, 2012). Bio-invasions refer to species that are able to establish functional communities and disrupt native ecosystems (Perrings et al, 2002). Much like the Arctic, the Antarctic has also experienced changes in maritime traffic due to increased transoceanic transport and research (Clarke and Harris, 2002; Smith, 1990). An increasingly observed taxonomic group of bio-

invader in both the Arctic and the Antarctic is the order Decapoda (Legezyńska et al., 2020). Although there is abundant literature on decapod invasions in each region, there is a gap in the knowledge regarding decapod invasions in polar waters as a whole (synthesising information from both the Arctic and the Antarctic). This is further exacerbated by the lack of citizen science in both regions due to low permanent human populations (aside from those stemming from industrial and research activities). Citizen science has become an increasingly useful tool for monitoring invasive species as it often provided the first accounts of new invasions (Howard et al, 2022). As polar regions are remote and do not boast large populations, the use of citizen science is limited thus contributing to the lack of knowledge of invasive species in both regions. This review aims to compare and contrast decapod invasions in both polar regions in order to provide a comprehensive understanding of decapod invasions in polar waters as a whole. This review uses term decapod in reference to lobsters, crabs, and shrimps as a whole with little regard for the taxonomic specificities (such as genus and family) within the order (Abele and Kim, 1986).

II. BIO-INVASIONS AND POLAR WATERS

Bio-invasions have been an increasing cause for concern globally as invasive (alien) species have become more frequent as a result of a

rapidly globalising world (Facon et al, 2006). Bio-invasions refer to the establishment of a species outside of their native range and often have dire consequences on the invaded ecosystem as they have the potential to cause local biodiversity loss and disrupt ecosystem functions (Facon et al, 2006). Marine ecosystems are particularly vulnerable as a majority of the world's cargo is moved transoceanically and thus offers an easy vessel for organisms to migrate over natural barriers such as ocean currents, and differing water temperatures (Anil et al, 2002; Van der Weijden et al, 2007). Polar marine ecosystems, in particular, are vulnerable to bio-invasions as organisms in these regions are highly endemic and adapted to extreme temperatures and seasonality (Ranjith et al, 2012; Bastos and Junqueira, 2010). Polar in this context refers to the Arctic ($>66.5^{\circ}\text{N}$) and Antarctic (also known as the Southern Ocean; $>60^{\circ}\text{S}$) oceans, respectively (fig.1). While both oceans are characterised by extreme temperatures, sea-ice, and the same methods of invasion (due to increased anthropogenic activity and consequent shipping activity), these oceans differ dramatically in many ways. Mainly, the Arctic Ocean has its own restricted circulation and is surrounded by various land masses that support extensive industrial activities (Clarke and Harris, 2002; Smith, 1990). In contrast, the Antarctic Ocean has a circumpolar current and human activity is predominantly in the form of research and an emerging tourism sector, thus making it less accessible than that of the Arctic (Clarke and Harris, 2002; Smith, 1990). While bio-invasions are generally low in polar regions due to ecological resistance (low temperatures, high salinity, high magnesium), these regions are becoming increasingly susceptible to bio-invasions, understanding different components such as vectors and pathways of invasions, invasive species, and determinants are crucial for a holistic management of bio-invasions in polar waters as a whole and predict patterns for future invasions, and decapod invasions in particular.

III. DECAPOD INVADERS

As climate change and shipping has driven an increase of bio-invasions in polar waters, there has been a steady increase of the amount of alien species introduced over recent decades. In the Arctic alone, 34 marine invasive species were recorded between 1960 and 2015 (Chan et al, 2018). Of these, 6 were decapods that were recorded to have established communities; the Red King Crab (*Paralithodes camtschaticus*), Chinese Mitten Crab (*Eriocheir sinensis*), European Brown Shrimp (*Crangon crangon*), Snow Crab (*Chionoecetes opilio*), and the Atlantic Rock Crab (*Cancer irroratus*) (Chan et al, 2018). In contrast, while there

have been a considerable number of studies carried out on the likelihood of invasions in the region, there has been no concrete record of invasive decapod species in the Antarctic ocean specifically, to date (Frenot et al, 2005; McCarthy et al, 2019; McCarthy et al, 2022; Bastos and Junqueira, 2010). The lack of any official records may boil down to semantics as the terms “alien” and “invasive” has been used interchangeably in many studies.

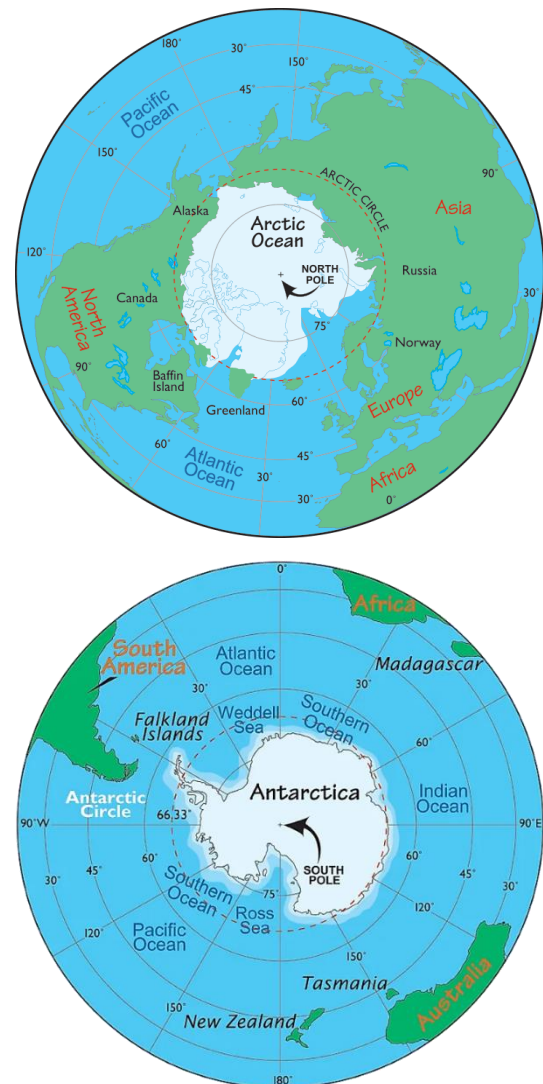


Figure 1. Diagram of the Arctic Circle (top; shown by the red dotted line), and the Antarctic Circle (bottom; shown by the red dotted line). Source: WorldAtlas.com, 2022

In the studies that suggest that there are no invasive species in the Antarctic (Hughes et al, 2020 most notably), the term “invasive” has been used to describe a non-native (alien) species that have an observable negative impact on an ecosystem

(Hughes et al, 2020). Due to the nature of polar regions, long-term impact assessments of alien species will most likely take an extensive amount of resources and time before being able to make a CONCLUSION on whether species are non-native (little to no negative impact) or invasive (high negative impact). As such, many researchers choose to use the term interchangeably in order to provide an overview of alien species as a whole. This review uses the terms “invasive”, “non-native”, and “alien”, interchangeably.

As such, evidence does suggest that there are, in fact, invasive decapod species in Antarctic waters (Thatje and Fuentes, 2003). Research suggests that low decapod richness is a result of Antarctic cooling during the Miocene epoch and the emergence of the Antarctic Circumpolar Current (ACC) (Thatje and Fuentes, 2003). While the literature does not explicitly confer the status of non-native species as invasive, the following decapod species have been confirmed to have established populations thus marking them as invasive according to this review's understanding; common name unknown (*Paralomis birsteini*), Subantarctic Stone Crab (*Lithodes murrayi*), common name unknown (*Neolithodes capensis*), Flattened Crab (*Halicarcinus planatus*), Spider Crab (*Hyas araneus*), common name unknown (*Neolithodes brodiei*), common name unknown (*Neolithodes yaldwyni*), common name unknown (*Paralomis birsteini*), as well as an emerging population of the genus *Pinnotheres* (based on larval stage characteristics, species remains unknown) (García Raso et al, 2005; Thatje and Fuentes, 2003; Innes, 2016; Thatje et al, 2005). In both regions, decapod species have been introduced via anthropogenic means, namely through ballast waters, hull fouling, and range expansion (Hughes et al, 2020; Innes, 2016).

IV. VECTORS OF INVASION

While the Arctic and the Atlantic have varying degrees of human activity (substantially more in the Arctic than the Antarctic), they share common methods of invasion as many invasive alien species are predominantly introduced through direct anthropogenic activities such as shipping and indirect climate change. Whether done for tourism, economic, or research purposes, shipping is considered one of the most significant vectors for invasion as organisms are easily transported through the ships ballast water or attached to ship hulls (hull fouling) (Lewis et al, 2003). As climate change has caused disproportionate warming of polar waters, sea-ice melt has opened new pathways for transoceanic movement (Anil et al, 2002; Van der Weijden et al, 2007). In turn, shipping has increased

commercial tourism as well as research tourism in polar regions where the Arctic and the Antarctic are becoming more desirable to everyday travellers as well as those interested in researching previously inaccessible (or poorly accessible) regions (Hall and Saarinen, 2010; Stewart et al., 2005). Ballast water has been documented to contribute to approximately one third of all known marine bio-invasions (approximately 237 species) due to increased maritime traffic along with the industrial shift from ‘dry’ ballast vessels to more efficient water ballast ships in the late nineteenth century (Davidson and Simkanin, 2012). Ballast water is fresh, or saltwater taken in by ships at the start location, to provide stability and maintain manoeuvrability throughout the journey, that is later discharged at the destination (Jing et al, 2012). The amount of ballast water transported is significant as it has been estimated by the International Maritime Organization (IMO, 2016) that 10 billion tonnes are discharged globally every year (Awad et al. 2014; Tamelander et al. 2010). This water often contains many organisms such as bacteria, phytoplankton, zooplankton, as well as the larval forms of many species from the origin location (Anil et al, 2002; Davidson and Simkanin, 2012). As such, ballast water creates an effective mechanism for bio-invasions as it allows species to overcome their natural distribution ranges and emigrate over biogeographic barriers (Chan et al, 2015). Similarly, hull fouling, a method predating ballast water, has also been observed to effectively facilitate bio-invasions via shipping. The term fouling refers to species that grow on rigid substrates such as rocks (Tempesti et al, 2022), and in areas with high maritime traffic, the hulls of ships (Lewis et al, 2003). While hull fouling has been observed to contribute significantly less to bio-invasions than ballast water, it still plays a considerable role in dispersing alien marine species globally (Lewis et al, 2003).

In areas with a lot of sea-ice specifically, hull fouling is particularly potent as the sea ice results in the abrasion of the hull thus effectively dislodging fouling species and removing anti-fouling paints and epoxies which in turn results in increased opportunities for new fouling species to become attached and transported (Lewis et al, 2003). Modern shipping practices generally spend less time docked at ports thus resulting in less fouling assemblage build up, however, in the Antarctic specifically, this is nearly negligible as shipping occurs on a seasonal basis with much longer docking (usually months) in the winter due to sea-ice (Lewis et al, 2003). Previous research suggests that decreased docking time may be negated in the Arctic as well due to an overall increase in commercial maritime activity (Lewis et al, 2003; Miller and Ruiz, 2014). Additionally, bio-invasions via shipping has become an increasing threat to polar

regions in particular as sea-ice melt, as a result of climate change, has opened new channels for ships which has consequently increased maritime traffic in the Arctic and Antarctic oceans (Ho, 2013). Decapods, specifically, both the larval and adult forms, have bypassed biogeographic barriers via unintentionally exploiting movement through both ship hulls and ballast waters (Innes, 2016). Additionally, some decapod groups (such as lithoid crabs, in particular) have been observed to be invading polar waters though geographic range expansion (Innes, 2016). Range expansion occurs when biogeographic and physiological barriers are either overcome by organism, or are broken down (increasingly observed due to climate change) thus allowing them to move into new and previously uninhabitable ranges (Hellberg et al, 2001; Renaud et al, 2015). The lithoid crab (*Neolithodes brodiei*), first recorded in Balleny Islands (Southern Ocean), is the first observed case of invasion of polar waters via range expansion by decapods thus showing that biogeographic barriers and physiological determinants are becomingly increasingly less hostile for alien species (Innes, 2016).

V. BARRIERS, CLIMATE CHANGE, AND PATTERNS:

Fossil records show that Antarctic waters were rich in decapods pre-Miocene (5 mya) due to temperate waters (Thatje and Fuentes, 2003). As the Antarctic Circumpolar Current formed, decapods were subjected to increasingly colder temperatures and thus eventually faced extinction in region (Thatje and Fuentes, 2003). This is due to a combination of polar temperatures being below the thermal tolerance of decapods, oxygen limitations, and high concentrations of magnesium [Mg^{2+}] present in the Antarctic and Arctic ocean (Thatje and Arntz, 2004; Pörtner, 2002). In temperatures below their thermal tolerance (differs for different decapod species; average lower limit unknown), decapods face the freezing of bodily fluids and thus a lack of oxygen needed to carry out respiration (Pörtner, 2002). This is especially true for polar benthic ecosystems as they are much colder than surface temperatures and organisms at those depths will be more susceptible to changes in temperature (Renaud et al, 2015). As such, although there is a gradual increase of decapods in polar regions, some parts (such as polar benthic zones) are still below the thermal tolerances of invading decapod species and have therefore not been subjected to decapod invasions as yet. The freezing of bodily fluids in decapods occurs due to high concentrations of Mg^{2+} (Pörtner, 2002). As magnesium acts an antagonist for calcium at neuromuscular joints, decapods experience a numbing effect thus further restricting metabolic functions needed for survival (Pörtner,

2002; Thatje et al, 2005). As a consequence of the lack of magnesium regulation, the effects of oxygen limitations are further exacerbated thus creating a hostile environment for most decapod species (Pörtner, 2002). An exception to this observation are lithoid crabs (*Anomura*) who appear to have the expected low magnesium regulation, however, are still abundant in subpolar, and, increasingly, polar regions (Thatje et al, 2005). There is not enough known on this group as yet to infer why or how they have been able to cope with these physiological barriers in polar waters, however, it can be speculated that the reasons behind the increase in invasive decapod species is twofold. The first reason can almost certainly be attributed to vast increases in temperature in polar waters creating a less hostile environment for species with higher thermal tolerances such as decapods (Aaronson et al, 2014). The second reason being that some decapod species may have adapted their life history strategies (such as being able to regulate magnesium better, or lowering their lower thermal tolerance limit) in order to overcome the physiological constraints of polar regions thus allowing them to successively recolonize polar waters (Aaronson et al, 2014). Studies have predicted that bio-invasions as a whole will continue to increase in the coming years as the intensity, distribution, and frequency of anthropogenic activities increase in polar regions (Hughes et al, 2020). This is mainly due to a disproportionate increase in polar waters due to sea-ice and snow melt resulting in a decrease in solar radiation being reflected back into space (Dai et al., 2019). As temperatures continue to increase around the globe, it can be predicted that the rate of bio-invasions in polar waters will continue at an unprecedented rate and more decapod species will continue to overcome the geographic and physiological boundaries of polar regions. It can be predicted from current knowledge that in the Arctic, decapods may invade at an increased rate throughout the entire region due to vast amounts of sea-ice melt which would result in lower Mg^{2+} concentrations thus creating a less hostile environment to colonize. In the Antarctic it can be inferred that while there may be an overall increase (on par with the current decapod colonization trend in the Southern Ocean) in decapod invasions, this may be sporadic throughout the region due to disproportionate sea-ice melt and formation in different areas. In contrast, more countries are introducing ballast water management schemes that may reduce the amount of ballast water being exchanged as well as where ballast water is being exchanged which may be able to minimize the amount of alien species being introduced into new ecosystems. For example, Canada has introduced ballast water regulations where ballast water can only be discharged at certain points that meet particular depth and latitude/longitude requirements (Transport Canada,

2021). Ballast water management plans similar to this may help reduce the predicted patterns of decapod invasions in polar waters.

VI. CONCLUSION

Decapods pose as interesting case as fossil records show that they once occurred abundantly in polar Antarctic waters yet did not occur at all in polar Arctic waters. After their extinction in the Antarctic during the Miocene epoch (~5 mya), presumably due to the dropping temperatures exceeding their thermal limits, decapods have not been observed in polar regions – until recently. Evidence suggests that the combination of climate change and increased anthropogenic activity have aided this taxonomic group in returning to the Antarctic and appearing for the first time in the Arctic. As polar waters face rising temperatures due to climate change, consequent increased sea-ice melt has opened new pathways for transoceanic movement thus increasing maritime traffic and the movement of decapod larvae and adult through ballast water and hull fouling, respectively. Climate change has also been observed to aid in breaking down physiological and geographic barriers thus allowing certain decapod species to expand their ranges into previously uninhabitable territory. Due to climate change and a rapidly globalising world, decapods have been able to successfully colonize or recolonize polar waters in both the Arctic and the Antarctic, respectively. While there is still a large gap in the knowledge surrounding bio-invasions in polar waters, this review aims to provide a comprehensive understanding and overview of decapod invasions in polar waters for further research.

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Kurhula Laphondo

Podcasts have gained popularity. I week does not go by without one of my friends asking me if I have listened to an episode of a certain podcast already. But what are podcasts, and why the sudden popularity?

Well, a podcast is a collection or series of digital audio files, focused on a certain topic or theme that is made available for listening or downloading via the internet. These are used for educational and entertainment purposes. They are also used to connect like-minded people. There are different types of podcasts, and they include nonfiction narrative podcasts, hybrid podcasts, interview and/or conversational podcasts, and scripted fiction podcasts. Podcasts are very convenient for the audience as they just have to play and listen, which does not require all of their attention. This can be done while someone cleaning, cooking, jogging, or driving, allowing multitasking, and saving time.



Podcasts have this amazing availability of relaying information to the audience in a way that books usually fail to do so. Educational podcasts are very important as they cater for all kinds of people, both young and old, and also educated and uneducated. This is very important for science, as most scientific topics may be hard to comprehend. Listening to science podcasts would then make it easier for people to know more about science in a more easier way than having to read a complicated scientific paper, which in most cases even educated people have to pay careful attention to detail in order to understand what the paper is all about.

This piece will focus on reviewing five different podcasts. They will be rated according to their Credibility ★ Engagement with audience, ★ Context of topic or theme, ★ Excellent discussion, ★ and Captivating ★

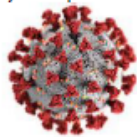


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UNIVERSITY OF MINNESOTA

It is exactly two years ago when out of nowhere there appeared a very tiny organism that is not even visible to the naked eye, which had a great impact in the lives of many people across the globe. This is a virus, called the Corona virus, also known as Covid-19. Covid-19 had a lot of devastating impacts including economic and social problems, affecting tourism, and job losses. But importantly, most people lost their lives battling the disease caused by the virus. In the very first episode of the Osterholm update podcast, which was made available three months after the first outbreak of Covid-19 in China, Dr. Osterholm and co-host Chris Dall discuss how the situation got to that point, and the expectations for the upcoming weeks and months. Moreover, they discuss the possibility of immunity after Covid-19 infection, shortages of testing and lastly how can people offer support to one another during such hard times.

Dr Osterholm explains that the reason why we are here is because there was an emergence of a Corona virus that transferred from an animal species to a human, which then spread to a lot of people within a small period of time. He explains the Covid-19 cases as a rolling set of outbreaks, because by the time a case outbreak is detected, there is already a great amount of infections in that area. The discussion then moves to talking about testing for the infection by the Corona virus, which Dr. Osterholm predicts that it was going to be a huge challenge considering the fact that the virus is spreading all over the world and the population is very massive. He comes up with suggestions of how this can be carried out while protecting or front line workers, especially doctors, nurses, and caregivers. This included that covid-19 patients can test each other as it was discovered that an average person can do the test as efficiently as a qualified doctor.



Osterholm Update: COVID-19

Episode 1: How We Got Here

Host: Dr Osterholm

Co-host: Chris Dall

Duration: 46:02

<https://www.cidrap.umn.edu/covid-19/podcasts-webinars/episode-1-how-we-got-here>



Dr. Osterholm briefly mentions that there may be a need for vaccines and that the government should act fast in this regard. He further says that nothing is certain, and no one is really sure of what is to happen, but we can only speculate. Dr. Osterholm emphasizes that even though we are speculating, we should do so like a chess board master, and not say things that do not have the evidence as a supporting tool. The episode was closed by words of comfort from the Dr. He said a lot of naked truths which definitely will be a hard pill to swallow for most people, but he ends up by saying we will get through this.

Review-★★★★★

This podcast was recorded just in the beginning of the global pandemic. No one was sure of what would happen and it is amazing how looking back at what has already happened and what is still happening now, almost all of Dr. Osterholm's predictions happened. This podcast was interesting, and worth listening to. It is a perfect podcast that I would recommend to my little sister who is 2 years old now in the future if they want to know more about what happened. The host asked relevant questions and were answered very well by the Dr. This podcast receives all the five stars

Podcast: Trust Me, I'm An Expert: forensic entomology, or what bugs can tell police about when someone died

Host: Sunanda Creagh

Co-host: Professor James F Wallman

Duration: 25:41

<https://theconversation.com/trust-me-im-an-expert-forensic-entomology-or-what-bugs-can-tell-police-about-when-someone-died-124416>

**Trust
Me,
I'm An
Expert.**

This episode begins with a small clip of an episode that was recorded a short time ago. It is an interview between the host Sunanda Creagh and a forensic scientist, Maiken Ueland, at the Australian Facility for Taphonomic Experimental Research (AFTER), also known as a body farm in books or movies. They discuss about the interesting findings about how donated human bodies decompose in Australian conditions. One of the most interesting discoveries according to Ueland is that they find that instead of the body decomposing to skeletal state, it mummifies and there be preserved. When moving to the main topic of the episode, host interviews Professor James Wallman, who also has an undergoing project at AFTER, discuss about insects and how they can help police tell when someone died.

Prof Wallman's job is to unpack the small clues that insects leave behind when a person dies or in a crime scene, thereby helping police solve crimes. Bugs are usually perceived as annoying, ugly, and pests. Many people do not like them. But Prof Wallman explains here that insects, including the so much hated blowfly, have a profound influence on decomposition. This is because insects such as maggots have a huge appetite for decaying matter. Prof says that a world without these insects would be full of dead bodies including those of humans (victims of murder). Imagine that!



So how do they know when a person died? Prof Wallman explains that the adult fly is able to quickly detect a dead body just when it is about to decay. After finding the body, the flies mate and lay eggs which will then hatch to feed on the dead body. When a body is found, the police can then collect the insects that are on the body and send them to a forensic entomologist who then identifies the insect and uses available scientific data to work out how long the insect would have been there by the time it was collected. Prof Wallman further busts myths associated with insects including that blowflies are dangerous or harmful, but the truth is their danger is understated and they are more good than harmful. In closing, the episode talks about the importance of people donating their remains at the AFTER facility for research purposes. He sheds light on how this can be very important for sciences and lastly, the last part of the conversation with Ueland is continued.

Review: ★ ★ ★ ★

'This episode was informative. It unpacked forensic entomology in a way that is easy to understand. It made me want to hear more and most importantly, consider donating my body to science after death. However, even though, the previous episode was similar to this one, I do think that they could have just referred to it and not have to include half of it again. It is a waste of time and would be boring to people who actually listened to the episode. Because of this, it loses one star. Otherwise it was credible and discussion was well broadened. Lastly, the use of sounds such as fly sounds as they talk about them is really awesome.'

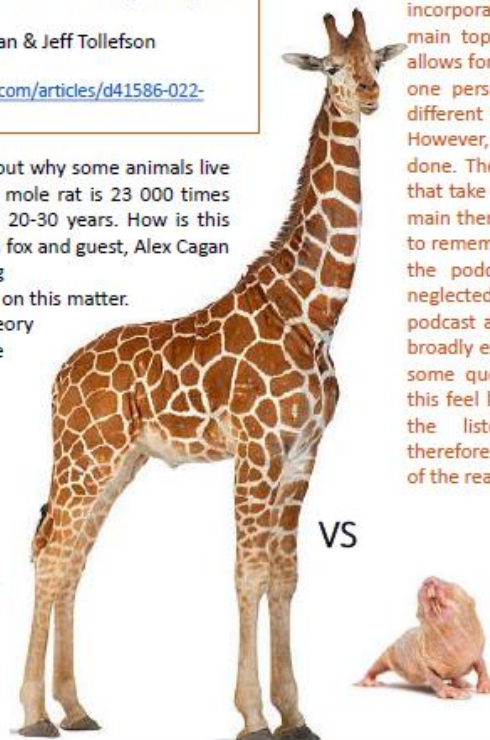


Nature podcast: Why do naked mole rats live as long as giraffes.
 Host (s): Nick Petrić Howe Dan fox, & Benjamin Thompson
 Guest (s): Alex Cagan & Jeff Tollefson
 Duration: 28: 47
<https://www.nature.com/articles/d41586-022-01062-x>

Review- ★★ ★

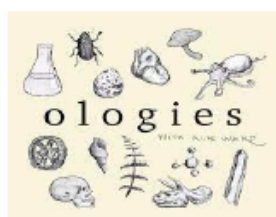
This podcasts, unlike most, incorporated more than just one main topic. This is very good as it allows for different audiences or just one person can learn about many different things in just one episode. However, I do think this was over done. There were many newsletters that take a long time in between the main themes which makes it difficult to remember what the main topic of the podcast was. I felt that this neglected the main purpose of the podcast as the main topic were not broadly explained and I was left with some questions still. All this made this feel long and can actually make the listener loose interest. I therefore give it only 3 stars because of the reasons mentions

Biologists have always tried to figure out why some animals live longer than others. For instance, the mole rat is 23 000 times smaller than the giraffe, but the live 20-30 years. How is this possible? In this podcast, the host Dan fox and guest, Alex Cagan discuss this. Alex begins by mentioning the theories that are already available on this matter. This include the somatic mutations theory which says that mutations accumulate in our DNA as we age. Fox explains that somatic mutations usually cause cancer, and this can reduce life span, especially in larger animals that have many cells to mutate. But this is not the case in the animal kingdom. In a research conducted by Fox and his team the actually discovered that somatic mutations are a cause of ageing, not a result of it. They show that long-lived animals get mutations at a slower rate, and this aids to explain why cancer risk does not scale with lifespan



I almost fainted when I saw the price of cooking as I was doing my groceries last week. I could not believe how the prices doubled up like that. What in the world happened?

In the second half of the episode, host Benjamin Thompson and guest Jeff Tollefson discuss about the war between Russia and Ukraine as it was beginning and predict the potential impacts that this would cause to both countries, other countries that depend on Russia for goods and the whole world. Tollefson mentions that energy markets would be most affected and would affect the economy and would hit poor people very hard all over the world. This is because many countries depend on Russian oil and some on Russian gas, and with the war going on it is not going to be easy for these countries to keep going. But other than that, Tollefson explains that the war boots emission which is not good a all for our atmosphere. He further talks about oil as a global commodity and predicts that prices would spike up and the coming weeks. And let me tell, I saw this with my own eyes.



Podcast episode- Is your carbon footprint bs? How to Save a Planet Takes Over Ologies.

Host: Alie Ward

Co-host: Alex Bumberg

Guest: Dr Catherine Wilkinson

Duration:46:59

<https://open.spotify.com/episode/6B0vzPGvmWmMo2zhGDqQfe>

In this episode, the host, co-host, and guest respond to the listeners more frequently asked question. One of the listeners, Anna summed it well, which is basically an argument between her and her brother about climate change and carbon footprint. Anna is optimistic that our individual efforts to combating efforts matter and that if we change our lifestyle in many ways, we can actually solve this. His brother, however, thinks that this is a huge systems and policies issue and our efforts do not really mean anything.

In the first half of the episode, the hosts with the help of Dr Wilkinson argue the side of Anna's brother. Dr Wilkinson says the reason why our individual effort does not matter is that the biggest sources of greenhouse gas emissions are things that we do not have control over. These include burning fossil fuel for electricity, agriculture, deforestation, industry, transport, and building, which all together contribute to over 90% or 50 billion tons of the world's total carbon emissions. On the other hand, an average American only contributes 16 tons of the 50 billion which equates to 0.0000000005%. Mathematically, this means that our efforts do not matter.

However, in the second half of the episode, Anna's side is debated. Wilkinson mentions ways in which we can contribute to solving the problem. The most important one is to not focus only on our own carbon footprint but communicate and teach other. She further says the top five things that once can do is to have less babies, drive less, fly less, be energy efficient and change to plant based diet. Lastly, they emphasize that our efforts matter, as the little things matter too. An example of how European people now fly less was discussed in closure.

Review



This podcast was very informative and great to listen to. I loved the fact that the topic that was discussed was taken from the listeners, therefore it definitely gets a star for engaging with the audience. It is credible and both sides we well debated without being biased to one side. It is very long but not boring at all because the presenters are lively, joke around making it even more captivating. As a result it receives 5 full stars



CARBON FOOTPRINT



Overheard at National Geographic

Host: Van Wallis

Co-host: Prof. Kang Lee

Duration: 16:07



<https://open.spotify.com/episode/1i44z9o7R7xn1SV18uWi8Q>

Review ★★★★★

This episode is my personal favourite. It was very interesting to listen to. When I look back at my childhood and remember how much I used to get a whipped by my mom for lying makes me a bit sad though ♥, ha ha. I mean she supposed to be happy that I am lying in the first place. Jokes aside. Both the host and guests were very fun to listen to, they share jokes here and there, engaging with the audience and also sticking to the topic. It is the shortest, yet the discussion was broad and straight to the point. Lastly, the episode is credible and attention seeking from the beginning to the end.



Lying can actually get someone in jail or paying a fine if not both, because it is a crime in all parts of the world. But according to science, it may actually be a good thing for a child's development. Huh? How though? How do kids lie and how can we know what is happening in their little brains?

In this episode, Dr Kang Lee, a professor of child psychology explains what happens. He spent the last 20 years getting more than five thousand children to lie to him. He mentions that he has heard all sorts of lies, both well crafted and very bad lies, and can actually tell if a child is lying or not. In the past, children below the age of seven were thought to not be able to lie intentionally. Even Charles Darwin was concerned when his son first lied to him at his very young age. He thought it could be a bad indicator. But according to Dr Lee, kids start their first lie as young as age two. How crazy is that? And that is perfectly fine.

Van then recorded some of his co-workers childhood lies and plays them for Dr Lee for analysis. Dr Lee says that the way we lie follows a pattern that is predictable, he can tell a lie about a kid's development just by analysing the kind of lies they tell. The different types of lies include the clumsy cover-up and white lies. The latter is told just to be polite

Dr Lee further says we need to shy away from the mentality that all lies are bad. He says it is not possible to tell the truth all the time and if one does tell the truth always, they are unlikely to survive in this society and will be very lonely as you cannot always tell people that they look horrible or say you do not like the gift you gave me as this will hurt their feelings.

Lying is actually good for development because it is hard, imagine knowing the truth, suppressing it having to create a new truth that will be convincing. Finally, kids get better at lying as they grow, and they learn that lies are okay but carefully observing their own parents.

But can Dr Lee tell if his 2 years old is lying? Ha ha NO! very interesting....

Just to add on the reviews, Dr Lee's first language is not English, the host then edited the podcast to ask the questions better and simplifies some concepts that Dr Lee may have found difficult to do so. This is amazing.

Podcasts are often best enjoyed using a
podcast app

Get entertained, updated and learn more about **science** through
podcasts



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ON

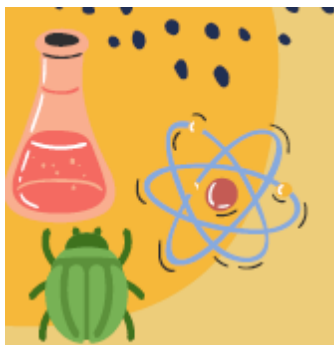


LISTEN ON
YOUTUBE



Listen on
Google Podcasts





Science Podcast Review

Marelise Faul

What are podcasts & why are they important in science?



Podcasts are a means of providing the internet audience with information through a digital audio format. With the increasingly busy lifestyles of most people today, podcasts can be a relaxing and very informative way of communicating information. In recent years, the importance of citizen involvement in science has become a hot topic as scientific literature in it's most common form is inaccessible to most people. Podcasts, if done in the correct manner, can bridge the ongoing gap between scientists and the general public. This can facilitate the integration of scientific evidence into education, policies and the general lives of many people to make informed decisions.



RATING PODCASTS

I will be rating 5 science episodes based on the rating system on the right. Podcasts will be rated on an "atom" system, where they will be given a score out of 5 atoms depending on the quality of the podcast (5 being the highest score). All episodes will be from different podcasts including; "Science weekly" by the Guardian, "Ologies" by Alie Ward, In Our Time by BBC Radio, "For The Wild Podcast" and "What You Do Counts" by National Geographic.



Rating



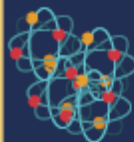
Interviewed a scientist



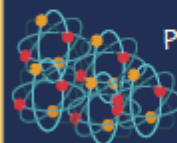
In- depth but understandable/engaging



Provided accurate information



Ethics: Avoid shaming a social group



Provide unbiased facts to audience



Podcast 1:

Why aren't women getting diagnosed with ADHD?

ABOUT:



This 15 minute long episode is about the underdiagnosis of attention deficit hyperactivity disorder (ADHD) in women. Madeleine Finlay interviews Jasmine Andersson concerning her experience being undiagnosed for the majority of her teenage life. Next, Prof Amanda Kirby speaks on the reasons for the condition not being recognized historically in women and girls involving behavioural and hormonal differences



Why aren't women getting diagnosed with ADHD?

It's estimated that a million women in the UK could have attention deficit...

Google Podcasts /

RATING: 3/5

This episode was very engaging and easy to follow. It provided interesting and understandable facts through the dialogue with a specialist in ADHD. However, the podcast only interviewed one woman who has suffered as a result of late diagnosis, which gave a very limited point of view on how it influences all different types of women. This episode could have been improved by interviewing at least 3 women from different cultural and socio-economic backgrounds, as well as different age groups. Furthermore, the professor did not provide a lot of information of the specific differences amongst ADHD in women and men.

Rating



Interviewed a scientist



In- depth but understandable/engaging



Provided accurate information



Ethics: Avoid shaming a social group



Provide unbiased facts to audience



Available at: Episode 3 Science weekly

<https://podcasts.google.com/feed/aHR0cHM6Ly93d3tudGhlZ3VhcmBpYw4uY29tL3NjaWVvY2Uvc2VyaWVvL3NjaWVvY2UvcG9kY2FzdC14bWwvepisode/Nj13YmQxNDY4ZjA4M2I4NTYzNzA4NWVvY29aZX8vvd=QCAcQkFYCahgKEwja9az8z-73AbVAAAAAHQAAAAAQnAE&hl=en-ZA>



Podcast 2:

Attention-Deficit Neuropsychology (ADHD) with
Russell Barkley

ABOUT:



This hour and 40 minute long episode is extremely informative on Attention-Deficit Neuropsychology (ADHD). It is easy to follow as the interviewer, Allie Ward speaks in a very colloquial manner despite the episode having in-depth details of biological terms and statistics concerning ADHD. She interviews Dr. Russell Barkley, a psychologist that has studied ADHD for about 40 years, and goes through a step by step explanation on treatments, most recent literature, underdiagnosis in women and social issues and stigmas concerning the disorder.



**Part 1: Attention-Deficit
Neuropsychology (ADHD) with
Russell Barkley**

alieward.com



RATING: 5/5

This episode was fascinating and very easy to understand. Neuroscience can be a difficult field to explain, but evidence concerning mental disorders is very important to share with the general public due to many people that lack diagnosis and live immensely difficult lives as a result. This podcast is brilliant as it provides unbiased facts as well as a very humane approach to the stigmas surrounding the disorder. It includes the most recently studies treatments available to those with ADHD as well as a more holistic and practical approaches to go about the disorder. The episode accurately provides facts and informs the viewer on where information concerning the disorder can be found.

Rating

-  Interviewed a scientist
-  In- depth but understandable/engaging
-  Provided accurate information
-  Ethics: Avoid shaming a social group
-  Provide unbiased facts to audience



Available at: Part 1 Attention-Deficit Neuropsychology (ADHD) with Russell Barkley

<https://www.alieward.com/ologies/adhd>



Podcast 3:

Corals

ABOUT:



This 52 minute long episode by the BBC is about the coral reefs, their importance in marine ecosystems, their biology, how they shaped Charles Darwin's first book "The Structure and Distribution of Coral Reefs" as well as the effect of climate change on them.



RATING: 3/5

This episode, although definitely informative was not very engaging. The interviewer Melvyn Bragg as well as the 5 experts being interviewed including Gareth Williams (associate professor in Marine Biology) and Nicolla Foster (Lecturer in Marine Biology at the University of Plymouth) have in depth coverage of the reefs. However, the manner in which the research is shared is not particularly easy to understand for someone that is not an expert in the field. The episode would be improved by a more colloquial use of language.

Rating



Interviewed a scientist



In- depth but understandable/engaging



Provided accurate information



Ethics: Avoid shaming a social group



Provide unbiased facts to audience



Available at: BBC Radio 4, Corals

<https://www.bbc.co.uk/programmes/m0010xnr>

Podcast 4:

MIKE PHILLIPS on Gray Wolves and the Vitality of Death

ABOUT:



This podcast is about the gray wolves that were historically found in high numbers in Northern America in a wide range stretching from Florida to Colorado. Ayana Young interviews Mike Phillips that served as the Executive Director of the Turner Endangered Species Fund. Mike discusses the importance of apex predators in an ecosystem, their spiritual importance to people as well as the colonial roots of their removal from the environment.



For The Wild Podcast- MIKE PHILLIPS on Gray Wolves and the...

Ayana and Mike's conversation touches on the history of cattle ranching and grazin...

FOR THE WILD / Jan 3, 2020



RATING: 4/5

This episode was very interesting as it involved the scientific, social, political and economic aspects of conservation. This is important for both scientists and the general public as the importance of multidisciplinary action is often overlooked in the scientific community, and policy makers. The information is explained in a very simple manner that is easy to understand the importance of predators in an ecosystem such as top-down effects. The episode also highlights how colonialism in the US influenced a hatred and fear towards wolves due to the introduction of livestock. Mike explains why it is importance to recognise the history of policies as well as provide easy solutions to keeping wolves away from livestock. However the episode could have benefitted by talking to a normal citizen (such as a farmer) and their experience with the wolves).

Rating

- Interviewed a scientist
- In- depth but understandable/engaging
- Provided accurate information
- Ethics: Avoid shaming a social group
- Provide unbiased facts to audience



Available at: Episode 152 For the Wild

<https://forthewild.world/listen/mike-phillips-on-gray-wolves-and-the-vitality-of-death-152>

Podcast 5:

Amazon

ABOUT:



This 13 minute long episode in Spanish is about the intricacies of the ecosystems found within the Amazon rainforest. This entire episode series is focused on how we are destroying the world around through contributing to climate change us and what we can do to make a difference. This episode looks at the effects on the Amazon and interviews Nobel Peace Prize winner Fernando Trujillo and the indigenous leader of Uitoto, Fany Kuira Castro as they discuss deforestation.



RATING: 4/5

This episode, although being very short it was very informative and enjoyable. Similar to the Gray wolf podcast, it highlighted the importance of incorporating a more social aspect of science by including the views of people indigenous to the area, their experience with deforestation and how it has influenced their community as a whole. Furthermore, the scientist that was interviewed was very easy to understand and did not use scientific jargon that often makes it harder to understand a topic.

Rating

Interviewed a scientist

In- depth but understandable/engaging

Provided accurate information

Ethics: Avoid shaming a social group

Provide unbiased facts to audience



Available at: Episode 152 For the Wild

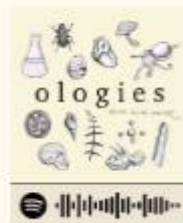
<https://www.nationalgeographicla.com/loquehacescuenta>



Final thoughts:

PODCAST REVIEWING:

Listening and reviewing the above mentioned podcasts was a very enjoyable and important exercise as it has highlighted how much people can differ in their way of giving out information. There is a great need to improve scientific communication to the general public, especially in our current state of climatic concern. Only if educational resources such as podcasts are available and easy to understand to a wider audience will they be considered in both policy makers and individuals.



Sheyenne Samy

G19S1481

Podcast Review

With the rise of social media, podcasts have become an effective way to communicate research to a wide audience on various platforms. Podcasts create a way for researchers, scientists, professionals, enthusiast, and everyone in between to publicise their work to a range of people. This method of communication allows people to phrase their work in a way that can reach a more general audience thus eliminating the common top-down method of communicating science which often leads to miscommunications, misunderstandings, and a general disinterest for science. Instead, podcasts use a more discussion based and informal approach which is much more effective as it communicates directly with the general public in a way that is engaging, fun, and interesting. Here are some of the podcasts (good and bad) that I've come across this year rated based on the following criteria; approach (was the approach consistent with the target audience?) , accessibility / language (was the language used uncomplicated and lacking science jargon thus making that research more accessible?), accuracy (was the information correct?), and ethics (was the information conveyed in a way that did not exclude certain groups?).



Conservation Tribe Podcast

Manta Ray Conservation

- ~ Host: Blane Edwards (Earth Offline)
- ~ Guest speaker/s: Simon Hilbourne (Communications manager at Manta trust)
- ~ <https://open.spotify.com/episode/6RE3jNXchEpbaqGpikpxe?si=c53e1b25b7ff4871>
- ~ Length: 33 minutes
- ~ Theme: Conservation
- ~ Rating: 4/4

The Conservation Tribe podcast is a weekly podcast with aims to educate the general public on environmental issues to inspire action to protect ecosystems, species, and biodiversity. Conservation Tribe does this through interviewing conservationist and environmental enthusiasts such as student, scientists, ecopreneurs, and innovators.

This episode aimed to cover a range of topics including:

- Major threats to Manta Rays
- Major roadblocks for addressing the impacts of fisheries
- Conservation actions needed to protect Manta Rays
- Impact of ecotourism in Manta Ray conservation
- Ethical, sustainable, and responsible ways to swim with manta rays

Overall, this episode was incredibly interesting. The style in which this interview was conducted was informal yet educational and engaging which made it easy to follow and understand. The language used throughout the interview was easy to follow as scientific jargon was used sparingly and explained where needed which made it accessible a larger audience. The interviewee was enthusiastic about the subject and approached the topic in a way that promoted community engagement and the use of citizen science. The information was also conveyed in a way that did not exclude particular groups but rather constantly brought attention to how the public can contribute to Manta Ray conservation regardless of academic or social background. As someone not involved in Manta Ray research or conservation, I found this episode to be engaging, interesting, and well conducted.

Approach: 1
Accessibility: 1
Accuracy: 1
Ethics: 1



The Biome Podcast

Coral Restoration

- ~ Host: Kate Sheridan
- ~ Guest speaker/s: Amélie Carraut
- ~ <https://open.spotify.com/episode/3q52rXq2e9BCjMxLWdYC?si=acb19824d8d54de6>
- ~ Length: 32 minutes
- ~ Theme: Conservation
- ~ Rating: 4/4

Hosted by ecology and wildlife students Robi Watkinson, Kate Sheridan, and Emma Hodson, - the Biome podcast is a bimonthly podcast dedicated to discussing everything relating to the natural world from species reintroductions to global conservation through interviews, discussions, and ramblings about their own research and interests.

This episode aimed to cover coral reef restoration and conservation in the Maldives through and interview with Reefscapers conservationist Amélie Carraut.

This episode was an interesting and fun one to listen to as the host was incredibly enthusiastic and shared her own experiences in conservation and research. The style in which this interview was conducted was more on the structured side but was still interesting which made it easy to engage with. While this was more of a structured, formal interview, the language used was still easy to follow as there was no confusing scientific jargon, it was more of an educational talk on coral restoration. The interviewee was enthusiastic about the topic and provided lots of information on what coral reefs are, their role in the marine ecosystem as well as their role in the environment as a whole. She also approached the topic in a way that promoted community engagement regardless of academic or social background. A really great aspect to this episode (and the podcast as a whole) is that it brought attention to how to get into different types of conservation regardless of what you may be specialised in as well as different career journeys and opportunities that are available to start a career in conservation.

Approach: 1
Accessibility: 1
Accuracy: 1
Ethics: 1

Seacreatures Podcast

Epaulette Sharks

- ~ Host: Matt Testoni
- ~ Guest speaker/s: Dr Jodie Rummer (Associate Professor at John Cooke University and leader of the Physioshark project)
- ~ <https://open.spotify.com/episode/2wt7g4OyxHoYruaP3b5RPV?si=a9695f7a8f404072>
- ~ Length: 35 minutes
- ~ Theme: Climate change, physiology
- ~ Rating: 4/4

seacreatures



a podcast about the amazing animals of the ocean

by Matt Testoni 



Hosted by marine biologist Matt Testoni - the Biome podcast is a bimonthly podcast dedicated to discussing different types of seacreatures. Every week the podcast discusses a different animal with a guest who has spent time or done research with them. Guests range from biologists, citizen scientists, divers, researchers, and more.

This episode aimed to look at the Epaulette Shark with Dr Jodie Rummer, a lead researcher on the animal. This episode dove into the shark's physiology and their ability to live in environments with almost no oxygen, as well as their observed and predicted responses to climate change.

This episode was one of the most interesting episodes I've come across. There was so much information on the shark itself as well as the research being on it that is not widely known. The style of this interview was both structured yet informal and interesting which made it accessible to a wide audience and not just those in the field of marine biology. Dr Jodie Rummer was excited to discuss the topic and provided lots of fun facts and information on what Epaulette Sharks are, their environments, their behaviour, and their predicted responses to climate change in a way that was engaging and uncomplicated. She also approached the topic in a way that fostered an interest in the audience as she brought attention to the more unknown side of sharks and shark behaviour. In addition to this, the episode also brought attention to the Physioshark project and shark research as a whole which was great as it allowed those interested in shark conservation and research what opportunities are available to get involved in.

Approach: 1
Accessibility: 1
Accuracy: 1
Ethics: 1



MCCANN DOGS

The McCann Dog Training Podcast

Will socialization help?

- ~ Host: Shannon McCann
- ~ Guest speaker/s: none
- ~ <https://open.spotify.com/episode/6FxlYkdpkxPqaoCRP7nVN?si=9381e8b4e333470a>
- ~ Length: 8 minutes
- ~ Theme: Dog behaviour, dog psychology
- ~ Rating: 2.5/4

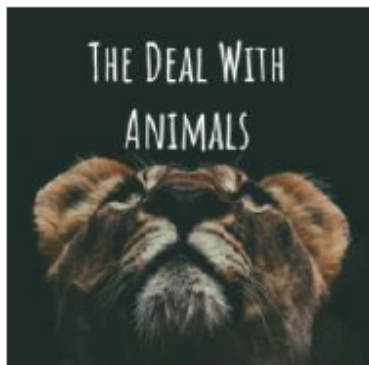
The McCann Dog Training podcast aims to help people understand how dogs think and learn. The podcast covers a broad range of topics relating to dog behaviour, dog training, and tips and tricks to fix common dog owner problems. Through conversations and educational talks, The McCann Dog Training podcast helps dog owners have a well behaved four-legged family member with weekly episodes.

In this episode, Shannon McCann tackled the common notion of socialization in dogs as a way to fix dog aggression and reactivity.

Overall, this episode was not entirely helpful or informative. As the episode was only 8 minutes in length, it was not expected to be incredibly in-depth or informative. The episode lived up to expectation as there was a brief overview on dog socialization and aggression as well as a brief mention on the psychology behind the behaviours commonly associated with reactivity. The second part of the episode was dedicated to ways to mitigate the behaviour, however, they were not entirely useful as it redirected the audience to seek the help of formal behaviourists rather than any ways to actually train or curb the behaviour (as expected from a dog training podcast). While this is good advice for certain dog owners, for a large group this is not an affordable or accessible solution, hence, I didn't think that this episode was that good. The style of the episode was easy to follow and understand and the language used throughout uncomplicated. While the information was ethically sound, the episode as a whole was plain and unengaging. The information was also conveyed in a way that was quite unenthusiastic and boring. Overall, this podcast is incredibly informative, however, this episode lacked the usual tone and creativity.

Approach: 0
Accessibility: 1
Accuracy: 0.5
Ethics: 1

The Deal with Animals Podcast



One Health: Our Pets and Covid 19

- ~ Host: Marika S. Bell
- ~ Guest speaker/s: Dr. Julianne Meisner (University of Washington)
- ~ <https://open.spotify.com/episode/6lBuiWK0c503JNBuFkoWQr?si=d6de27d49b884a71>
- ~ Length: 37 minutes
- ~ Theme: Epidemiology, Virology, Zoonotic Diseases
- ~ Rating: 3/4



Hosted by Marika S. Bell, The Deal with Animals podcast discusses everything in the field of Anthrozoology. Marika's aim is to make science and research more accessible to the general public while bringing lived experiences with human connections to animals to academia.

In this episode, Marika interviewed veterinarian and epidemiologist Dr Marika S. Bell on her research with companion animals. More specifically, this episode was focused on the ability for companion animals to contract Covid 19 from their human caretakers.

Overall, this episode was interesting yet lacked any real information. The episode was sufficiently long yet lacked the coverage one would expect from an episode of this length. There was a lot of interesting background on the topic as well as on the anthropological responses to companion animals being able to contract the virus. While the background was interesting, the interview was focused mainly on research that was still in the very early period and hence there was not a lot of information regarding the status of companion animals and Covid 19. As such, there was a lot of confusing information being passed on due to the nature of a low sample size and even fewer results. While there was a brief mention of how to handle having Covid 19 while also having a pet, it was quickly negated by the interviewee stating that this advice was her own opinion and not based on their research. The style of the episode was more complicated to follow and understand. While the information was ethically sound, the episode as a whole was confusing and contradictory. Overall, this podcast is incredibly informative, however, this episode lacked the usual tone and creativity.

Approach: 0.5
Accessibility: 1
Accuracy: 0.5
Ethics: 1

Overall, I think podcasts are an effective and convenient communication tool for those with busy, on-the-go schedules as well as the general public. The above podcasts showcase the pros and cons of podcast and show how this tool can be used to effectively communicate exciting research to a wider audience. They also show how communities and individuals can get involved in science wherever they are while also bringing to light some of the opportunities available to aspiring scientists as well as those with a general interest in science.

Quiet down, listen and you shall learn: The podcast reviews.

By: SIFUNDO NGXEKISA.

Introduction.

Podcasts are audio or video programs that are made available for watching, downloading, and listening to over the internet. Initially, podcasts were only recognized as audio programs only but over time things have evolved and changed and they can be classified or can come either as audio or video. Podcasts are usually made into series programs consisting of episodes. They are typically hosted by an individual or many individuals who lead and coordinate the conversation being had and sometimes even report news or share their own personal experiences if the conversation leads them there. Most of the time the host or hosts then have guests who are either experts on the topic being discussed or public people to get their views. Podcasts are basically a way of spreading or conveying information to people in a much more interesting and easy-to-follow way rather than them having to fish for the information from books and journals. There are many different styles of podcasts namely:

Interview podcasts involve the hosts speaking with one or more guests usually asking them questions and trying to get their view on things. This requires a bit of research on the topic of discussion.

Conversational podcasts are very similar to radio shows, they involve entertaining conversations about specific themes and topics.

Monologue podcasts involve a single host speaking for the whole episode on a specific

subject or matter, usually hosted by experts on those subjects.

Storytelling/Investigative podcasts are usually where one or more hosts use the whole episode of the podcast to share or tell a story.

Roundtable podcasts are podcasts where a rotating or consistent group of hosts provide commentary and conversation around a specific topic.

Theatrical podcasts can either be a single host narrating a story or an entire cast of performers and sound engineers.

Repurposed content podcasts are podcasts in which stories available in other podcasts are reiterated.

Hybrid podcasts are podcasts that mix and match different podcast formats to create their own style.

Podcasts are likely to be beneficial to the dissemination of science to a broader audience they stimulate conversations in the public and also get a lot of people engaged rather than just the experts.

Scientific podcasts are very effective as they sort of help summarise and simplify much broader and hard subjects to learn. Yes, I have listened to many podcasts before this assignment.

StarTalk Podcast: Science of the brain

Host: Neil deGrasse Tyson.

Co-host: Jackie Hoffman (comedian).

Guest: Heather Berlin (Neuroscientist at Mount Sinai).

Duration: 47min15sec.

Link address:

[https://www.youtube.com/watch?v=yRxyCoquhmI&list=PLzkV1bBuYxaUpoF6gv7paaKS3egoRIue&index=1.](https://www.youtube.com/watch?v=yRxyCoquhmI&list=PLzkV1bBuYxaUpoF6gv7paaKS3egoRIue&index=1)

Points out of 5:

❖ Audience range	4
❖ Topic context	4
❖ Topic covered broadly	3
❖ Credibility	4
❖ Public participation	4
❖ Topic ethically presented	4

Critique:

Most of the points or questions brought by the audience were covered properly in a deep and informative manner besides the ones that they addressed towards the end of the podcast as they were pressed for time. The guest speaker properly covered and addressed the questions and she stuck to context and covered everything that was asked making general examples in-between. Credibility was not undermined as the two hosts invited an expert in the field and the points were both addressed from an expert's perspective and from a general



person's perspective. The expert that was invited has been to the podcast multiple times and can therefore be seen as a friend by the greater community.

The podcast addressed questions from the public that were sent through social media platforms. The topics or questions addressed were very open-ended and they even used other experts' views that have been discredited as an example.

Episode summary:

The episode begins with the guest Heather Berlin addressing if there are any neurological differences between women and men and she shares that men might have a bigger brain than women but they are all equally capable but there are a few neurological and anatomical like the one that women have a more emotional memory than men and that in men language is lateralized meaning it is stored in one side of the brain whereas in women it is stored in both. The guest then goes on to address that psychedelics affect the brain by lowering activation in the frontal lobe of the brain which causes the brain to create its own internal world as it goes to a state that is more like a dream. The guest then explains

how everything that humans experience is basically a figment of their imagination based on the external data that the brain receives and processes. She then said that it is possible for a blind person to create an image based on senses from other senses of the brain depending on when they became blind and how their brains can redirect information. She also addressed that mental science is very subjective therefore many people tend to question its validity and some in society are still not accepting the fact that mental illness is a thing which is therefore why it is still viewed as a stigma. The other viewer mentioned that she has had a brain tumor but it did not affect their motor function, speech, memory, etc, and asked how is this possible? The guest speaker addressed this by saying that the effect and damage of brain tumors depend on their location and if the area they are in is upper sensitive or not. She also addressed that dreams are not a sighting into the future but images or information that has been fed to the brain but not been processed. She also mentioned that it is possible for one to dream and not be aware. The host also mentioned that one does not need a brain to sense pain if they have a functional nervous system. The host here then questioned her on the role of the brain in processing signals. They then went on to say that focusing is when the engaged part of the brain is filtering out certain information. To close off the two hosts and the guest debated on whether a brain transplant will ever become reality with the main host saying that it is possible while the guest disagreed and stated that would take hundreds of years.



Born Free Podcast: #1 Wild animals as exotic pets.

Host: Sarah Locke

Guest: Dr. Chris Draper (Head of captivity
and animal welfare).

Duration: 26min16sec.

Link:

[https://www.youtube.com/watch?v=8M3d1Nfwlwk
&list=PLzkV1bBuYxaUpoF6rv7paaKS3egoRIue
&index=2.](https://www.youtube.com/watch?v=8M3d1Nfwlwk&list=PLzkV1bBuYxaUpoF6rv7paaKS3egoRIue&index=2)

Points out of 5:

❖ Audience range	3
❖ Topic context	4
❖ Topic covered broadly	3
❖ Credibility	4
❖ Public participation	2
❖ Topic ethically presented	4

Critique:

The points that were brought up by the host and the guest Dr. Chris Draper were covered properly but the audience range was not that inclusive as they only discussed some of the points from scientists' viewpoint. The topic was effectively covered. The podcast episode's credibility might be questioned as the guest is a conservational scientist, therefore, he was only covering or viewing things from the scientist's point of view there was not an ordinary person who spoke for the greater public. No questions or comments from the public community

were included in the discussion and the topic was properly framed as it is not biased.



Episode summary:

The podcast begins with the host asking the guest if he thinks zoos should be licensed and the guest then expressed that he does not like zoos and how they keep animals, and he thinks that the licensing process of the zoos should be very strict in protecting mostly the freedom of animals. Dr. Chris Draper then went on to talk about indoor zoo keeping and how cruel it is, comparing it to keeping animals in a shopping mall. He added to say that nowadays especially in the United Kingdom people do not go to zoos to see animals, but they go there to kill time. He mentioned that most people do not actually care or know about conservation in dept of what it entails. The guest speaker also questioned the host on how much she thought people learn and changed behavior after visiting a zoo. After the host could not answer that question, he addressed that a study that was done in the UK shows that there was no change in behavior towards animals and conservation after visiting a zoo. He addressed that people only changed behavior or attitude toward recycling which is not the goal they are trying to achieve. They then went on to talk about the legality of people keeping

animals & exotic pets with the guest saying that in some places it is legal to have a wild animal as a pet. Dr. Chris draper also went on to address that legislation of this is mostly to protect the public and less the animals. If one has the right paperwork and permits, they can own a wild animal as an exotic pet. The guest then went on to criticize social media's influence on people wanting to have wild animals as exotic pets saying that celebrities and adverts drive this a lot. He explained that this drive and increase in demand for exotic pets make it easier for people to sell the animals illegally on the dark web. He then advised that people should not look at the availability of an animal as suitability as a pet and that they should consider the fact that these animals are designed to live in the wild, not in cages. He went on to say that we do not have enough knowledge of what wild animals need to keep them as pets. He also added that it is sad that the illegal trade of animals sources them from the wild and that the animals are not protected enough making an example using lion cubs that were found in the UK saying that eventually when they take them back to the wild they are going to be vulnerable as they already suffered from nutritional and health issues. He closed by advising that people should be aware of the damage caused by celebrities and that it is very hard to conserve animals and protect them due to inconsistencies in laws.



Bill Nye Has a plan to solve climate change: Klepper podcast.

Host: Jordan Klepper.

Guest: Bill Nye.

Duration: 16min28sec.

Link:

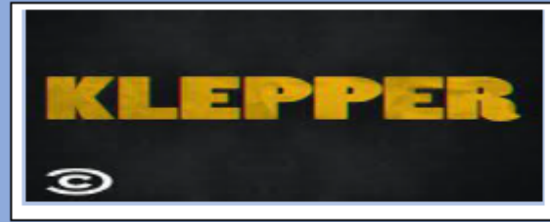
<https://www.youtube.com/watch?v=cuDoQxvnaUY&list=PLzkV1bBuYxaUpoF6rv7paaKS3agoRlne&index=3>

Points out of 5

❖ Audience range	5
❖ Topic context	4
❖ Topic covered broadly	4
❖ Credibility	4
❖ Public participation	2
❖ Topic ethically presented	4

Critique:

The points were properly addressed for a wide audience as they were very general and showed that everyone can take action. The context and the topic were properly covered making a few examples. The guest's point was from his own view of what he thought could solve climate change no other person's views were included. The guest seemed like a personal friend of the host and they started out and closed by advertising or promoting the guest's new podcast. The podcast involved no questions or views from the public. The topic was fairly framed in an ethical manner.



The episode begins with Jordan Klepper asking Bill Nye about his new podcast and what the audience can expect from it. Klepper also asks Bill Nye if he is an activist, and he answers by saying that he is an environmentalist, therefore, making him an activist. Bill Nye then goes on to say that the most effective way of fighting or enacting climate change is to vote. They then go on to talk about how Klepper was exposed to a group of activists but within that group there was diversity as some of them play by the book and the others do not but believe in enforcing stuff by taking action but at the end of the day, they all fight for change. Bill Nye reiterated that if we want to see change and fight climate change people should vote and put into the government people who are going to act and address climate change. He also mentioned that everyone wants the same things which are renewable energy and clean water, and we should all act together to drive change. He suggested that eventually, everyone should be using renewable energy and that an electricity grid that can move energy around like phone calls is needed. But then stressed that society is very fractured due to differences in views and that for now, climate change does not seem like a big enough issue to bring them together. He also stated that there is a weird anti-science movement in society that was introduced by the fossil fuel industry and that this began because they said they were making a lot of money from the effects they have on the environment. But he disagreed by stating that there is a smaller benefit to damage to the environment. Bill Nye also stated that the internet needs to be regulated so that people do not just post anything like false stuff.

Rewilding: Lynx, wolves, and taking nature back! - the conservation Is'podcast.

Host: Laura Roberts

Co-host: Stuart Collier

Guest: Abby Woodbridge

Duration: 46min13sec.

Link:

https://www.youtube.com/watch?v=ppVIQslm-PU&list=PLMXicKFWFwxcgaKXtyCG_hsTn1K9v01RO.

Points out of 5:

❖ Audience range	4
❖ Topic context	4
❖ Topic covered broadly	4
❖ Credibility	4
❖ Public participation	2
❖ Topic ethically presented	4

Critique:

The points were well addressed and driven across to the audience. The topic was well covered in a simple and easily understandable manner. The credibility was well respected as they included both positive and negative impacts of rewilding and also opened the floor for people who are not for rewilding to voice their concerns. There were times when they used the podcast to advertise a documentary that the guest is working on. The podcast did not include any questions or comments from the public but did open

for a further conversation with the public through their social media platforms. The topic was ethically framed and addressed.



Episode summary:

The episode begins by discussing what rewilding is in the context of reintroducing extinct species or restoring habitats to their original states and making the earth green again. The co-host and host describe how the United Kingdom has become a sterile country due to land conversion for agricultural purposes. They address that people in the UK see the little wildlife that is still left and think that they do not need change and do not realize how much has been lost. Stuart Collier mentions that the United Kingdom is stagnant and that they only have about 10% tree cover in the whole country. The three then went on to talk about how habitat fragmentation is the biggest threat to biodiversity loss as it divides populations and makes it hard for species to interbreed. They then go on to talk about the possibility of the reintroduction of beavers in the UK and they mention that these animals are very important for the ecosystem, and they have not been in the UK for a long time. They talked about how they're the successful and unsuccessful reintroduction of species in has been the past and that it is hard to sell rewilding to the public as it might affect people and because the people who put these species back into their habitats are not really

affected by them, but farmers and communities are affected.

They also discussed the fact that in the UK most of the land is privately owned and not much land is government land and how that has become an issue when it comes to rewilding. They mentioned that they have to sell the story that rewilding is to the greater benefit of everyone as it will mean that in the future there will be better food, cleaner water, and better homes. It was talked about how the adults influence kids when growing up and that for children to care about nature they have to be exposed to it from a younger age. They also talked about the fact that for rewilding to happen the redd light people or people who are against it have to get involved in the process. The other way that came up to sell the story or rewilding was the fact that greener earth means improved mental health. Another factor that makes rewilding hard is that people do not like change but maybe if they are shown successful rewilding projects like that of a Pine martin they will be open to the idea as it shows how nature can restore itself if we help a bit like people. The guest talked about how rewilding is scary from a political point of view and how the public usually thinks scary stuff when they hear about rewilding. They touched on the fact that deforestation is one of the major problems to species going extinct and that greener spaces need to be reintroduced as this will also help in the quest to combat climate change highlighting how alarming it was the UK government missed its tree replanting target by 70%. This led them to the fact that another approach to try and turn the tide would be to educate the public.

The last part was about the possible reintroduction of wolves to the UK and how people are divided by it since it would affect some negatively. They mentioned that lynx is a possible alternative to this as they are solitary animals and do not usually go to

spaces where they might be exposed to humans. They suggested that for an improved environment people should be willing to take risks and that the government should be prepared to compensate parties affected by rewilding. They also talked about how people should be able to tolerate the wild taking from them as they also have taken a lot from it. They closed off by mentioning that for rewilding to occur at a faster rate a story to sell is needed.



Episode #23: Nuclear Power- what are the risks? - GIMLET podcast

Host: Wendy Zuckerman

Link:

<https://www.youtube.com/watch?v=Ff5CU3hF68M&list=PLzkV1bBuYxaUpoF6tv7paaKS3egoRlue&index=6>

Duration: 39min26sec.

Points out of 5:

❖ Audience range	4
❖ Topic context	4
❖ Topic covered broadly	4
❖ Credibility	3
❖ Public participation	2
❖ Topic ethically presented	4

Critique:

The podcast was well presented as it catered to a wide audience range and was not too scientific or too general. The topic was well covered as they addressed many dangers of nuclear waste. The information presented and the podcast have credibility as the host was discussing the raised dangers with experts in the field. The podcast did not really cover comments or questions from the public, but the host did direct a few questions to the experts that she met with.

Episode summary:



The episode started with the host discussing how dangerous nuclear power is and stating that there were three accidents of nuclear plants blowing up in the past decade alone. The host then went on to mention that this causes people to fear nuclear power even though it might be useful as an energy source. She covered that nuclear power has upsides that it does not release greenhouse gases into the atmosphere when used for electricity production. To prevent the feared explosions, she said that the nuclear uranium core should always be surrounded by water. She then went on to talk about the most concerning the danger of nuclear power, which is radiation, she mentioned that radiation is prevented from escaping by very thick protective walls of steel and concrete and that there is also a rod that sucks up all the neutrons. In cases when radioactive particles escape, they enter the body and break down DNA which causes some cells to be cancerous. She explained that studies done in places that have nuclear powerplants show no spike in or scientific increase in the overall number of people that have cancer and that the only type of cancer that increase is thyroid cancer. She went on to emphasize that the risk of getting cancer exists, but it is very small and to prevent accidents from happening in these powerplants they use a probability safety analysis, and that people are screened before leaving these areas for radioactive particles. In closing, she explained that the safest way to store radioactive waste to date is by digging holes in rocks and burying but if there are cracks this might lead to particles escaping to the soil.

Final thoughts:

The episodes that I listened to or watched were actually very informative and I learned a lot from them even though some of them were too scientific and sometimes hard to follow. Another thing that I did not like about some of the episodes was the duration of time as they were too long, and I got tired and lost interest in listening halfway through the podcasts. In my opinion, I think scientific podcasts should not go over 30 minutes, this will leave the listeners interested and make them want to engage and listen more, but when they go for over an hour and people do not even finish them. In general, I enjoy listening to podcasts as opposed to reading as I struggle a lot with reading and can barely finish a page in ten minutes. If I were to start my own podcast, I would make sure that they kept very short and that they engage people from different fields and the public as a whole. Another thing I would make sure of is that they are kept interesting and are not too scientific that I lose some of the audience halfway through an episode.

PODCAST REVIEWS

BY GARY KYLE NICOLAU

WHAT ARE PODCASTS

Podcasts consist of digital online media that address a particular topic. This allows a listener to download and listen to whenever they like. These address a wide variety of topics ranging from hobbies, entertainment and politics to literature, science, and mechanics to name a few. They can involve interviews, key speakers or a range of different hosts. The methods of podcasts vary significantly, and the audience is generally based on personal interests and preferences.

SCIENTIFIC PODCASTS

Podcasts can benefit/hinder the dissemination of science to a broader audience in various ways. Podcasts are the ideal tool to break complex and essential science to others in the field or the general public. This can be done efficiently through hosts that understand the topic and have good scientific communication skills. However, podcasts are not peer-reviewed and accepted or declined, this means anyone can create one. Therefore, those that do not understand or have other intentions can purposely or unintentionally misinform the public and their listeners. It is essential that podcasts are critically reviewed and that listeners can know the accuracy of the content of a podcast.

Scientific podcasts can be an effective and powerful tool to bridge the gap between the general public and science. This, however, is not restricted to the general public but is an excellent outreach among scientists. I enjoy scientific podcasts and have been listening to them for a few years now, predominantly in political science and herpetology. I have learnt a significant amount from some podcasts, while others help calm the mind in the otherwise intense system we live in.

I will address five different podcasts in the field of herpetology. Focusing on the methods and content addressed, and lastly, rate these based on criteria I feel are important for a good podcast.

PODCASTS ADDRESSED

HERPETOLOGICAL HIGHLIGHTS: *"In Herpetological Highlights we will explore the recent scientific discoveries in the field of Herpetology. We'll cover everything from lizards to frogs, snakes, and toads. Every episode we'll be digging into the biology and ecology of these reptiles and amphibians in an attempt to disentangle the science."*

SNAKE TALK: *"Confronting a world that instinctively persecutes snakes, Dr. Chris Jenkins speaks with the foremost experts in their field regarding snakes and the issues they face. The discussions range widely, including topics such as snake ecology, snakebite, and captive breeding, but all the conversations have the same goal – to help the listener increase their knowledge and leave behind their irrational fears."*

LIFE, DEATH & TAXONOMY: *"Learn about all things animalia in this family friendly podcast. Each week Joe and Carlos dive deep into nature to find animals that have at least one amazing talent they use to survive. Gather up some animal knowledge and crazy nature facts to drop on your friends at parties with each new episode of Life, Death, and Taxonomy."*

MORELIA PYTHON RADIO: *"A talk show that focuses on the keeping, breeding and current events of the world of morelia and other pythons of the world."*

DUDE! NATURE: *"Dude! Nature is a podcast about the natural world for those of us who got kicked out of science class for throwing pencils at the ceiling. It's hosted by the talented and dashing Adam Weitzman and his brother creature Noah Weitzman."*





Podcast name: Herpetological Highlights
 Episode number: 081
 Episode name: Is that even a frog
 Episode duration: 01h: 06min: 49sec
 Hosted by: Tom Major and Ben Marshall.
 Link to episode: <https://herphighlights.podbean.com/p/contact-1495888146/>

DESCRIPTION

In this episode, as with the majority of Herpetological Highlights episodes, Tom Major and Ben Marshall address two papers on a particular topic. While doing so, they discuss, break down, and humorously critique each paper while ending off the episode with the "Species of the Bi-week", which essentially addresses a newly described species relating to the episode topic. This episode, "Is that even a frog," is dedicated to a Patreon request and focuses on the African group of rain frogs from the genus *Breviceps*. The papers selected in this episode are from Brom et al. (2020), Skliba et al. (2016) and lastly Neilsen et al. (2020).

The first paper, Skliba et al. (2016), talks about *Breviceps mossambicus* and *B. poweri*, specifically about the behaviour of these frogs and how they utilise the mounds and burrows of moles. It was particularly interesting as it was somewhat speculative regarding the trophic interaction. This is generally not the type of papers they address in the podcast.

The second paper was that of Brom et al. (2020), this paper looks at Amphibians in the City of Cape Town and the general public views on them. It showed that the public had preconceived ideas about frogs and that their opinions on them differed among taxonomic affiliations. Many members of the public did not think that rain frogs were even frogs, and many described them as "gross". I, and many like me, however, disagree and think rain frogs are some of the most adorable animals out present day organisms have to offer.

Finally, they address a new species of *Breviceps* by Neilsen et al. (2020). This species, *B. ombelanonga*, was discovered in Angola and is thought to be endemic to the country. The species is related to a central African species *B. poweri*. It is boldly patterned with striking colours. This section is my personal favourite and is a great way to end it off the podcast.

RANKING

5/5

This is a very entertaining and funny episode. The hosts have a good understanding on the topic and explain the science well. They do a good job at drawing in the listener and keeping them interested, by breaking down the science of the papers with humour. I highly recommend this podcast.

Ranking criteria

Appropriate for audience	= 2
Accurate science	= 2
Entertaining and easy to understand	= 2
Content is informative and interesting	= 2
Quality of podcast	= 2





Podcast name: Snake Talk

Episode number: E23

Episode name: Dr. Emily Taylor on Research, Service, and Woman in Herpetology

Episode duration: 01h: 13min: 54sec

Hosted by: Dr. Chris Jenkins

Link to episode: <https://www.oriannesociety.org/initiatives/snake-talk/?v=400b9db48e62>

DESCRIPTION

This episode is an interview by Dr. Chris Jenkins with Dr. Emily Taylor, who runs the Physiological Ecology of Reptiles Lab at the California Polytechnic State University. Dr. Taylor is an accomplished reptile physiological ecologist that has dedicated her career to studying aspects of snake ecology, such as thermal biology and physiological adaptations to survive in desert environments. They cover various topics in this episode, from her past studies and how she got into herpetology, to her postgraduate studies on rattlesnakes. They then move on to how to deal with studies, research and students as a professor, and the concerning lack of jobs within herpetology, despite an increasing number of people with PhDs in the field. Dr. Taylor then talks about her current position, which is her dream job, and explains the active research projects within her lab. Chris and Emily discuss snake translocation, stress ecology, and community engagement, some of Emily's primary fields. They also go into Central Coast Snake Services, Emily's company, that provides communities with tools to help live safely in snake country without harming the snakes. Lastly, they end off on what is arguably the most crucial topic. Women in herpetology. They talk about Dr. Taylor's recent paper on the topic and that the issues raised are not only restricted to herpetology but are present throughout science, technology, education and mechanics.

RANKING

5/5

Extremely informative and accurate information given in an ethical manner. They engaged with one another and were open to debates. The episode followed a structure and topics were appropriate to the audience. From this episode I highly recommend this podcast.

Ranking criteria

Appropriate for audience	= 2
Accurate science	= 2
Entertaining and easy to understand	= 2
Content is informative and interesting	= 2
Quality of podcast	= 2





Podcast name: Life Death & Taxonomy
 Episode number: 202
 Episode name: Golden-tailed Gecko: The Sticky Sprayer
 Episode duration: 00h: 39min: 00sec
 Hosted by: Joe Raspolich and Carlos Rivera
 Link to episode: <http://taxonomy.com/podcast/golden-tailed-gecko>

DESCRIPTION

This episode focuses on the Golden-tailed Gecko from Australia. They address the nomenclature of the species by quizzing one of the hosts. They give a range of information on the Golden-tailed Gecko, which is found in Australia. It is approximately 7 cm in size and live between 4-8 years. This gecko has striking colouration seen in the images on this page, displaying the silver and black scales across the body, orange/bronze coloured eyes, and a golden stripe down the tail. The fascinating part of this episode is when they talk about the geckos' defence mechanism. When threatened, the Golden Tailed Gecko sprays a sticky and smelly substance from the top of the gecko's tail. Should the gecko lose its tail, the regenerated tail does not have the same rupture zones and can no longer use this defence mechanism. This could potentially come at a cost to the gecko. For the length of the episode, there is not much that is addressed, and although accurate information is given, this is not done professionally.



RANKING

3/5

There is distracting music which diverts the listener from what is being said. I found this review incredibly tedious and only moderately informative. A considerable amount of time was spoken about irrelative things. This being how many geckos will fit into the largest turkey in terms of their weight. The majority of the podcast episode went on like this. It was far from what I thought the podcast would be about given the name.

Ranking criteria

Appropriate for audience	= 2
Accurate science	= 1
Entertaining and easy to understand	= 1
Content is informative and interesting	= 1
Quality of podcast	= 1





Podcast name: Morelia Python Radio
 Episode number: 081
 Episode name: The Complete Knob-Tailed Gecko with Justin Julander
 Episode duration: 02h: 14min: 52sec
 Hosted by: Eric Burke
 Link to episode: <https://share.transistor.fm/s/f90227c2>

DESCRIPTION

In this episode, the host, Eric Burke speaks to Justin Julander about his new book, the Complete Knob-Tailed Gecko. This book addresses the husbandry and captive care of these geckos by addressing the species' natural history. The book broke down the papers and research on these geckos to date and put them into an easy to read format. The Knob-Tailed Gecko is a unique looking group of geckos. Julander speaks about his early interests in these geckos, the species he has bred and how he did this. Justin gives his opinion on why they have a knob-tail, a topic of debate—being either a sensory function or defence mechanism for plugging of burrows.

They talk about the methods of accessing papers, precisely the difficulties of accessing some of the information needed to write these books. However, if the research authors are still alive, how they will definitely be happy to send papers to those who ask. They talk about how he started his books and specifically the limitations and processes of writing books on rare and understudied species. Additionally, the difficulties with breeding geckos in captivity. The new book was reviewed in the journal Herpetological Review, which received good comments. These were the main aspects and topics of the episode, and overall it was informative.

RANKING

4/5

Overall the interview was interesting. The information was informative, and the episode was interactive and fun. Justin Julander was an excellent interviewee. He is very knowledgeable, spoke well and informed the listeners. Some sections of the episode had terrible sound and was rather distracting.

Ranking criteria

Appropriate for audience	= 2
Accurate science	= 2
Entertaining and easy to understand	= 2
Content is informative and interesting	= 1.5
Quality of podcast	= 1.5





Podcast name: Dude! Nature
 Episode number: Mar 10 2021
 Episode name: THE NITTY GRITTY: Tom Major
 Co-Host Of The Herpetological Highlights Podcast On
 Snakes, Venom, Snake Reproduction and American
 Accent
 Episode duration: 00h: 35min: 23sec
 Hosted by: Noah Weitzman
 Link to episode: https://www.podbean.com/media/share/dir-u5qwu-d5ddf7a?utm_campaign=w_share_ep&utm_medium=dlink&utm_source=w_share

DESCRIPTION

This episode of the Dude! Nature podcast has Noah Weitzman as the host, where he interviews Tom Major, herpetologist and the co-host of the podcast Herpetological Highlights (reviewed above). The podcast introduces Tom, speaking about a range of intriguing reptile facts. They then move on to a natural history paper highlighting the behaviour of a snake feeding on toads with high levels of bufo toxins and precisely how the snake can overcome these. They then move back to Tom's personal life and talk about his PhD, which focuses on studying the ecology of an invasive snake in northern Wales, the Aesculapian snake, how they survive in such a harsh environment, and their movement patterns.

The next topic for this episode is the work of Tom Major's supervisor, Wolfgang Wuster. This work looked at the evolution of spitting cobras and the possible coevolution between the ability of spitting and hominids. Lastly, the two talk about rapid evolution witnessed in an Anolis species, specifically how natural selection favoured individual lizards with longer fingers after a hurricane hit the island. The information given was great; however, the host interrupts and appears lost throughout the podcast.

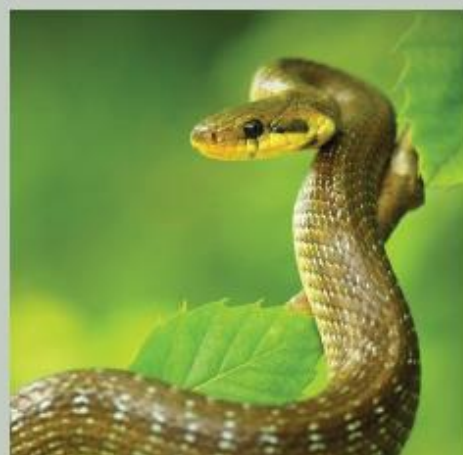
RANKING

2.5/5

Overall the interview was a discussion of random topics in herpetology. Personally, I am not too fond of the name of the podcast nor the flow and design of the episode. The host does not seem to listen to the answers given in the interview and will often repeat what has already been said. The episode rapidly changes between topics. However, the information provided was still informative. They engaged with one another, expanded on the research, and ethically presented the episode. I think the interviewee made this podcast, and if it were not for Tom Major, I would not listen to this podcast again.

Ranking criteria

Appropriate for audience	= 1
Accurate science	= 2
Entertaining and easy to understand	= 1
Content is informative and interesting	= 1
Quality of podcast	= 0





Overall I enjoyed listening to the podcasts. It does not give you the information you would get from reading the paper. However, if it is not something you are actively researching or a topic you are focused on, it is nice to get the broad information through a medium such as podcasts. It takes little effort, and you can be proactive while listening

It is important to know that, although scientists are leading the discussion in these podcasts, they can misinterpret information, and thus, one must still listen with a critical and open mind.

From the episodes I listened, the styles varied greatly, thus it does not give a comprehensive idea on how broadly podcasts are run. It is therefore difficult to say what is lacking from this medium, however, live podcasts would be great, allowing more interaction from the listeners.

If I had to do my own podcast, I would do it similar to Snake Talk and Herpetological Highlights. Both these were informative and gave a comprehensive breakdown of science accurately and entertainingly. I would ensure that I did not stray too far from the topic and ensure that the podcast was of high quality and done professionally, but in a laid back format.

References

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- Nisbet, M. C. (2016). The ethics of framing science. *Communicating biological sciences*, 51-74.
- Škliba, J., Jirků, M., & Šumbera, R. (2016). Burrow systems of mole-rats as refuges for frogs in the Miombo woodlands of south-east Africa. *Journal of Tropical Ecology*, 32(2), 158-161.



The Podcast Reviews

By Erin Boon



Are new to the world of podcasts? Podcasts are essentially an audio episode that is focussed on a certain theme or topic. Podcasts cover every single topic you can think of! Ultimately, podcasts are a fantastic way to indulge in knowledge without staring at a computer screen. There is an array of podcast styles. For example, formal podcasts tend to more of an interview style or one host whilst less serious podcasts tend to have co-hosts or have multiple guests. Podcasts are a great method for conveying scientific topics to a broader audience as the general public do not tend to read scientific articles but find podcasts, even with scientific topics, to be engaging and understandable. Before this assignment, I would call myself a 'newbie' to podcasts but now I question how I ever lived without them? This article reviews five podcasts with the common scientific theme throughout the various topics and ranked between 0-5 with a star symbol.

Over-Informed on IPM

Setting the stage for classical biological control of BMSB (part 4)

Host: Dr. Anna Kate Wallingford

Guest: Nicole Quinn

Duration: 14:59



This episode dives into the basic theory of how insects become invasive pests in introduced region. The BMSB (brown marmorated stink bug) is native to Asia but is an invasive pest throughout the United States. The podcast spoke about the importance of surveying and host-specificity testing for the viability of a biocontrol (it in pests host range and introduced range). Usually, biocontrol agents are brought to the region of the introduced pest to control it. Uniquely enough, the parasitoid Samurai wasp was surveyed in the pest's introduced range and was found to be highly successful. Its is not known how this parasitoid arrived in the US but they found that the wasp populations expanded to surrounding states. Ultimately, the podcast talks about the importance of biocontrol for agricultural practices as there will be less pesticides going into the environment.

Points to chew on

Issues regarding invasive species and using biological agents to control them is an extremely relevant topic now! I found this podcast to be informative as it touches on the benefits of using biocontrol for sustainable agricultural practices which is a major need today. Even though the podcast is scientific, I found that the nature of this podcast accommodates for a wide audience range through the casual conversation interview style between the guest and the host.

Rankings:

Dialogue	*****
Accuracy	*****
Communication of values	*****



<https://soundcloud.com/over-informed/027-setting-the-stage-for-classical-biological-control-of-bmsb-part-4>

TEDTalks

A climate change solution that's right under our feet

Host: Asmeret Asefaw Berhe

Duration: 13:33



Asmeret Asefaw Berhe climbs straight into a fundamental climate change mitigation strategy that should be employed: "stop treating soil like dirt". So, why is soil important in this global challenge? She emphasises that there is two times more carbon in soil than all the world's flora and the atmosphere combined! Soil represents a long-term storage strategy as it can hold carbon for 100-200 years. Unfortunately, anthropogenic pressures drive soil degradation through practices such as deforestation and erosion. This ultimately compromises the ability of soil to provide food and resources for all living organisms. This is not the only problem that the speaker points to, soil erosion has releases more carbon into the atmosphere compared to the current rate of carbon emissions. So, what are the solutions to soil degradation? Climate smart land practices, such as replanting forests and adding carbon back into soil. This will maximise the amount of carbon stored in soil.

Points to chew on

Climate change is a global issue and should be communicated to in a manner that could be understood by people with various background knowledge. However, climate change is an extremely complex phenomenon. Asmeret Asefaw Berhe was highly credible in delivering this talk as it is scientific but explained in such a way that someone without a scientific background would, somewhat, understand. I do, however, feel that minimising the use of jargon would make this episode more appealing to people without scientific backgrounds. If you are new to the concept of climate change, I would not recommend this episode as its main focus may be a bit too complex to grasp. An audience with a scientific background would find this episode appealing as it is academic and interesting.

Rankings:

Dialogue	***
Accuracy	*****
Communication of values	****



https://www.ted.com/talks/asmeret_asefaw_berhe_a_climate_change_solution_that_s_right_under_our_feet

60 Second Science

Chimps apply insects to their wounds

Host: Christopher Intagliata

Guest: Simone Pika

Duration: 2:20



Chimps closely resemble humans, such as using tools and showing emotion, but chimps also use resources for medicinal reasons. Catching flying insects and applying them to wounds is a relatively new behaviour that has been observed. But what are the benefits to this? Christopher Intagliata brings on guest Simone Pika, part of the biology team that studies these chimps. She refers to the possibility that they are using insects for medicinal reasons as some insects have antibacterial properties. Unfortunately, this has not been confirmed as the type of insects being used has not been identified yet. So, could this just be a social behaviour with no obvious benefit?

Points to chew on

I found this podcast to be enticing from the get-go. It appears that this behaviour in chimps is relatively new, so it was amazing to find the link to the journal article about this topic at hand. Even though this podcast was relatively shorter than others, the speaker got the main points across and even increased the podcasts credibility by getting the biologist studying these chimps to talk about her observations. This podcast would appeal to all audiences as it makes direct comparisons between the behaviours of chimps and humans.

Rankings:	
Dialogue	*****
Accuracy	*****
Communication of values	***



<https://www.scientificamerican.com/podcast/episode/chimps-apply-insects-to-their-wounds/>

Science VS

Blue Balls: Are they real?

Host: Wendy Zukerman

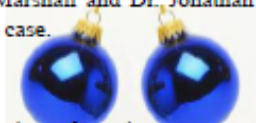
Guests: Sam Pierstorff, Dr. Jamin Brahmabhatt,
Professor Caroline Pukall and
Dr. Jonathan Chalett.

Duration: 37:01



The host immediately jumps to the main question, 'blue balls, medical mystery, or just a pressure mechanism? Wendy Zukerman brings on guest Sam who explains his personal story of feeling 'blue balls'. The host then brings on guest Dr. Jonathan Chalett whom elaborates on his medical literature showing one unique occurrence of this 'blue' case. 22 years have passed with no research published on 'blue-balls'. The host points to the fact that men's medical issues are more extensively researched than women's. Neurologist and sexual wellness expert Dr. Jamin Brahmabhatt is then brought on to explain the science behind this question. Sperm may take 2 months to fully develop, and your body stores it. So, he explains that blue-balls is definitely not a build of sperm in the testes. Guest Professor Caroline Pukall is then brought on to explain another theory behind the discomfort experienced. Blood leaves the phallus but may remain within the testes, as they have different nerves and blood supplies. 12000 men with male genitalia were surveyed online and only 60% reported discomfort but 7% of this only said they experienced severe aching or pain when they were aroused but had not 'finished'. So, Dr. Jonathan Chalett insinuates that that the reason that this topic may not be extensively researched is because it is an extremely rare phenomenon or because the treatment is pretty straight forward... 'a safe, effective and an enjoyable treatment,' he states. So now the new question is, does the female genitalia experience the same physiological response of this 'build up'. The host brings on multiple doctors at this point that prove that this is true! Females experience 'pink puffy pelvis', the new term that the host, Sarah Marshall and Dr. Jonathan Chalett decided on. Basic anatomy proves that the wiring is pretty similar in this case.

Points to chew on



Comedic delivery of this topic is intriguing as it can be quite a controversial one. One thing that struck me is that all the female doctors were extremely excited to just be speaking about a phenomenon that women experience too. This was extremely refreshing! The fact that they compare this event to a "stuffy nose" further contributes to the comic delivery. But not only does it make it laughable, it makes it extremely human and emphasises the point that all humans, of both sexes, experience this feeling. The catchy music also added to the overall enjoyment of this podcast. The accuracy of information was exceptional as the host brought on multiple doctors with diverse knowledge backgrounds regarding this 'ball-busting question.'

Rankings:

Dialogue	* * * * *
Accuracy	* * * * *
Communication of values	* * * * *

<https://gimletmedia.com/shows/science-vs/1lh2l14/blue-balls-youre-wrong-about-them>

Short Wave

What Octopus Minds May Tell Us About Aliens

Hosts: Aaron Scott and Emily Kwong

Guest: Dominic sivitilli

Duration: 15:05



Evolution of octopus is perfect candidate to demonstrate intelligence on other planets as they evolved completely separately to humans. Hosts Aaron Scott and Emily Kwong on guest Dominic sivitilli, a PhD candidate in astrobiology and behavioural neuroscience at the University of Washington. He goes into his work, explaining that each sucker on the animal's arms processes information. The guest highlights that humans evolved to have central intelligence (the brain) whereas octopuses evolved to have distributed intelligence. This highlights that there are different types of intelligence. Ultimately, Dr Domick's research demonstrates that there is potentially a wide variety of intelligence in the universe.

Points to chew on

This podcast has a bit of a blasé attitude to it. It was not overly attention grabbing. It was interesting but that was mainly due to the topic. This is an extremely interesting topic that various scientist would be interested in. Unfortunately, the delivery of the podcast potentially reduces the credibility of this topic. The information delivered appears to be accurate as the host bring on a scientist in this specific field. However, I do see this episode appealing to a wide audience range as no scientific jargon was used.

Rankings:

Dialogue	***
Accuracy	***
Communication of values	***



<https://www.npr.org/2022/03/30/1089748972/what-octopus-minds-may-tell-us-about-alien>

Final Thoughts

I thoroughly enjoyed this medium learning and listening and found it to be exceptionally useful. After searching through the wide range of topics with the common scientific theme in mind, I realised that there are podcasts on literally every topic you can think of. I do understand how some people may struggle with podcast as this mediums only downfall is that there are no visual aids accompanying the talks. However, I found it effective in the dissemination of science as most of the podcasts avoided technical jargon and accommodated for people without scientific backgrounds. Which is basically the point, isn't it? I mean what's the good in finding or doing something scientific if you can't explain it to the general public. People with extensive scientific backgrounds is a small target audience. If I were to start my own podcast production I present my dialogue in a manner that accommodates for mainly all audience types. It would definitely have a smidge of catchy music at the start and the end of the episodes. I would not put music in all the time as I found that to be severely distracting. Out of all the podcast platforms I found Science Vs to be the most intriguing as the topics covered are relevant but always delivered in comprehending manner.

REAL VIEWS MAGAZINE.



MAY 2022

BY S. MANQELE

CONTENT

<i>What are podcasts?</i>	
<i>Types of podcasts</i>	
<i>Benefits of podcasts</i>	
<i>Authors views on podcasts</i>	
<i>Review of Top 5 Scientific Podcasts</i>	
<i>Podcast score card appendix</i>	
<i>Podcast apps</i>	

What is a podcast?



A podcast is a digital audio/video program made available to listen to or download over the internet. The program is made of episodes that are made available daily, weekly, or monthly depending on the podcast host. Episodes can be accessed free or by subscribing to the podcast with an app on your phone or computer and they can be listened to anytime. The duration of podcast episodes varies from 1 minute to 4 hours. It is entirely dependent on the podcast host.

Types of podcasts

Personal, professional, and public are the three types of podcast (Panday 2009). The main difference between the three types of podcasts is their content and target audience. **Personal podcasts** aim to share the experiences of the podcaster with individuals close to them. They are like a photo album, but they are in an audio format. **Public podcasts** discuss generic

topics like celebrity news, which are easy to understand. **Professional podcasts** are educational, and their primary purpose is to educate basic knowledge of content is necessary as specific jargon may be used throughout the episode (Panday 2009).

Benefits of podcasts

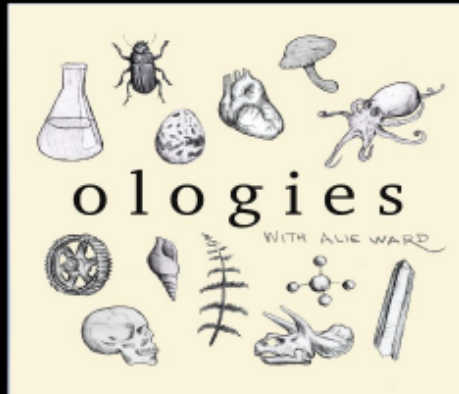
Podcasts are essential in spreading science to a broader audience as they are easily accessible to everyone. They do not require visual attention, which means they can be listened to while doing other tasks like driving, working out etc. Podcasts bring up-to-date academic discussions to a broader audience. Depending on a person's interest, one can find the channel that suits them. Podcasts are another effective way for students to refresh what they learned in class as there are podcast channels that focus on the topics they have covered in class, this gives teachers from all over the world an opportunity to teach students from other countries, breaking the barrier that exists in the quality of education learners get.

My opinion on scientific

Scientific podcasts are an efficient way to bring different scientific topics to me. My interest is in thermal physiology and how climate change affects the physiology of animals. There are several papers on the topic, but I cannot read them due to time, and podcasts have been a solution to that. They have given me a chance to listen while I work. They are very effective in presenting the topic in a single episode, there can be more than experts who discuss the topic. As a listener, I can send questions to them. This form of communication is not possible with journals or books as with the two you need to email the author and wait for some time before they respond. The podcast community is significantly helpful in the comment section. I can find like-minded people to engage with about the topic.

Before the assignment, I had listened to entertainment podcasts but not scientific podcasts. The assignment has allowed me to be exposed to a range of scientific podcasts I have my favourites that I listen to frequently e.g. Science Friday, Science Rules, Podcast & Chill to name a few.

TOP FIVE SCIENCE PODCASTS 2022



About Channel: Ologies is a comedic science podcast hosted by Alie Ward, a science correspondent of CBS. She is a writer, presenter, and actor. The podcast ologies was named top 50 podcasts in Time magazine.

<https://www.alieward.com/ologies/entomology?rq=insect%20>

Duration: 55 minutes 45 seconds

Presenter: Alie Ward

Guest: Lila Higgins

Episode No: Ten

Episode Name: Entomology (INSECT) with Lila Higgins.

Episode Description

In this episode of Ologies Alie ward interviews a self-professed "bug geek" Lila Higgins an entomologist and a citizen science manager at the Natural history museum in Los Angeles. The episode starts with how Lila got to be an entomologist. She details what is fulfilling about her job, naming the bugs she likes Hercules beetle. Lila gives insight on how one can be an entomologist, subject

to study in school, and the different fields within entomology. She details the project she did on the biological control of glassy-winged sharpshooter pest of citrus and grapes. The host pauses the interview during the episode and explains terms for the listeners as Lila uses scientific jargon like proboscis, aspirator, and others. The guest gives the listener the different ways to deal with insects at home, garden, and in their insect collection, which is very important for everyone. She also debunks the myths about insects. Before the podcast ends, the host fires questions from her listeners to the guest. One question was what the difference between dragonflies and damselflies is? *If you want to know what entomology this is the episode to listen to!!*

From the episode I learned what citizen science is and why it is essential for science research from the episode. The websites that I can be part of for citizen science, the guest, gave sites for entomology projects and others for science like <https://www.zooniverse.org/projects>. The guest advises future entomologists on how they can choose a path they may love within entomology

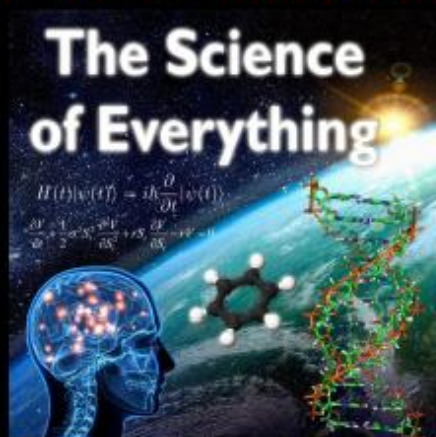
Critique

The episode is 55 minutes short, I say this because it is presented in a comedy type of way, the host is interviewing the guest in a semiformal way this makes it super chilled. The they make jokes and explain stuff using simple examples I enjoyed this episode.

Episode score

OUT OF 5

Attention grabbing ... 
Continuity..... 
Audience..... 
Public participation... 



About Channel: This podcast channel discusses various natural and social sciences topics to better understand the world around us. The podcast is hosted by James Fodor, a student from Australian National University in Canberra, Australia.

<https://open.spotify.com/episode/77KS78wP4FWQbmH1Vq8Xa5>

Duration: 32 Minutes

Presenter: James Fodor

Guest: No guest

Episode No: 10

Episode Name: The cell

Episode Description

James starts the episode defining a cell as the smallest unit that can carry the fundamental processes of life. He gives a brief history of the cell and how the cell was discovered by Robert Hooke, a scientist who first used a microscope and came up with the word cell. He details the different types of cells and answers why cells are so small? Eggs are the biggest type of cell, making ostrich eggs the biggest cell on the planet. James then gives a detailed description of the function of cells, which is mainly protein and energy production that occurs within the nucleus. Lastly, he explains how energy is made within the cell by mitochondria and chloroplast. He explains the theory of endosymbiosis, which explains how the two came about. From the episode you gain the basic theory of a cell which is a refresher for those who know it. The episode is very useful to students doing cell biology.

Critique

The episode is "lecture period" long. It does not use any soundtracks during the episode the presenter talks from start to finish without engaging with anyone. The sound and production of the episode is high quality.

Episode score

OUT OF 5

Attention getting 
Continuity..... 
Audience..... 
Public participation... 



About Channel: The podcast covers developments of science worldwide, presented by science journalists and using experts to give details of what is being discussed in a news bulletin fashion. The episodes are concise, covering the most important themes of the topic.

Duration: 4 minutes.

Presenter: Journalist

Guest: No (interview recordings)

Episode Name: Climate Change Is Shrinking Animals, Especially Bird-Brained Birds.

<https://www.scientificamerican.com/podcast/episode/climate-change-is-shrinking-animals-especially-bird-brained-birds/>

Episode Description

This episode covers why climate change is causing birds to shrink in size. Scientists observed that most different animal species are decreasing in size. Observation of songbirds in Michigan helped in testing this observation. In 2019 about 70 birds died after hitting

windows they were and measured compared to the bird species 40 years ago. The study found that the birds in 2019 had decreased in size, and the data showed that some species were shrinking at different rates. The presenter supports the claims made by the study by playing voice recordings of the interview she did with the authors of the study. They suggest that the change in body size of birds is due to their behaviour, were species with bigger brains are smatter and can change their behaviour based on their environment, this means that they are buffering them self from the increasing temperatures, by staying spending more time in the shade which may result in their size-changing slower. Inclosing of this episode, they suggest that smaller brained birds may be more vulnerable to climate change than other birds. From the episode learned how climate change affects the size of birds which is something I did not know about the effects of climate change.

Critique

The episode is very short, but it goes straight to the point of what is being discussed and they support their findings with expert opinions which makes the episode more reliable. If you are a person who wants detail on why climate change is shrinking animals the episode will leave you wanting more as they do not answer all questions about the topic. The style in which they present the episode is effective

for people who do not have time to listen to long episodes.

Episode score

OUT OF 5

Attention getting 🍌 🍌 🍌 🍌 🍌

Continuity..... 🍌 🍌 🍌 🍌 🍌

Audience..... 🍌

Public participation... 🍌 🍌



About Channel: This podcast channel aims to connect as many people as possible with science. This is achieved by getting scientists to talk to the public about their work to develop critical thinking and a spirit of exploration to young people through their work. The episodes are recorded at the Royal Institute and the audience is given a chance to ask questions during the recording.

Duration :1 hour 16 minutes & 33 seconds

Presenter: Sue Black

Guest: no guest but the audience

Episode No: 55

Episode Name: Using Forensic Anthropology to solve crimes

<https://open.spotify.com/episode/736VEEUchdFdLd3SmhYRfO>

Description

This episode is about forensic anthropology. Presented by Sue Black, a forensic anthropologist who has worked on various crime cases worldwide and helped identify the victims and perpetrators of various conflicts. Sue starts the talk by asking her audience a question, "Who are you?" and how do you know who you are. The question sets the foundation of her talk. She gives real-life examples of how she took a molester to court on rape cases using vain patterns shown on a skype camera of a victim her evidence was heard in court this was the first case in the UK to be heard using evidence of this type. The episode gives insight into the importance of forensic anthropology in solving crimes.

Critique

The episode was presented in an auditorium, the presenter was interacting with the audience during the talk. The sound of the episode was at the right pitch for the audience. The episode had no soundtrack and it is very long close to one hour, when listening to the episode you might need to take breaks to understand what the episode is about.

Episode score

OUT OF 5

Attention getting 🍌 🍌 🍌 🍌 🍌

Continuity..... 🍌 🍌 🍌 🍌 🍌

Audience..... 🍌 🍌 🍌 🍌 🍌

Public participation... 🍌 🍌 🍌 🍌 🍌



About Channel: The podcast explores various questions kids ask about science presented by two vibrant hosts, a science journalist, and a teacher, in a kid-friendly way to be enjoyed by everyone in the family.

Duration : 13 minutes 52 seconds

Presenter: Lindsay Patterson & Marshall Escamilla

Guest: Molly

Episode Name: The Science of smell

<https://open.spotify.com/episode/2Dk3csgtz3B3dxdvfxzgzzg>

Description

The episode looks at why foods smell. They answer this question by interviewing Molly, who went on a quest to find answers to this question. She pieced the science of how humans process smells by talking to smell scientists who try to understand human process of smell. Who detailed that when you smell something, you inhale and aromatic compounds enter the nose where neurons exist and send the chemicals to the brain, which interpret the message allowing us to interpret the smell. They further explain that each thing has

different compounds that allow the brain to understand different smells. The podcast teaches the science behind smell interpretation by the nose.

Critique

The episode uses simple language that kids can understand, they make examples to explain better what is being said. The soundtrack used for the episode is similar to those in cartoons which makes the episode similar to what kids are used to watching. The duration of the episode is long enough for kids' attention span. The two presenters interact in a "play" like form during the episode, making the episode more interesting. Everyone in the family can enjoy the podcast

Episode score

OUT OF 5

Attention getting 🍌 🍌 🍌 🍌 🍌
Continuity..... 🍌 🍌 🍌 🍌 🍌
Duration..... 🍌 🍌
Public participation... 🍌 🍌 🍌 🍌

SCORE CARD APPENDIX

Each podcast is scored according to the following categories

1. Attention getting
2. Continuity
3. Duration
4. Public participation
 - Attention getting does the podcast grab the attention of the listener in the first 3 min
 - Continuity does the host stick to on topic or is the episode disorganised
 - Duration is the podcast long enough to keep the listener entertained
 - Public participation is the public allowed to get involved in the podcast in any way e.g asking questions

References

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APPS TO LISTEN TO YOUR FAVOURITE PODCASTS



Talking Science.

Wade Sexton

Podcasts are audible episodes whereby hosts discuss a particular topic of interest for an audience. Podcast channels are almost always attached with a string of episodes ready for an audience to enjoy. It is up to the producers and hosts of the podcast channel as to what their podcasts deal with in terms of discussion topics. The tone of these episodes vary from channel to channel and different hosts may take a different approach to exploring ideas as well as having different interview styles. It gives this form of communication a personal touch. Podcast episodes are mostly in the form of a relaxed discussion and becomes a place where people come together to have honest and carefree discussions about current issues or topics.

Podcasts have become a great learning tool and can help adapt a person's listening skills. Apart from this, many people listen to podcasts while they are multi-tasking, for example many people enjoy listening to a podcast whilst cooking or driving. It takes no effort and often people take in more content in these scenarios, because they are relaxed. This makes podcasts a unique and fantastic approach to disseminating science to the general public. People in the science industry often use podcasts to explore ideas that may be too complex for the general public in order to help people better understand scientific ideas and research. Apart from disseminating science to the general public, some podcasts often explore controversial scientific topics or even just use podcasts as a way for other scientists to talk about their research.

As someone who is a scientist in training, it becomes clear that the work we do as scientists is to better our community and create a better world of understanding. We need to be able to engage with the public about our work and do it in such a way whereby the public understand the importance of our work and how it fits the bigger picture. Mediums such as podcasts is an avenue scientists need to explore and use to their advantage.



In this space, we will look at five different episodes from five different podcast channels. We will look at how the host engages with the topic and to see whether they are able to allow the viewer to digest the content in a healthy and easy way We scored the podcasts on different categories. We scored each category out of 3 stars. These include content which was scored and how interesting the topic is. The host was scored according to their engagement with the listener and/or their guest. Audience range was scored based on whether the episode is accessible and easily understood by the general public. Clarity was scored based on how well the host spoke and if they had a strong verbal presence. And because we are scientists, we will be giving them a rank out of 5 (just because).

Episode name: Climate change... the Apocalypse?

Host: Wendy Zukerman

Time: 43:07 minutes

<https://open.spotify.com/episode/686MnqkthXNWKtr6Zxlmh7q?d8793133Ad7c4d71>



Rank: 1

Content: ★ ★ ★

Host: ★ ★ ★

Audience range: ★ ★ ★

Entertainment: ★ ★ ★

Clarity: ★ ★ ★

About the podcast:

A Spotify original, Wendy Zukerman and her team create multiple podcasts around hot topics in science and discuss them in a fun and charismatic way in which few podcast series do. The aim of course is allowing the general public to understand and enjoy the topic in a manner that keeps the listener tuned in as well as creating an understanding of the topics. The exuberant energy and well-timed background music allows for a very enjoyable listening experience.

What's it about?

Without a doubt climate change is one of the most talked about issues of our time. Most people have not perused a career in science and therefore the issue does not smack them in the face the same way it does to scientists. So, then what exactly is climate change? What has caused it? How serious is it? Are we all going to die? The answers lie within this podcast. This episode takes a very good approach to take you back to the basics as to why and how scientists started talking about climate change. We hear interviews from scientists with data through 60 years and scientists who specialize in climate modelling. This episode talks about the main drivers of climate change like the burning of fossil fuels, the rising of our oceans, the rising of our average global temperatures as well as talking about some of the skepticism surrounding climate change. This episode gives an honest review of some of the certainties of climate change as well as some uncertainties, because the fact of the matter is, science isn't perfect, and we are still learning about how this will affect us in the future as well as how quickly. So, all in all, if you are someone who wants to understand the basics of climate change or if you are someone who is a skeptic, or if you simply want to gain more knowledge on the topic to help educate your peers, tune into this podcast

What did I think?

In short, I thought this was a very well worked and well thought out episode. The exuberance of the host really comes through as she interviews multiple scientists with a strong background in climate change research. This podcast tackles a very serious issue in a light and easy to follow way with good use of sound effects to create a picture. The topic of climate change in our current society is one that is frequently discussed. However, the issues around these discussions are that people who aren't involved in the scientific field often don't understand the science or rather don't understand the bigger picture as to why climate change is happening. What we see in today's society is that when people think of climate change, they think that the planets temperatures are going to rise exponentially when really climate change is happening because of a variety of factors and some information out there is over

Episode name: Ep. 7. Dr. Anjel Helms, Chemical cues in plant defenses

Host: Erfan Vafaie

Guest: Dr. Anjel Helms

Time: 47:01 minutes

<https://open.spotify.com/episode/0aku9g4H4MoAuWIMIEScOc?si=3d8a0b11d51e4686>



Rank: 2

Content: ★★★★★

Host: ★★★★★

Audience range: ★★★★★

Entertainment: ★★★★★

Clarity: ★★★★★

About the podcast:

The host of the podcast is an entomologist who works with Texas A&M AgriLife Extension service which is based in Texas. Erfan created a podcast to engage with the public about integrated pest management and to explore good research with his audience in the hopes to create a better understanding of the importance of insects in agriculture. It is an inspiring series for any young entomologist and anyone else who enjoys listening to interesting research involving these fantastic and interesting little critters.

What's it about?

I find that almost every accomplished entomologist says, "I had no idea I would become an entomologist", and in Dr. Helms's case, she initially found it to be "nerdy". It's in podcasts like these that we understand more that well established academics had to start somewhere and that we tend to view these academics as very intimidating characters. After we get to know the speaker a bit more, she begins talking a bit about her research. She begins explaining that insects and plants are constantly at chemical warfare. She introduces some theory to us that chemicals are given off from plants once insects begin feeding on them in order to induce certain defenses that help defend the plant. She explains that in her research she wanted to see whether inducing these chemicals could help aid in a biocontrol context for agricultural plants. In her experiment they used Potato plants and the Colorado potato beetle. Her results showed that leaves had a heightened defenses and reduced Beetle larvae performance when exposed to chemical signals. This study shows that plants are constantly "aware" of their environment and can prepare more readily from herbivores above and above ground.



What did I think?

The intro led really set a smooth tone for the rest of the podcast. However, the interviewer did initially speak quite quickly but as the episode went on, the episode was very well paced and had a great flow. It was also nice to get some background into the guest speaker and where she came from, however more than twenty minutes were dedicated to this. The host of the podcast has a good background knowledge of Entomology which allowed him to engage well with the speaker. The guest, Anjel Helms described her research in a way that anyone could understand. She gave good quick introductions to topics related to her research as well as keeping the audience engaged. She gave very useful analogies and anthropomorphized many of her details so that the audience could view the plant and its interactions in a unique way. This podcast is perfect if you are fascinated by insects and their relationships with plants. This one is a goodie!

Episode name: Mycology Education Host: Seth Shostak, Molly Bentley Guest: Merlin Sheldrake Time: 52:40 minutes https://open.spotify.com/episode/1rwcKQy3nbDevGoVsVNeXx?si=c4f9407547f44f6c		Rank: 3 Content: ★ ★ Host: ★ ★ ★ Audience range: ★ ★ Entertainment: ★ ★ Clarity: ★ ★
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About the podcast:

Produced at the SETI institute, astronomer Seth Shostak and Journalist Molly Bentley combine forces to make an interesting and unique podcast series. The hosts use this space to discuss science, technology and any advancements that interest them. Their episodes are dedicated to diving headfirst into topics that broaden the mind and get into discussions with well accomplished guests. The experience of the hosts makes for a enjoyable and delightful experience.

What's it about?

The episode deals with the broad world of mycology. Guest Merlin Sheldrake is one of the worlds leading Mycologists and is the author of "Entangled life: How fungi make our worlds, change our minds and shape our future". He talks about how intertwined fungi is in our everyday life and how it has helped shaped the evolution of plants. The episode is a perfect mixed of science and philosophy, with intriguing discussions. Questions like, "Can we consider Fungi as intelligent?", are just some of the topics discussed here. If you want to look further than the mushrooms on your pizza, this podcast is the one for you.

What did I think?

I think most people can admit their knowledge on Fungi is limited. The world of Mycology and fungi is globally underappreciated, and it seems it is the most important link to our survival. This podcast really brought to life this visualization of a complex and incredible world that lives beneath our feet. The guest had the listener at the edge of their seat and spoke so elegantly that it is hard not to be interested in what he was saying. The hosts did a great job at keeping up with the intellectual discussions and asked very formative and relevant questions. The use of analogies allows us to visualize these complex systems in a simple way. There are so many open large spaces that we are yet to learn more about, if you are fascinated by the vast unknown and want to learn about an entire world beneath your feet, tune into this episode!



Episode name: Gene therapy

Host: Kira Deneen

Time: 26:03 minutes

<https://open.spotify.com/episode/3QDvGHqntjfrWP2fskD5X?si=26e6d1003e4141ee>



Rank: 4

Content: ★ ★

Host: ★ ★

Audience range: ★ ★

Entertainment: ★ ★

Clarity: ★ ★

About the podcast:

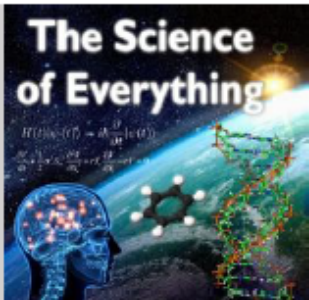
This podcast is dedicated to giving detailed yet understandable talks on hot topics surrounding genetics. The target audience is to the general public and includes guests within the genetic based sciences. It tries to engage with the audience as much as possible on the extraordinary field of genetics. Kira Deneen and her podcast has been nominated three times in the podcast awards and most recently has won the award for Best 2020 medicine and science podcast award.

What's it about?

Gene therapy can be somewhat of a controversial topic, for those that don't know what it is about. The episode begins with a brief introduction about what gene therapy is and why it is necessary. To add more context, she gives us a background into some of the genetic disorders and diseases that have proved to be difficult to cure by scientists and doctors. Kira Deneen speaks about how beneficial this could be to us as a human race. Gene therapy could hold the secrets to curing cancer, cystic fibrosis, diabetes and several more disorders and diseases. So, what's the problem? Why is this not becoming readily available to the public? Kira Deneen discusses some of the controversy around gene therapy research as well as some of the costs and benefits associated with gene therapy. She gives an honest account on her beliefs and views on the topic and aims to bring the listener to better understand the discussions around gene therapy.

What did I think?

People tend to steer away from these "complicated topics" for fear of not understanding. Some scientific concepts and discussions can be dealt with slowly and in a manner that the common person can understand them. Kira Deneen explains the theory in a tasteful way. However, I do feel that in this particular podcast, Kira Deneen takes an almost aggressive approach to her topic. She did well to set the scene and give a discussion, however because of her pace I feel it may be harder for someone without a scientific background to understand everything so quickly. You also miss out on the discussion aspect of the podcast without there being a guest. However, that being said, the topics discussed were very interesting to those that listen to it, and she gave a good outline of how the field of gene therapy is perceived.

Episode name: Episode 10: The Cell Host: James Fodel Time: 31:18 minutes https://open.spotify.com/episode/77KS78wP4PWQbmH1Vq8XaS?si=ebbad564f4344a21		<u>Rank:</u> 5 <u>Content:</u> ★ ★ <u>Host:</u> ★ ★ <u>Audience range:</u> ★ <u>Entertainment:</u> ★ <u>Clarity:</u> ★ ★
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About the podcast:

James Fodel has created a podcast channel for those looking to learn about biology. His passions for explaining certain scientific topics for the general public to understand is most inspiring. This channel can serve as a great diving off point for anyone looking to broaden their understanding on scientific or biological topics and is especially useful for those who are auditory learners.

What's it about?

This episode is all about cells. Cell history, cell biology, cell origins and everything cells! The podcast begins by the host giving an overview of what a cell is and what they do. He begins by talking about some of the significant historical moments in history with regards to cells. He talks about the scientists who described them as well as why and how they were discovered. The podcast gives an incredibly broad view of cells within animals and plants. This podcast can be a useful tool for any person looking to understand more about cells as well as university or high school students who need to understand cells for their courses.

What did I think?

The host has a great voice and speaks fluently which makes it easy to hear some of the complex cell names. As someone who has done a course in Cell Biology, I may have found this useful as an additional learning tool. However, for someone who is more of a visual learner, I found this podcast to be rather difficult to understand without a visual aid. In my opinion, learning about cells and their functions is really only possible with a visual aid. Without a visual diagram in front of you when certain cells are being described I personally feel it becomes more difficult and overwhelming to visualize what is happening in the bigger picture. However, the host does not only talk about cells and their functions, but also has some very cool facts about how cells were discovered and other cool facts. As much as this episode did not tick all my boxes, I do feel that it could be a great learning tool for someone who is an auditory learner.



Wrapping up...

Engaging with the public about scientific related topics is an incredibly useful tool to improve our field and the scientific industry. As scientists, we are in a privileged position to be well educated in our prospective fields and it becomes increasingly important in this day and age to help find new prospective students to join the scientific field. We need to inspire the youth to pursue a career in the sciences and not just that, but also to allow the general public to have a good perspective on certain scientific topics.



This is why it is important to use as many trending mediums as possible to discern our science to the public. In a world where people are becoming more and more on their technological gadgets, we need to find ways to reach people through these means. The use of podcasts as well as the people listening to podcasts has increased drastically over the last couple of years.

Through listening to these podcasts, it is very clear that the use of podcasts to educate the public about science can prove quite useful. Being able to hear the voices of some scientists and to break down the wall that most of the public see scientists in. These episodes allow us to see scientists as a normal human being rather than a super intellectual human being that are separated from the public in their own world. Science is an important part of our lives, and we need to try and curb the idea that science is “hard” or “boring”, but rather use our skills and improve our skills on using creative ways to engage with our world in a way that is palatable and exciting for the general public. This is our responsibility, not a burden. I hope to keep listening to podcasts through my career and as I do, I hope to keep gaining skills and ideas of what the best ways to keep an audience engaged in.

One of the most important skills a scientist can have is the ability to communicate what they think, what they do and and what they know to different audiences. The scientific paper is aimed a very specific audience, which is other scientists – people with the training to interpret the format and the language used. Reading long technical texts is mentally taxing, and short articles written in direct language – such as those in print and online magazines – are more accessible to the layperson and the rushed academic alike. Even better are those formats that don't require reading at all: the educational video and the podcast, which scientists are relying on more and more to communicate their thoughts and knowledge. This series of podcast reviews is intended to sharpen the reviewer's, and the reader's, skills in interpreting audio communication by evaluating those who have done it already. The following episodes, all defining themselves as science podcasts of various stripes, were chosen at the reviewer's discretion.

Guide to review scores:

5 = outstanding

4 = good

3 = mixed

2 = poor

1 = atrocious

Key Criteria: Quality of presentation – Scientific communication – Ethical framing of science

The ideal podcast has a high quality of presentation (including not only technical aspects like sound quality but also the unique style of the presenter(s) and appropriateness for the subject and target audience); communicates science effectively, being neither overcomplicated nor superficial in its presentation of science; and frames science in a way that is ethical and not misleading or damaging to public perception of science*.

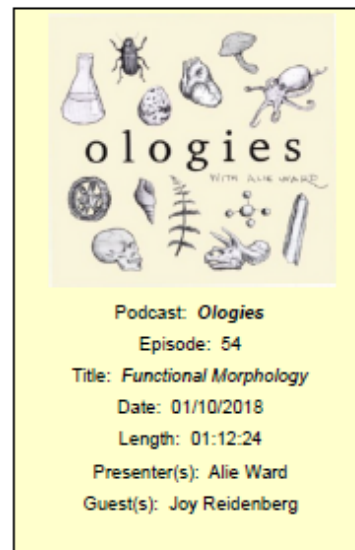
* Nisbet, Matthew. 2016. 'The Ethics of Framing Science' in *Communicating Biological Science*

Ologies

Functional Morphology

3/5 "Worth a listen"

- Interesting speaker
- Humorous tone, uniquely quirky and personal
- Editing used to good effect
- Didn't go deep into the science
- Ethical framing of science



This episode was my personal favourite. In each episode of *Ologies*, the presenter, Alie Ward, invites a specialist in a different field of science (usually suffixed with "-ology") as guest speaker to talk about their career. This episode's guest speaker is Joy Reidenberg, a comparative anatomist / functional morphologist. Ward's wry sense of humour takes a bit of time to get used to but mixes well with Reidenberg's personality, and the personal, one-on-one style of interview is a good fit with the nature of the podcast. The guest speaker's story of her childhood and her initiation as a young scientist starts slow, but is described in such colour (with a dash of self-deprecating humour) that it held my attention well past the half-hour mark. The presenter allows the guest to do most of the speaking, with enough engagement to keep the show from becoming a monologue. I appreciated the use of editing to inject sound effects, explanations and brief diversions from the main topic at several points during the episode, which helped to break up the sense of monotony that usually arises in podcasts of this length and pace.

One of the biggest strengths of this episode is that you do not have to know much about science to enjoy it. It is a human story, not a science lesson (although you will learn some science along the way). There is a downside to this. The presenter did not prompt the speaker to delve very deeply into the science of functional morphology, its history or how it is done – a major weakness considering the podcast's stated focus of exploring a different field of science in every episode. An issue in the ethical framing of science appears in Reidenberg's account of an incident where, as a postgraduate researcher, she is stopped on the road for breaking the speed limit and then bends the truth about her research in her conversation with a police officer. Admittedly this was a very unusual situation – the young Reidenberg found herself at risk of being wrongly identified as a suspect in a serial murder case while en route to a whale dissection. If your curiosity has suddenly been piqued, listen to the episode.

The cherry-on-top was the end credits theme, which the presenter reveals was done by Nick Thorburn,

frontman of the Canadian band *Islands* – whose music I have enjoyed for years. Despite its shortcomings, I enjoyed the quirky, intimate character of this low-budget podcast; it falls just shy of a "good" rating.

Episode link: <https://www.alieward.com/ologies/functional-morphology>

The Naked Scientists

Xenotransplantation

5/5 "Factual and flawless"

- Quality editing
- Good pace; gets to the point
- Covers a range of breaking topics in science
- Fields listener questions
- Nothing really



I had the privilege of attending a show by *The Naked Scientists* when I was a boy about ten years old – and you can bet it involved liquid nitrogen. The British duo are true gurus of popular science and are masterful at making chemistry, biology and technology accessible and exciting for a general audience.

Episodes of the TNS podcast focus on the latest developments in science, technology and medicine, digested into a format that's less than an hour long. This episode is strongly focused on medicine, opening with a story electric stimulation to alleviate paralysis, covering a study on the effects of COVID-19 vaccinations on menstruation, and delving deep into the headline story on xenotransplantation – the transplantation of animal organs into humans. Sidetracks include a report on the discovery of a new type of behaviour in chimpanzees (incidentally also related to medicine). Every bit of terminology that might not be familiar to a non-scientist is explained by the presenters in a quick and simple way that leaves you confident that you understand what the speakers are talking about. There is also a segment where the presenters field questions from listeners.

The podcast creators prove to be conscious of the ethics of science. A balanced perspective is presented, acknowledging multiple ways of framing a contentious issue. This episode's story on the use of facial recognition technology by law enforcement – reported by South African-born Robert Spencer – concludes that there is a difficult trade-off to be made between the convenience and security that facial

recognition affords and the risk of its use a tool for private companies and nation states to invade individuals' privacy and suppress dissent. The technology is also framed as a diversity issue, with a guest speaker implying that facial recognition algorithms tend to suffer from inaccuracy as a result of being trained mainly on young, white, adult male faces.

As long-time professionals in the presentation of science to the public, Smith and Murphy have more money and a larger team of skilled people at their disposal than the average podcaster. It shows. The episode is perfectly paced to remain interesting from start to finish and neatly edited to eliminate superfluous content. Both the main presenters and the reporters speak clearly and to-the-point, and they have access to relatively high-profile scientists working on the cutting edge of their respective fields. Advertisements are few and brief and do not really interrupt the momentum of the episode.

This episode is technically flawless, true to the podcast's stated intention, and frames science in an accessible and ethical way. I give it an outstanding rating.

Episode link: <https://www.thenakedscientists.com/podcasts/naked-scientists-podcast/xenotransplantation>

Cinema Science

*Gattaca's Designer Baby Astronauts:
Editing the Murder Gene*

2/5 "Sloppy"

- Connects the movie with real-world science
- Discusses ethical implications
- Fields listener questions
- Playful tone
- Guest speaker hasn't seen the movie
- Initial explanation drags on a bit
- Too heavy on banter
- Needs editing



Podcast: Cinema Science
Episode: S1E02
Title: *Gattaca's Designer Baby Astronauts*
Date: 16/10/2018
Length: 00:51:27
Presenter(s): Heidi Febinger, Anne Gibson
Guest(s): Sasha Luks-Morgan

Cinema Science is a podcast that discusses the science behind the movies. This is a very "chatty" podcast, full of playful banter between the presenters – all of whom are university students. At times the banter becomes too heavy and goes off-topic, making the episode seem very disorganized. It does not help that the guest speaker, who is invited onto the show as the expert on this episode's movie,

Gattaca, announces up front and with some embarrassment that she has not actually seen the movie. Almost half of the episode is taken up by presenters explaining to the guest what the movie is about.

It gets much better when we do finally get to the connection between the movie and the real-world science. The scientific inaccuracies of the movie are discussed, as are the basics of genetics, the practicalities of gene editing using CRISPR-Cas9 and the ethical implications of gene editing and embryonic selection. The presenters have a balanced view on the ethics of gene editing: they are welcoming of GMO crops and say such technology should not be stigmatized, but also warn that human modification is a slippery slope that could take us from curing genetic diseases, such as sickle cell anaemia and Huntington's, to wealthy parents picking their children's eye colour. They point out that there is a great need for scientists to get involved in policy-making and the drafting of scientifically sound regulations on genetic technology, since the current decision-makers are not necessarily scientifically literate. They also encourage making scientific information and science education more accessible to the general public. The presenters field some entertaining listener questions via *Yahoo Questions* – such as: why can't we splice animal DNA into our genomes to give ourselves superpowers? – and answer them succinctly.

The podcast itself could have used some "gene editing" to cut out the off-topic diversions and to enhance the spotty audio. The presenters could have done themselves a favour by sparing their listeners their more juvenile comments – like their odd criticism of Uma Thurman's eyebrows. In other places, they could have said much more. Considering that the guest speaker is a postgraduate in a genetics lab, the scientific knowledge that was communicated was bit outdated, leaving out recent developments regarding human gene editing, and omitting the entire field of epigenetics.

This episode has its strong points, but is brought down by sloppy presentation. I give it a poor rating.

Episode link: <https://www.podbean.com/site/EpisodeDownload/PB9C806BT32XA>

The Skeptic's Guide to the Universe

Canyons on Mars

3/5 "Serious listening"

- Impressive pace and focus
- Covers a wide range of topics
- Critical and ethical discussions
- Good balance of serious and light tone
- Not strictly science-focused
- Spends too much time on some topics
- Less accessible for non-scientists



The Skeptic's Guide to the Universe styles itself as a science podcast for (obviously) the skeptical listener and critical thinker. This episode kicks with a couple of minutes of political commentary and the weekly pandemic update, and quickly gets into the latest COVID-19 science. I was very impressed by the ability of the presenters to maintain pace and focus despite the length of the podcast, the range of topics and the number of presenters. There is some friendly banter that is quite welcome in a podcast of this length, but it is kept to a happy minimum until the end, when the episode shifts to a more playful tone. The main topics come from many different fields of science – you will about hear new insights from orbital photos of vast Martian canyons; the ethics of brain-machine interface technology which allows people to write Tweets with nothing but their thoughts; learning styles and why they are a myth; and the ongoing debate as to whether ravens are smarter than apes.

The presenters clearly have a strong scientific background, and sometimes discuss science in a way that might be less accessible to non-scientists. However, *SGU* is not strictly focused on science, branching out at times into current events, policy and politics, education, lexicology and online security. A little bit too much time is spent on the presenters' opinions on the failings of learning style theory, clearly a hot-button issue. After a solid hour of scientific topics, the presenters get on to a half-hour of fun and informative science quiz games – "Science or Fiction", "Plant Evolution" and "What's That Noisy Noise", with some audience participation in the latter. There are some advertising breaks where the presenters spend some minutes plugging online courses and (amusingly) socks.

As a science podcast this episode had its fair share of strengths and weaknesses; I give it a mixed rating.

Episode link: <https://www.theskepticsguide.org/podcasts/episode-809>

PBS Eons: Mysteries of Deep Time

*The Extinct Human Species Discovered
Deep Within A Cave System*

4/5 "Story time"

- Great sound and narration
- Descriptive style
- Good example of a South African discovery
- Could have gone deeper into scientific context



You might be familiar with *PBS Eons* on YouTube, a channel that presents natural history in an accessible and aesthetically-pleasing format. *Mysteries of Deep Time*, the new Eons podcast launched earlier this year, looks set to carry on that tradition. The quality of the narration and sound strikes me immediately in this episode on human evolution. The gentle music sets the atmosphere and without being distracting and is the perfect backdrop to a highly narrative and descriptive summary of the South African discovery of a new human species, *Homo naledi*. This is a fascinating story on its own, but is especially well-told here. The centre-point is the mystery of how and why eighteen hominids found their way deep into a cave system, where their bones were preserved in a pristine state for hundreds of thousands of years. All hypotheses are carefully explored on the road to the most likely conclusion: *Homo naledi* went out of their way to dispose of their dead and dying. The odd features of this species and the fascinating fact that at one time they shared the Earth with *Homo sapiens* (recalling Yuval Noah Harari's book *Sapiens*) are included in this well-paced story, which ends on a profound and satisfying note. It has implications for how we consider our extinct relatives and ourselves as intelligent, spiritual beings.

The only issue I take with this episode is that more time could have been spent on contextualizing the discovery of *Homo naledi* by relating it to the history of other hominid findings. Otherwise, I was impressed by the presentation of the episode and found it informative and thought-provoking. I give it a good rating.

Episode link: <https://beta.prx.org/stories/404073>

Bonus Mini podcast, mini review!

Scientific American 60-Second Science *Sonified Bugs*

4/5 "Short and sweet"

- Fascinating topic
- Quality editing
- A bit overcondensed



This bite-sized episode is about a fascinating topic that lends itself well to the podcast format: sonified chemistry, the art of representing molecular information with sounds. The professional sound editing immediately caught my attention from the intro and held it throughout. In keeping with the brief nature of this podcast, all of the key information is communicated in just seven minutes. The episode is a bit too short for my liking – the briefing from the guest speaker and the science behind sonification and its use in insect pest control are all reduced to a few soundbites. Nonetheless it also manages to squeeze in the "musification" of DNA as a tool for geneticists and even the psychological effects of chemical sounds on willing human test subjects, which can be used to intuit what effects these chemicals might have on animals without exposing them to these chemicals. I give this episode a good rating.

Episode link: <https://www.scientificamerican.com/podcast/episode/these-bugs-produce-smelly-defenses-that-need-to-be-heard-to-be-believed/>

Thanks for reading!

PODCAST REVIEWS

In collaboration with,



Salmaan Banoo

PROLOGUE

What are podcasts?

Podcasts were formally digital audio files that were freely available for listening and downloading through the internet, but in recent times, podcasts have taken on several different forms and can now be accessed through a vast range of platforms. Podcasts are presented in various formats, ranging from audio-only to audio-visuals, as well as videographic recordings. There are several different types of podcasts, from interview- and conversation styles, to monologue- and storytelling types, and almost everything in-between. Basically, podcasts are classified as any collection of episodic media that can be listened to. The main purpose of podcasts is to convey a message to an audience, and this can be seen as a way to learn new things, to keep up to date on certain topics, to get inspiration, or to simply keep oneself entertained.

Authored by,
Salmaan Banoo
Hons Student, Rhodes University



Scientific dissemination

Podcasts can be highly effective in conveying scientific information, when done right. This media format can be beneficial to scientific dissemination if it is sufficiently backed by scientific knowledge. To adequately meet that criterion, podcasts should be run for a reasonable period of time, so that factual information can be passed on thoroughly and truthfully. If podcasts veer away from these guidelines however, the message portrayed may be partial and fragmented, ultimately hindering the dissemination of science.

Writer's opinion

As mentioned before, podcasts need to fulfil specific criteria in order for them to be effective, but, conclusively, I think podcasts can be influential and constructive when presented in a manner that appeals to any given audience. Prior to this set task, I occasionally listened to podcasts in times of leisure, yet never took the opportunity to enrich my scientific understanding. That is sure to change now!



Climate Change

Adaptation of Plants to Climate Change



Duration: 11 Minutes

Presenter(s): Janet Emmanuel

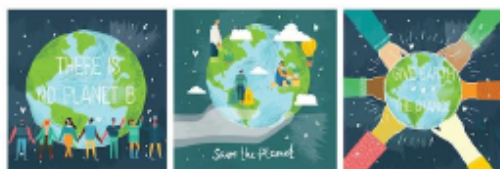
Guest(s): Michael Donoghue (Professor of Ecology and Evolutionary Biology)

Style(s): Interview, Discussion

<https://podcasts.apple.com/za/podcast/climate-change/id387959113?i=1000086217170>

DESCRIPTION

The episode launches with host Janet Emmanuel introducing the podcast and outlining what will be covered in the episode. She then proceeds to ask Professor Donoghue to explain how he got started on his study, followed by an expansion of his research findings. Professor Donoghue answers this question by describing that his research looked at the evolution and adaptations of evening primroses that have been subject to significant climatic variation over the past 10000 years. The core focus of his research was the interactions and relationships between plants in the given environment, as well as the phenology (natural timing of events) of the plants, and how these factors influenced their ability to adapt to climate change. He sums up his findings by stating that in some cases, significant evolutionary shifts can happen over a short period of time, meaning that some species of plants could possibly adapt to rapid climate change within a few generational cycles.



CRITIQUE

This episode covered an interesting topic, however the overall delivery was rather underwhelming. Due to the short time frame, the expansion and explanation of the topic was limited, yet the intended message of the episode was clear. This episode scored poorly for audience participation, as there was no form of audience engagement. The episode was thought to be scientifically credible, as it was evident that the guest speaker was highly knowledgeable and well-read in his field. The audio quality and delivery for this podcast was fair. Though the guest speaker was enthusiastic and engaging, the host had a monotone and dull tone throughout.

RANKING

	Points scored: (0=bad, 1=fair, 2=excellent)
Interesting content	1
Consistency with topic	2
Degree of explanation of topic	1
Audience range	2
Public participation	0
Scientific credibility	2
Audio quality and delivery	1



60-Second Science

Chimps Apply Insects to Their Wounds



Duration: 2 Minutes

Presenter(s): Christopher Intagliata

Guest(s): Simone Pika (Cognitive biologist)

Style(s): Summary, Repurposed-content, Monologue

<https://podcasts.apple.com/za/podcast/chimps-apply-insects-to-their-wounds/id189330872?i=1000553328104>

DESCRIPTION

This short episode begins by briefly explaining how far chimpanzees have evolved, by stating that they have the ability to make tools, express emotion and even outsmart humans at memory games. The presenter, Intagliata, then goes on to state that researchers have recently observed never-before-seen behaviour in chimpanzees, a behaviour that is being described as a treatment for wounds! Chimpanzees have been seen to catch flying-insects and apply them to wounds. The episode then cuts to an audio of Simone Pika, a cognitive biologist researching chimpanzees in Central Africa. Intagliata then goes on to state the possible reasons for this behaviour could be of medicinal significance or simply a social practice. The presenter closes off the episode by stating that entomologists could be the key in identifying the flying-insects to learn more about the behaviour shown by the chimpanzees.



CRITIQUE

The episode is very short, yet interesting and easy to understand. It keeps with the topic and briefly outlines key points surrounding the topic. I found this episode to be too short, as there was insufficient scientific knowledge conveyed to the audience. This would make for a great full-length episode, as behavioural ecology is a fascinating subject! The audience range is good, as this podcast is widely available to a range of viewers across numerous platforms. The scientific credibility is fair, as it makes use of a recording of a cognitive biologist commenting on chimpanzee behaviour, but the factual science in the episode is lacking.

RANKING

	Points scored: (0-bad, 1-fair, 2-excellent)
Interesting content	2
Consistency with topic	2
Degree of explanation of topic	1
Audience range	2
Public participation	0
Scientific credibility	1
Audio quality and delivery	2



The Infinite Monkey Cage

What is Life?



Duration: 40 Minutes

Presenter(s): Brian Cox (Physicist) & Robin Ince (Comedian)

Guest(s): Jo Brand & Ross Noble (Comedians), Sir Paul Nurse (Noble Prize winner) & Prof. Aoife McLysaght (Geneticist)

Style(s): Comedy, Interview, Discussion

<https://podcasts.apple.com/za/podcast/what-is-life/id343580439?i=1000485541873>

DESCRIPTION

The episode kicks off with the presenters introducing each of the guests. The guests then introduce themselves and give a short description of what is their definition of life. The first subtopic discussed is what is a virus and whether or not it fits the description of a living organism. From viruses, the discussion shifts to DNA and replication, as well as traits and characteristic inheritance. The discussion then focusses on the chemistry of life and the chemical activity that takes place at a molecular level, with a touch on the hypothesis of the origin of life. The discussion also covers how vast and incomprehensible the origin of life actually is, and mentions that all attempts to explain the beginning of life are arguments backed only in theory. The discussion then flows into the three fundamentals of biology, the first being cellular organisation and purpose, the second being thought processing and consciousness, and the third being the origin of life. The discussion rounds off with each guest answering the question of "what is the best reason for not destroying all life on Earth?".



CRITIQUE

The podcast is easy to follow and understand. The guest comedians on the podcast bring a light-hearted spin to the discussion, with no clear scientific backing, while the scientists are far more factual and have a more direct approach to the topic. The topic is explained thoroughly throughout the podcast, and the presenters make good use of the time to discuss the topic. The presenters make a reference to earlier Infinite Monkey Cage podcasts in a joking manner, hinting at the expectation that the podcast has a wide audience that listen to it as a form of entertainment. The podcast has a significant degree of public participation, as it is recorded alongside a live virtual audience that are asked general questions and broadly, yet briefly, brought into the discussion. The public are also presented with a question prior to the podcast being recorded and asked to write-in with their responses. The content covered in the podcast is beyond interesting and is profusely discussed throughout!

RANKING

Points scored:
(0=bad, 1=fair, 2=excellent)

RANKING	Points scored: (0=bad, 1=fair, 2=excellent)
Interesting content	2
Consistency with topic	1
Degree of explanation of topic	2
Audience range	2
Public participation	1
Scientific credibility	2
Audio quality and delivery	2



Auto/Motive Podcast

Turbocharging, Explanation and History



Duration: 29 Minutes

Presenter(s): A/M

Guest(s): N/A

Style(s): Information, Monologue (Solo)

<https://podcasts.apple.com/za/podcast/turbocharging-explanation-and-history/id1516124279?i=1000512951197>

DESCRIPTION

The podcast kicks off by outlining what the episode will focus on, "the history, and explaining, TURBOCHARGING!". The presenter starts by saying that in order to understand how turbocharging works, one needs to understand how forced induction works, and in order to understand the physics behind that, one needs to have a basic understanding of how a combustion engine functions. The four basic requirements for an engine as stated by the presenter are air, fuel, compression and spark. The reactions and combustions that takes place can simply be summed up as the conversion from chemical energy to mechanical energy. The presenter outlines how engines receive air without forced induction, through atmospheric pressure. He also explains how altitude and temperature affect air density, and ultimately the air pressure available to the engine. He then goes on to explain how turbochargers function, with the main purpose being to provide more power, by recycling exhaust gases and repurposing that energy in the form of compressed air which is forced into the engine. The presenter then shifts the focus on to the different types of turbochargers and engine set ups, followed by the pros and cons of turbocharged engines. The episode is rounded off with a short summary of the history of turbocharging, dating all the way back to 1896!

CRITIQUE

This episode was thoroughly enjoyable! It was easy to follow and understand, however I think a member of the general public who does not have a passion for the automotive world may find this podcast a difficult one to follow. The presenter provided an ample amount of information on the functional basics of engines and induction, and exceedingly explained the topic given the short time frame. The audience range is fair, as the podcast is accessible on various platforms, however the topic is very specific and would only appeal to a focussed, target audience. The podcast had a bad rating for public participation, as there was no form of public participation or interaction. The scientific credibility was fair, as the sources used for the information conveyed were all provided for self-access, however, perhaps an expert in the field could have been invited as a guest.

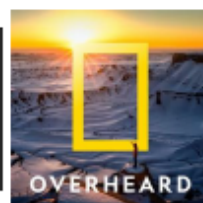
RANKING

	Points scored: (0=bad, 1=fair, 2=excellent)
Interesting content	2
Consistency with topic	2
Degree of explanation of topic	2
Audience range	1
Public participation	0
Scientific credibility	1
Audio quality and delivery	2



Overheard at National Geographic

The Secret Culture of Killer Whales



Duration: 29 Minutes

Presenter(s): Amy Briggs & Peter Gwin

Guest(s): Brian Skerry (National Geographic Photographer)

Style(s): Interview, Narration

<https://podcasts.apple.com/za/podcast/overheard-at-national-geographic/id1466697207?i=1000516934402>

DESCRIPTION

The episode starts off with guest speaker Brian Skerry narrating his experience as a diver looking for a pod of killer whales off the coast of Norway. Skerry then goes on to deliver his impression of orcas, stating that they are the most intelligent and unpredictable creatures in the open ocean. The show then jumps to a voiceover of host Peter Gwin, who describes orcas as “freakishly smart”, followed by a short description of their social and curious behaviours. Cue the music, Gwin then introduces the show, with a brief overview of what the podcast is all about. Once under way, the host asks Skerry to tell the story of how he got into diving and landed the job at National Geographic, which, for the record, is seriously out of the ordinary! Skerry then goes on to ardently narrate his experience of observing a pod of orcas hunting a school of herring while free-diving. He contrasts this experience by vividly describing the changes he saw and sensed in the orca behaviour when he switched to using scuba gear. The episode then progresses to the topic of feeding behaviour and dietary preference of different orca populations in different parts of the world. Skerry follows this up by detailing a marvellously close encounter he had with a female orca who offered him something to eat! The cheerful tone of the episode then drastically turns melancholic, as Skerry describes the emotion he felt when seeing what appeared to be a whale-mourning ceremony. This is followed by Gwin and Skerry briefly discussing the dangers and threats

that orcas are faced with across the globe, leading on to why Skerry believes his work is imperative in conserving and protecting these whale populations. The episode is rounded off by Gwin presenting Skerry with the question “what was his closest call?”. Thereafter, Gwin ends the podcast by providing listeners with information on where to find Skerry’s latest works, both in podcast format and as a published book.

CRITIQUE

This episode is justifiably the best! From the fervid emotional story-telling to the perfectly timed background noises, this episode is captivating and entertaining right from the off! The episode kept with and expanded on the topic throughout, with an in-depth, first-hand experience, which ultimately helped it score highly for credibility. The episode scored well for audience range, as it is widely available across a range of platforms. The episode did however rank poorly in terms of public participation, as there was no degree of public interaction during the interview.

RANKING

	Points scored: (0-bad, 1-fair, 2-excellent)
Interesting content	2
Consistency with topic	2
Degree of explanation of topic	2
Audience range	2
Public participation	0
Scientific credibility	2
Audio quality and delivery	2

PODCAST REVIEWS

In collaboration with,



EPILOGUE



Final thoughts,

Throughout this task, I listened to podcasts that ranged across a spectrum of emotion. Some were intensely engaging, some jovial and comedic, while others were rather dreary and boring. Yet all of these podcasts had one thing in common, they were easily accessible and convenient to listen to. My overall impression of podcasts is I find them to be great, as they're an opportune way of keeping entertained, staying up to date with the latest news, or enriching one's understanding of every topic, ever!



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